STRATEGY
Interoperability for crisis management

Facilitating EU pre-standardization process through streamlining and validating interoperability in systems and procedures involved in the crisis management cycle

D4.4 Final Evaluation and validation of crisis management solutions and pre-standardization items (from all user groups)

Former D4.4 Pre-Standardisation items report

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The STRATEGY project

STRATEGY will develop a pan-European framework of the pre-standardisation activities for systems, solutions and procedures, addressing crisis management, validated by sustainable tests and evaluation frameworks which will improve the crisis management and disaster resilience capabilities, including European Defence Agency (EDA) initiatives in the CBRNE area. Based on the needs identified from previous EU initiatives and the desktop research on the EU priorities, the project addresses eight streams within crisis management: Search and Rescue, Critical Infrastructure Protection, Response Planning, Command and Control, Early Warning Systems and Rapid Damage Assessment, CBRNE, Training and Terminology.

Aiming at selecting and implementing existing, evolving, and new standards within solutions, tools and procedural guidelines and recommendations STRATEGY will streamline and validate technical and organisational interoperability in a fully transboundary configuration with means of standards through the implementation of use cases involving EU and National Standardisation bodies.

STRATEGY aims at a consistent presence of all actively involved experts from security stakeholders throughout all stages of standardisation, from the preparatory work done in Programming Mandates until the concrete standardisation work in Technical Committees (TCs) and Common Working Groups (CWGs) to ensure the success of the standardisation actions of the EC.

The ultimate project goal is to strengthen the resilience of EU against all types of natural & manmade disasters (multi-hazard approach), by ensuring first responder (FR) safety and empowering their operational capacity through validating the standards of next-generation solutions and procedures, providing an effective and efficient collaborative response.

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Definitions and abbreviations

Table 1: List of Definitions

Term	Explanation
CBRN-E	Chemical, biological, radiological, nuclear and explosive threats
CBRNE	Chemical, biological, radiological, nuclear and explosive threats
CEN	European Committee for Standardization
CEN/CENELEC Workshop	working platform open to the participation of any interested parties for elaboration of CEN/CENELEC Workshop Agreements [SOURCE: CEN/CLC Guide 29, 2.1]
CEN/CENELEC Workshop Agreement (CWA)	CEN/CENELEC deliverable, developed by a Workshop, which reflects an agreement between identified individuals and organizations responsible for its contents, and which is made available by CEN/CENELEC in at least one of the official languages SOURCE: CEN/CLC Guide 29, 2.2]
Command & Control	Activities of target-orientated decision making, including assessing the situation, planning, implementing decisions and controlling the effects of implementation on the incident
Critical Infrastructure	Critical infrastructure is an asset or system which is essential for the maintenance of vital societal functions.
Discussion based exercise	A discussion-based exercise in response to a scenario intended to generate a dialogue of various issues to facilitate a conceptual understanding, identify strengths and areas for improvement, and/or achieve changes in perceptions about plans, policies, or procedures
Early Warning	Warning that tells in advance that something serious or dangerous is or might going to happen
End users	Standardization stakeholders who take advantage of standards in their daily operation, such as first responders and non-governmental organizations
European Standardization Organization (ESO)	The European Standardisation Organisations (CEN, CENELEC and ETSI) are officially recognised by Regulation (EU) No 1025/2012 as providers of European standards
Full-scale exercise (FSX)	A demonstration of the feasibility of standardisation items in minimum subset of the operational environment that is required to demonstrate critical functions of the pre-standard performance in its operational environment
KPI	Key Performance Indicator KPIs are the critical (key) quantifiable indicators of progress toward an intended result or other equivalent definition
Pre-standardization activities	Activities filling the gap between pre-normative research and innovation on the one hand, and formal standardization activities on the other hand
Technical Specification	Document that prescribes technical requirements to be fulfilled by a product, process or service [SOURCE: EN 45020:2006, 3.4]

Term	Explanation
	NOTE 1 A technical specification should indicate, whenever appropriate, the procedure(s) by means of which it may be determined whether the requirements given are fulfilled. NOTE 2 A technical specification may be a standard, a part of a standard or independent of a standard
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Table 2: List of Abbreviations

Abbreviation	Explanation
CBRN-E	Chemical, Biological, Radiological, Nuclear and Explosive
CEN	European Committee for Standardization
CENELEC	European Committee for Electrotechnical Standardization
CI	Critical Infrastructure
CIP	Critical Intrastuctures Protection
CIR	Critical Infrastructure Representative
CTP	Custody Transfer Point

Abbreviation	Explanation
CW	CEN/CENELEC Workshop
CWA	CEN/CENELEC Workshop Agreements
CWG	Common Working Group
dCoC	Digital Chain of custody
DoA	Description of the Action
EC	European Commission
EDA	European Defence Agency
EDXL-CAP	Standard of the EDXL family of standards (Emergency Data Exchange Language) for Common Alerting Protocol
EDXL-SitRep	Standard of the EDXL family of standards (Emergency Data Exchange Language) for the generation of Situation Reports
EMS	Emergency Medical Services
ESO	European Standardization Organizations (CEN, CENELEC and ETSI)
ETSI	European Telecommunications Standards Institute
EU	European Union
FR	First Responder
FSX	Full-Scale Exercise
KPI	Key Performance Indicator
LEA	Law Enforcement Agency
NCSl	Number of Concepts in the Semantic layer
NWIP	New Work Item Proposal
OASIS OPEN	Technical standardization consortium in charge of the EDXL family of standards
OGC	Open Geospatial Consortium: Technical standardization consortium specialized in geographical standards
PC	Project Coordinator
RAF	Rapid Assessment Framework
RTO	Research and Technology Organization
SA	Standardisation Activity
SAR	Search and Rescue
SMART	Specific, Measurable, Achievable, Relevant, and Time-bound
SMAT	Social Media Alerting Tool

Abbreviation	Explanation
SOS	Sensor Observation Service: A standard that defines the format of data transmitted by sensors from the OGC
TC	Technical Committee
TS	Technical Specifications
TTX	Tabletop exercise
WP	Work Package
XMT	eXercise Management Team

Executive Summary

In the framework of the STRATEGY project, crisis management solutions in the form of pre-standardisation and standardisation items (CWA and TS, respectively) are developed in cooperation with research institutes, technology providers, National Standardisation Bodies and practitioners. To ensure the quality of the solutions and their suitability to become a standard, many tabletop exercises (TTX), workshops, small scale validation events and a full-scale exercise (FSX) were performed to test the content of the documents. The evaluation exercises were performed in realistic use cases and allowed for the collection of valuable feedback from participants (see also deliverables of work packages WP2 and WP3). The evaluation of the solutions according to the results of the different events is subject of Task 4.3 and, therefore, of the assigned deliverables D4.3 [35] and D4.4. In that way the potential of the STRATEGY crisis management solutions to fill respective standardisation gaps and finally to become standards has been assessed.

The project set out to determine criteria, define Key Performance Indicators (KPIs) based on these criteria, and developed questions to evaluate the KPIs through data derived from questionnaires held during the exercises (TTXs and FSX).

Since the STRATEGY project covers eight different thematic streams, for each stream a different group of experts was established to perform a specific evaluation. In addition, a special group developed criteria and a questionnaire for formal aspects of the CWA/TSs. The whole process was guided and reviewed by a supervising group.

Whilst the status of the CWAs/TSs at the time of the related TTXs is reflected in an intermediate evaluation, described in D4.3 [35], the present deliverable D4.4 comprises the results of the FSX held in Gualdo Tadino, Italy, in March 2023, the interoperability workshops, the formal aspects evaluations and an update of a ResiStand Framework assessment for each CWA/TS, which was initially performed in WP2. During the different experts' sessions and practical exercises, notes were made by evaluation teams, feedback was collected in interviews, during small group and plenary discussions and by using stream specific questionnaires that were filled in by the participants. These data were analysed, qualitatively and quantitatively. Details of this analysis are outlined per CWA/TS in chapter 3 of the current deliverable. The final evaluation showed promising results for all cases. Almost all KPIs were evaluated positively in the questionnaires carried out during the FSX and in the questionnaires regarding the formal aspects of the CWAs/TSs. Lively discussions revealed strong interest in the results of the project and provided valuable feedback and suggestions for improvement, which could be integrated in the very last version of the CWAs/TSs. The good quality of the CWAs/TSs is underlined furthermore by the fact, that all of them are meanwhile published by CEN-CENELEC. Two CWAs (Stream 3 and 4) have been selected to be promoted as New Work Item Proposals (NWIP) and the two TSs (Stream 6,) are already submitted as NWIPs to CEN/TC 391 for voting.

1 Introduction

The present deliverable is a follow up to the STRATEGY deliverable D4.3 [35] “Intermediate Evaluation and validation of crisis management solutions and pre-standardization items (from all user groups)”. Originally it was planned to provide D4.4 as an updated version of D4.3 [35] containing the intermediate and – as a new amended part - the final evaluation. But it turned out, that the document would become that large and bulky that it was decided to conceive D4.4 as a part 2 with no repetition of the content in D4.3. [35]

An overview of STRATEGY Pre-Standardisation activities is given in the table below. In the STRATEGY project, Crisis Management related gaps were identified [1], solutions were proposed, and a considerable number of pre-standardisation documents, related to these solutions, were developed, for at least one of each of the eight streams of the project (see STRATEGY deliverables D2.1 [4] to D2.8 [11], D3.2 [26] to D3.9 [33] and D5.1[36] to D5.8 [43]).

Table 3: Overview of STRATEGY Pre-Standardisation Activities (cf. Table 3 in D4.3)

Stream	Pre-Standardisation Document	Standardisation Activity ID
1 Search & Rescue	CWA Requirements for acquiring digital information from victims during Search and Rescue operations.	**SA 1**
2 Critical Infrastructures	CWA Emergency management – Incident situational reporting for Critical Infrastructures	**SA 2.1**
	CWA Semantic layer definition and suitability of EDXL-CAP+EDXL-SitRep	**SA 2.2**
3 Response Planning	CWA Structuring an emergency response plan for crisis management stakeholders	**SA 3**
4 Command & Control	CWA Collaborative emergency response – Common addressing format and emergency identification protocol	**SA 4.1**
	CWA Management of forest fire incidents – SITAC-based symbology	**SA 4.2**
5 Early Warning	CWA Requirements and recommendations for social media early warning messages in crisis and disaster management	**SA 5.1**
	CWA Exchanging of building and infrastructure damage information with Common Alerting Protocol	**SA 5.2**
6 CBRN-E	TS Societal and citizen security — Digital Chain of Custody for CBRN-E evidence — Part 1: Overview and concepts	**SA 6.1**

Stream	Pre-Standardisation Document	Standardisation Activity ID
	TS Societal and citizen security — Digital Chain of Custody for CBRN-E evidence — Part 2: Data management and audit	**SA 6.2**
7 Training	CWA Specifications for Digital Scenarios for Crisis Management Exercises	**SA 7.1**
	CWA Evaluation of Exercises - Implementation Guidelines	**SA 7.2**
8 Terminology	CWA Guideline for the mapping of terminology and icons	**S A8**

Since testing and evaluating play a crucial role to ensure the relevance and suitability of the selected pre-standardisation items (solutions), many tabletop exercises, workshops, small scale validation events and a full-scale exercise (FSX) were performed to test the content of the documents in realistic use cases and to receive feedback from practitioners.

The evaluation of the solutions according to the results of the different events is subject of Task 4.3 and, therefore, of the deliverables D4.3 [35] and D4.4. Whilst D4.3 [35] refers to the status of evaluation after each pre-standardisation document had been introduced in a tabletop exercise (see D2.10 [12]to D2.18 [20]) and serves also as an intermediate orientation for the project activities, the present deliverable D4.4 contains the final evaluation of the solutions. It should be noted that the deliverables D4.3 [35] and D4.4 do not cover the evaluation of exercises from an organisational perspective, which is rather documented in the deliverables of WP2 (D2.10 -D2.17) and in deliverable D4.1 [34], but it covers the evaluation of the pre-standardisation drafts that were tested.

The intended readers of the present deliverable are primarily the project participants responsible for the different pre-standardisation documents (CWAs and TSs) and the solutions, which are proposed there. The results of the evaluation, the indication of strengths and weaknesses together with suggestions by experts and practitioners, were made available to this target group immediately after the FSX and were supposed to serve as a basis for final improvements. Furthermore, all those who are interested in the reflection and evaluation of the STRATEGY project solutions may read the present deliverable.

In the way described above, Task 4.3 and the content of D4.4 supports WP5 “Pre-standardisation implementation activities” with respect to final improvements. Task T4.3 builds mainly on the work of WP3 “Implementation of selected standards and integration testing in the framework of each scenario” and Task T4.1 “Full scale exercise addressing all interoperability streams”. Also results from Tasks T1.3 “Regulatory, political, and organizational compliance of pre-standardisation activities” and T1.4 “Ethical, Societal, and Privacy Issues in the design, development, and implementation of standards” were considered.

For the criteria and KPIs, which were identified to ensure the quality of the proposed solutions, described in the pre-standardisation documents, the reader is referred to deliverable D4.3 [35], chapter 3. Since the problems and solutions differ from stream to stream, each subchapter of the deliverable D4.3 [35] refers to a specific stream, namely subchapter 3.1 to stream 1, subchapter 3.2 to stream 2 etc. The last subchapter (3.9) lists formal criteria and KPIs which refer to the form, not the content of the pre-standardisation documents and should, therefore, be used for all of them. These formal aspects were not evaluated in the intermediate evaluation, but in the final one, and are therefore included in D4.4.

The present deliverable is structured in the following way:

Chapter 2 resumes shortly the evaluation concept and indicates some special features, where the final evaluation is different from the intermediate one. More details can be found in the chapter 2 of deliverable D4.3 [35].

Chapter 3 presents results of the final evaluation, structured in subchapters according to the streams, from 3.1 (stream 1) to 3.8 (stream 8), similar as in deliverable D4.3 [35]. The subchapters are structured for each stream in a similar way:

Clause 1, the first part of each subchapter outlines the basis of the final evaluation. It addresses the status of the CWA or TS after the improvements according to the intermediate evaluation and describes the data, which were used for the final evaluation. Clause 2 presents results from expert sessions and testing during the FSX. In clause 3 and in clause 4 the results from the questionnaires (participant feedback forms) and the results of the formal aspects evaluation are analyzed respectively. In clause 5 the analysis, with the help of the ResiStand Assessment Framework (RAF), which was done already in an early phase of the project, is reconsidered and updated. The subchapters for each stream are completed with stream specific conclusions in clause 6.

Chapter 4 provides general conclusions to the evaluation of the solutions tested during the STRATEGY Project.

2 Summary of Concept and Approach of Evaluation and Validation

2.1 Working Groups and Work Structure

As outlined in STRATEGY deliverable D4.3 "Intermediate evaluation and validation of crisis management solutions and pre-standardization items (from all user groups)" [35] the evaluation and validation of the crisis management solutions, which were developed in the form of pre-standardisation documents related to eight streams in the STRATEGY project (see *Table 1*), is a complex and demanding challenge. Therefore, it was necessary to integrate specialists with diverse backgrounds into the evaluation process. To this end, a special procedure was set up and several groups were established.

The concept and the overall approach, including the ordering of process steps, are described in detail in deliverable D4.3 [35] in chapter 2. The following figure is supposed to serve as a short reminder, showing the whole process and the different groups.

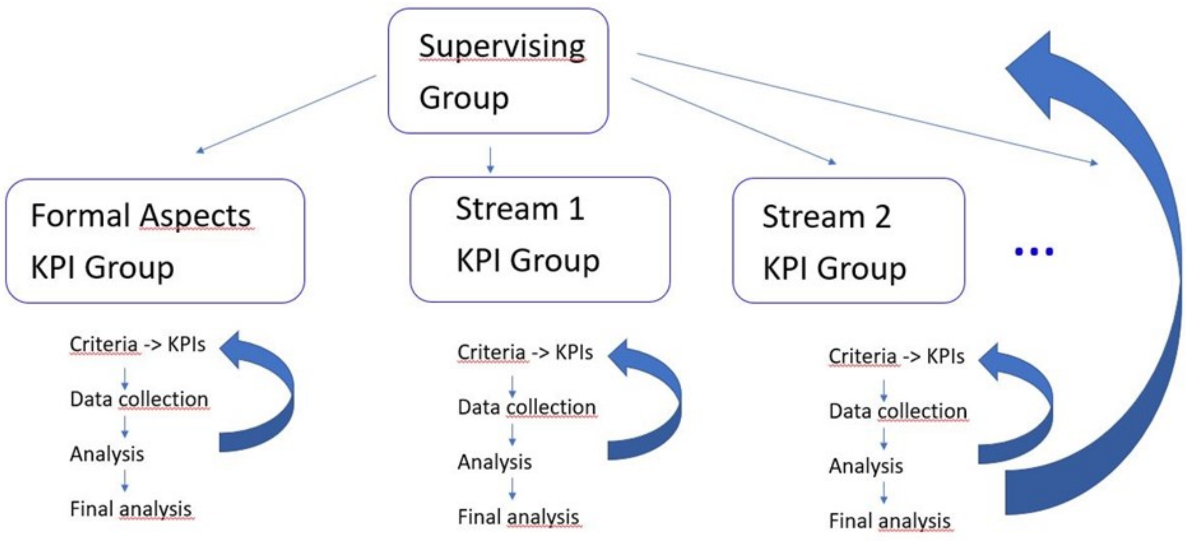


Figure 1: Overview of evaluation groups and process

The identification of the criteria and KPIs per stream and the related questionnaires (participants feedback forms) can be found in D4.3 [35] in chapter 3. Furthermore, the results of an intermediate evaluation are analysed in D4.3 [35], chapter 4. Here, in the intermediate evaluation, all data were taken into consideration that were collected before and during the TTXs, which were performed during the project and were focused on the different streams and their related CWAs or TSs.

The present document outlines the results of the evaluation process after the TTXs. Also, in this evaluation phase, the supervising group checked the progress, discussed overarching questions, and provided feedback on (preliminary) results of the groups.

2.2 Basis of the Final Evaluation in General

Based on the intermediate evaluation the pre-standard documents were further developed. In accordance with this development and based on the experience from the TTXs, the KPIs and the questionnaires were possibly modified (see blue arrows in the figure above). Then, the improved CWAs and TSs were evaluated and validated in a final evaluation process.

An important basis for final evaluation were the activities during the Full-scale exercise (FSX) in March 2023 in in the municipalities of Gualdo Tadino and Valfabbrica (Italy), hosted by the Italian National Fire and Rescue Service. During that event the solutions have been demonstrated, discussed, tested and evaluated by evaluation teams and participants. The aim of the FSX was to showcase all integrated systems and methodologies of the different pre-standardization items, evaluate the concepts of the CWAs/TSs in realistic operational conditions across different threat scenarios with simulation of fires, post-flood relief, and evacuation from sports facilities, including real actors who represented First Responders, Emergency Management Centres and Crisis Management Response Teams. Apart from this real exercise there were expert sessions scheduled during the week, dedicated to the different CWAs/TSs, where the participants could get familiar with the new developed solutions in concrete scenarios, where they could discuss advantages and disadvantages, and give feedback. The FSX is described in detail in STRATEGY deliverable D4.1 “Full scale exercise demonstration report” [34].

An overview of all data, which were used for the final evaluation of the content of the pre-standardisation documents, can be found for each stream in the present deliverable in chapter 3, from 3.1 (stream 1) to 3.8 (stream 8). Also, all organisations with attendants in the respective expert sessions and all evaluation team members are listed there. As for the TTX, also for the FSX, the announcement of an FSX data collection team (evaluation team) related to the respective stream with a dedicated lead evaluator was a task of the KPI groups. For the FSX evaluation teams a guidance was elaborated as internal document: “Guidance for Evaluation Teams appointed for the STRATEGY FSX in Gualdo Tadino 27.-30.3.2023”, see appendix 1. Also, other guidelines, which were designed for the different groups in the evaluation process as living documents, are attached in the present deliverable.

The formulas used for the quantitative analysis of the questionnaires and details about the elaboration of the questionnaires can be found in D4.3 [35] in chapter 2.

2.3 The Formal Aspects Evaluation in General

As stated in D4.3 [35] the Formal Aspects Evaluation was not part of the intermediate evaluation since the assessed CWAs/TSs should be mature enough for this purpose. Nevertheless, the Formal Aspects KPI Group elaborated criteria, KPIs and questions already during the intermediate evaluation phase so that the related questionnaires timely became available. More details to the criteria, KPIs and questionnaires can be found in D4.3 [35], chapter 3.9. It was decided to perform only one evaluation since the CWAs and TSs were formally well prepared towards the end of the project and a 2nd review (a repeated “data collection phase” in the above figure) could be omitted. For each stream, project partners from national standardisation bodies (NSBs), which were not authors of the CWA or TS, acted as formal aspects evaluator, filled in the related questionnaire and were on hand for clarifications. Moreover, the questionnaire for the formal aspects was further elaborated as a separate document. It would be an additional output of the STRATEGY project, which might help future authors and reviewers to generate pre-standard documents with high formal quality.

2.4 The ResiStand Assessment Framework Supported Evaluation in General

In addition, for each CWA/TS the ResiStand Assessment Framework (RAF) – provided by TNO – has been used to support the evaluation. The RAF is a tool which was developed in the ResiStand project [44] for assessing proposed standardization activities - for details see D1.2 [2] and D3.1 [21]. The RAF was designed for assessing the potential of a proposed new standardization initiative in the domain of disaster resilience and crisis management to be implemented as a new standard. It enables systematic mapping of potential benefits, impact, and feasibility of an initiative with the help of Excel tables. It also provides a tool for examining the extent to which the proposed standard has considered any essential ethical, legal and social issues, and to select the organizational conditions, under which the standard would be developed and implemented. To underscore the significance of the proposed CWA, comprehensive details on the anticipated advantages for end users, industry, and researchers are also presented. Specific questions on potential advantages for end users/practitioners and financial advantages/technological developments for industry and/or research organizations are answered on dedicated tabs within the Excel worksheet to undertake a thorough analysis of the CWA's impact. These inquiries aid in outlining possible advantages and thoroughly evaluating impact. The RAF tool was already used in WP2 to make an initial assessment of the standardization opportunities. Towards the end of the STRATEGY project these assessments were reconsidered, updated and included into the final evaluation.

3 Final Evaluation Results

3.1 Final Evaluation of Stream 1 Search and Rescue

3.1.1 Basis of the Final Evaluation

The Subject of the final evaluation in Stream 1 – “Search and rescue” was the concepts presented in the

> CWA “Requirements for acquiring digital information from victims during Search and Rescue operations”.

The evaluation of the standardization activities conducted within Stream 1 is mainly based on three pillars: TTX evaluation, FSX evaluation and comments received through the public. The results from the TTX dedicated to Stream 1 and conducted in June 2022 (23/06/2022) were already reported in the D4.3 [35] “Intermediate Evaluation and Validation of Crisis Management solutions”. A great deal of the suggestions made previously during the TTX and during other discussions with experts could be considered in the current version of the CWA. The final evaluation was performed on the base of the FSX conducted in March 2023 (27-31/3/2023).

Stream 1 participated in the FSX with the above-mentioned CWA, dealing with requirements of digital victim triaging systems, including also points of safety, security, data protection, usability and system compatibility. The CWA is intended to be used by Search and Rescue equipment developers, but since First Responders and medical personnel should benefit from the guidelines, they play a crucial role in the elaboration of the requirements. The affiliations of the participants regarding Stream 1 and their role in the FSX event can be seen in the table below.

Table 4: List of organisations with attendants in the FSX related to Stream 1.

Organisation Name	Type/Description
CNVVF (Italy)	End-user (Firefighter)
Valabre (France)	End-user (Firefighter)
Hellenic Police (Greece)	End-user (Law Enforcement)
Ministry of National Defence (Greece)	End-user (Armed Forces)
Prefettura – Ufficio Territoriale del Governo di Perugia (Italy)	End-user (Local Government Agency Prefecture)
STWS (Greece)	SME - Small and medium-sized enterprise
Carabinieri (Italy)	End-user (Law Enforcement)
Deep Blue (Italy)	RTO (Research and Technology Organisation)

Stream 1 involvement in the FSX took place in 2 phases. Firstly, on 28/3/2023, with a 4-hour slot, that was dedicated to present the current status of the CWA document, to demonstrate a potential solution adapted to the propositions of the CWA and to conduct an open discussion based on the end-user questions. In the end of the discussions, a questionnaire has been shared among the participants to provide input. Secondly, on 30/3/2023, Stream 1 led the first part of the exercise providing a solution to triage and track victims after a mass casualty incident. Details about the FSX can be found in D4.1 [34] “Full scale exercise demonstration report” and especially for the activities regarding stream 1 in D3.2 [26] “Standards selection, implementation, testing and interoperability assessment for Search and Rescue”.

To support the data collection and evaluate the SaR FSX results, a specific evaluation team was set up, see Table 5.

Table 5: Stream 1 FSX Evaluation team

Full Name	Affiliation	Role / Required expertise
Spyros Athanasiadis	ICCS	KPI leader
Luis MF Carvalho	CINAMIL	KPI team
Giannis Chasiotis	STWS	KPI team
Eleftherios Ouzounoglou	ICCS	KPI team
Panagiotis Michalis	ICCS	KPI team

3.1.2 Results from Experts Discussions and Testing during the FSX

As previously mentioned, during the FSX, a 4-hour slot was scheduled for a round table discussion with experts in the domain of tracking and triage victims, having on board First Responders, Civil protection, a disaster victim identification team and technology providers. Everyone seemed intrigued by the idea of a digital victim tracking and triaging system, as it would allow for a centralised system to track the triage that is done by different organisations as well as tracking patients as they move from the field to hospitals. It is important to distinguish that the CWA is not to promote specific solutions and minimizing the potential technical solutions could be used, but requirements on the using of digital triage systems based on 3 pillars: digital victim tracking system, digital victim software and the digital victim supportive services. Several specific questions and issues were brought up during the session. The main points are summarized below:

1. **Data traffic prioritization**: It could be that telephone lines are down or not available during an incident, it should be possible to have mobile devices that have a sim card with prioritization of traffic and that can be used under specific VPN of the organisation to be faster in transmitting data. Also, it is important to have specific procedures to operate the system even without connection and to enable first responders to login in the application also without connection.
2. **Code of the event definition**: It is important to determine who can define the unique code of the incident. Is it the first person on the scene? The incident commander? There should be specific rules in the software to define automatically the unique name of the event (maybe something like type of event + place + date). It should also be clear which code is the one for the current event, so that people don’t accidentally register tags under an incident that occurred in the past. It should

also be clear when an event has been created so that two different responders don't make different names for the same event.

3. **Rules and chain of command and data sharing**: Establish a chain of command and implement it in the software which is important to manage the event smoothly. So, it is important to be able to define different level of views for the data, and define carefully who can access the overall view of all victims, follow what are the regulations and laws of the specific country. First responders on the field will collect all required data. The system will apply the different filters to show the specific data to the different organisations. This is also important when sharing data with hospitals or other 3rd party organisations. Only certain data should be shared to maintain privacy and ethics regulations. The types of data that each organisation has access to, needs to be specified. Additionally, it should be specified who can see the information once it has been entered into the mobile application.
4. **Simplifying procedures**: When in the field, first responders should be able to spend less time possible to triage people in order to assist the ones more in need, so it could be provided some simplified procedure to register "green codes". Additionally, making sure none of the fields are mandatory will speed up the process.
5. **Command centre specifications**: It could be important to have in the command and control also a visualisation of the number of tags and the information collected that each organisation has and that are available and/or used. So, at the central level the use of the tags can be coordinated, and in case of need, requests could be sent to other organisations for providing more tags or exchange information.

Also, the CWA was stress tested during the actual exercise leading the first part of the scenario using 70 tags (50 commercial tags that are used in an EU hospital and 20 electronic tags/prototypes developed by ICCS in the INGENIOUS EU project) and tested by a large number of first responders.

Appropriate adaptations were made to reflect the evolution of the final draft version of the CWA. Using a tool adapted according to the CWA was helpful for the participants (external experts and operational stakeholders) of the FSX to understand in depth the gap identified and the solution by the current CWA to solve this problem.

3.1.3 Questionnaire Results

Based on the criteria and KPIs, which are listed in D4.3 [35] (cf. section 3.1), the questionnaire originally created for the TTX was updated with minor modifications according to the experiences gained at the TTX and according to the progress of the CWA.

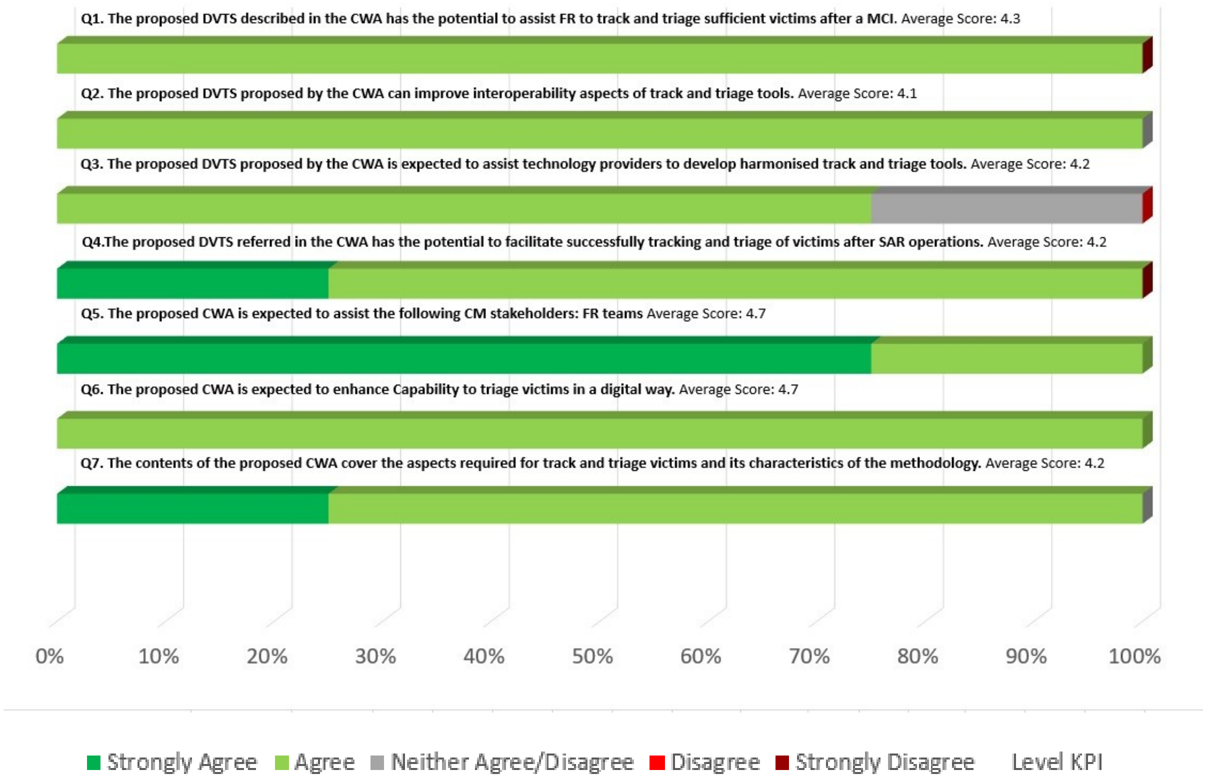


Figure 2: Scores' distribution obtained for all the statements in Stream 1 – Search and rescue questionnaire

The figure above summarizes the results of the evaluation received from the participants of the round table. The overall score (Likert scale) ranges from 4.1 to 4.7 (out of 5) and the average score is 4.2 which indicates that the participants considered the CWA work promising and well-structured. In general, the proposed standardization item provides recommendations on a digital victim tracking system covering the hardware (tags) and the software at the mobile and platform level. The proposed general solution is a guideline to develop technical solutions, without limiting the potential technological implementation ways. It was considered to be useful for response at mass casualty incidents, referring to a new technology to improve First Responders' efficiency and effectiveness. In the free text of the questionnaire concerns were raised about the topics of the unique ID per victim, about interconnection and sharing data with 3rd party systems and organizations, about permissions access and about view aspects of the user interface (UI) for the first responders in the field and at the command centre level. All these points were analysed and discussed again in the last workshop of the CWA; possible modifications for the final version of the CWA were identified.

3.1.4 Results from Formal Aspects Evaluation

The evaluation regarding formal aspects was conducted by Annika Almqvist (SIS) with the help of the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3 [35], chapter 2.1.9). The CWA "Requirements for acquiring digital information from victims during Search and Rescue operations" does not receive any major remark in all points, achieving highly score in all the questions, proving the overall good quality of the document is ready for publication at CEN/CENELEC.

3.1.5 Results from RAF Tool Analysis

The RAF tool was used in the beginning of the project for the most interesting standardization items identified in Stream 1. In this section, two out of seven tabs available in the excel file are listed. The file updated at the completion of the standardization activities of this Stream is based on the experience of the TTX, FSX and the actual document development and is published at CEN/CENELEC. The tabs below are the intake and the assessment sheet. In the assessment tab the score is:

- Feasibility: 3.8 out of 5
- Impact: 4.5 out of 5
- Urgency: 4 out of 5

Intake ResiStand

Title of the proposed standard:	Acquiring digital information from victims during SaR operations
Identification number:	SaR-6

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):	Organisation or project consortium	Stakeholder category
	STRATEGY	Hybrid
	ICCS	Research
	SIS	Policy makers
	SAMU	End-users
	CNVVF	End-users

Type of standard:	Workshop Agreement
Scope of the standard:	Main scope is standardization of casualties registration as well as standardization of the process of cross-border data exchange. Alongside this, the standardization of victim ID, the definition of the process for matching victims with incidents, as well as the exchange of data for searching for missing persons, particularly with regard to family reunification, are of great importance.
Example or illustration 1:	
Example or illustration 2:	

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Yes	
with national and regional legislation:	Yes	

arget groups for applying the standard	Target group	Free space for additional comments
First Responders:	Yes	Firefighters, Police, Crisis Management Operation Centers etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	No	
Industry/SMEs:	Yes	Technology companies developing search and rescue solutions
Consultancy organisations:	No	
Research institutes:	Yes	Research institutes developing new solutions for search and rescue operations
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protection Mechanism

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Early, unique identification, registration and following of victims during a disaster or MCI. Reducing under-triage and over-triage. Avoiding secondary transfers.
Industry:	Design of innovative interoperable systems for acquiring, managing and sharing digital information from victims.
Research:	Research and technology providers will further develop technologies and solutions towards effective acquiring of digital information from victims during SaR operations und innovative data management and cross-border
Policy makers:	Keeping track of the exact number of victims, their condition, the status of their transport, and their hospital destination. Interoperability and availability of adequate victim information to relevant authorities and participants
Citizens:	Avoiding preventable deaths.

Urgency of the standard: (when is it needed for implementation?)	High (within 1 year)	The proposed standard is expected to play a significant role in the process of aqcuiring digital information from victims during SaR operations and its outcomes potentially can drastically benefit rescue operations by reducing both loss of lives and

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	First responders will be the main actors using the proposed solutions for acquiring
Governmental organisations:	Yes	Main stakeholders organising and planning SaR operations
NGOs:	No	
Industry/SMEs:	No	Potential technology providers
Consultancy organisations:	No	Potential consultancy services to stakeholders
Research institutes:	Yes	Technology and research providers will enhance the process of acquiring and
Others:	No	
Preferred leading type of stakeholder:	First Responders	Firefighters, Police, Crisis Management Operation Centers etc.
Expected barriers and constraints:		

Other information

Additional remarks:	

Figure 3: Intake sheet of stream 1: General overview

Assessment

ResiStand

Title of the proposed standard: Acquiring digital information from victims during SaR operations

Identification number: SaR-6

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
STRATEGY	Hybrid
ICCS	Research
SIS	Policy makers
SAMU	End-users
CNVVF	End-users

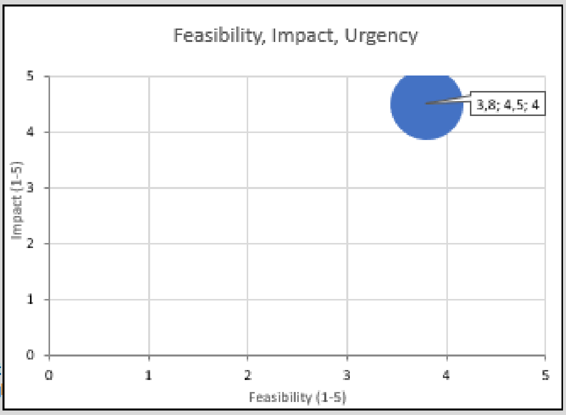
Scope of the standard: Main scope is standardization of casualties registration as well as standardization of the process of cross-border data exchange. Alongside this, the standardization of victim ID, the definition of the process for matching victims with incidents, as well as the exchange of data for searching for missing persons, particularly with regard to family reunification, are of great importance.

Type of standard: Workshop Agreement

Urgency (when needed?): High (within 1 year)

Overall impact: Great

Feasibility: High



Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very hig

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Great
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Moderate
Improvement of business quality management:	Moderate
Innovation progress:	Great
Improvement of business functions:	Great

Feasibility

Foundation:	Amply sufficient
Development perspectives:	Amply sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Not at all

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Completely
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	Earthquake Bomb attack -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Developments in disaster resilience and crisis management due to increasing involvement of society towards Network-Enabled Capabilities of emergency services and crisis partners, as well as increasing need for analysis tools.
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Technological progress in protection of the public and crisis logistics.

Figure 4: Assessment sheet of stream 1

3.1.6 Conclusions

The results show that the CWA “Requirements for acquiring digital information from victims during Search and Rescue operations” which was assessed in the final draft version is on the right track. Operational stakeholders, which participated in the round table presentation, demonstration and open discussion, considered the standardization as useful and promising to incorporate proposed technology at search and rescue operations after a mass casualty incident. The CWA was discussed from different angles such as the perspective of First Responders as users and the perspective of SMEs/RTOs as developers of similar solutions. Also, the CWA was tested in the first part of the scenario during the FSX by a large number of first responders.

The high average score of all questions indicates that the CWA objectives defined have been met, while the feedback from the key stakeholders assisted towards the next steps of the pre-standardization activities. The CWA document provides valuable requirements on the functionality for digital victim tracking solutions without limiting the potential implementation technologies or paths that could be followed to achieved them. This approach proposed by the NSB supporting Stream 1 is based on the general philosophy of the CWAs. The CWA is in context with different fields of standardisation as for example data traffic standards, data sharing and rules for triage, which were all topics that have been discussed in expert discussions and ben considered by the developers of the CWA.

The final step for the upcoming period is to submit the document for publication within CEN/CENELEC.

3.2 Final Evaluation of Stream 2 Critical Infrastructure Protection

3.2.1 Stream 2 Critical Infrastructure Protection CWA 1

3.2.1.1 Basis of the Final Evaluation

One subject of the final evaluation in stream 2 - Critical Infrastructure Protection (CIP) – was

> CWA 1: “Semantic layer definition and suitability of EDXL-CAP and EDXL-SitRep standards for crisis management in Critical Infrastructures.”

As the title of the CWA indicates, existing standards EDXL-CAP and EDXL-SitRep are relevant for the CWA and are considered in it. Many suggestions made previously during the related TTX and during other discussions with experts were considered to improve the CWA since the intermediate evaluation.

The final evaluation concerning this CWA was performed mainly on the base of events during the week of the STRATEGY FSX. Details about this can be found in STRATEGY deliverable D4.1 [34] “Full scale exercise demonstration report” [34]and especially for the activities regarding stream 2 in D3.3 [27] “Standards selection, implementation, testing and interoperability assessment for CIP”.

For the stream 2 of CIP, the objectives of the FSX were distinct from the objectives of the TTX and the evaluation, their corresponding KPIs and the questions varied accordingly.

The objectives of the TTX included:

a) Checking the proper interpretation of data from sensors using the semantic layer stated in the CWA
b) Checking the completeness of the concepts included in the semantic layer
c) Defining the operational needs of the responders

However, the FSX followed a different approach. It was much more focused on demonstrating the operations to the users than in checking the validity of the technical proposals, including the semantic layer. For that reason, the semantic layer was used only in the beginning of the field exercise to demonstrate the proper reception of the data from a crack meter and its subsequent triggering of an alarm, and the codification of the data used in the use case of damage assessment from buildings.

For that reason, as the FSX showed almost no detail of the standards developed in the stream 2 during the exercise, the evaluation of the CWAs of the stream 2 was carried out during a presentation made to a reduced set of reviewers the day before the exercise. This presentation gave an overall view of the semantic layer to the reviewers, its philosophy and the concepts included in it. The presentation and therefore the questions about it were focused on the concepts included in the semantic layer. The proper interpretation of those concepts was not tested because it was already demonstrated and validated in the TTX.

For that reason, the KPIs from the TTX presented in D4.3 [35]

- KPI 1 Number of concepts whose meaning could not be stated by the information system and
- KPI 2 Proportion of concepts whose meaning was successfully stated by the information system were not used in the FSX.

But CWA 1 includes a chapter on operational needs of users for evaluating the suitability of standards. For that reason, as the TTX was used to extract those operational needs of the users from a scenario presented in it, the FSX was used to validate the final set of operational needs during the discussion with the responders and the conclusions of the evaluation. The validation consisted in checking whether there was any operational need pending to be supported and the operational needs already included in the document. To support this, three new KPIs was introduced:

- KPI 5 Proportion of operational needs of responders missing in the document
- KPI 6 Proportion of operational needs that had to be amended
- KPI 7 Proportion of the conclusions of the evaluation of the suitability of EDXL-CAP and EDXL-SitRep for the automatic processing of data during a crisis agreed by the experts

For that purpose, the form included three new open questions, which were:

- Q5: 'Did you miss any important operational need that you think that should be included in the list of operational needs to be supported for the evaluation of EDXL-CAP and EDXL-SitRep?
- Q6: 'Do you want to amend any of the existing operational needs? If so, please indicate the amendment'
- Q7: 'Do you disagree with any of the of the conclusions of the evaluation of EDXL-CAP and EDXL-SitRep? If so, please indicate the disagreement and the reason'.

The reason why the questions were open was to let complete freedom for the attendees to provide any level of detail. For that reason:

- KPI5 is calculated as the sum of all operational needs that the attendants missed in the meeting, unless those needs were really already covered in the document, divided by the total of operational needs. Leaving that question empty means that the attendee did not miss any operational need and therefore it counts as zero.
- KPI6 is calculated as the sum of all operational needs finally amended by all attendees according to question 6, divided by the total of operational needs. Leaving that question empty means that the attendee did not amend any operational need and therefore it counts as zero.
- KPI7 is calculated as the sum of any disagreements expressed in Q7 divided by the total number of conclusions expressed in the document multiplied by 100, because it is a percentage.

During the expert session of the second day, the scope of the two CWA was presented to a reduced set of stakeholders. For the case of the CWA1 end users, research organisations and industry were included, as can be seen in the table below. The presentation comprised a thorough rationale of the gaps covered by the CWA and its impact, plus a discussion on the concepts included in the FSX.

Table 6: List of organisations with attendants in the expert sessions of the FSX for stream2 CIP, CWA 1

Organisation Name	Type/Description
Hellenic Police (Greece)	End User (Police)
CNVVF (Italy)	End user (Firefighter)

Organisation Name	Type/Description
Portuguese Army (Portugal)	End user (Military)
INOV	RTO
SINTEF	RTO
Trilateral	Industry / SME
ATOS (Spain)	Industry / SME
LEA - Law Enforcement Agency; NSB - National Standardization Body; RTO - Research and Technology Organisation; SME - Small and medium-sized enterprise	

Several data were collected before and during the FSX as a base for the final evaluation:

- Discussions during expert session
- Individual interviews
- Notes collected by evaluators
- Registration forms with information on the respondents
- Questionnaires, filled in by the participants remotely after the event.

To support the data collection and evaluate the results for stream 2, CWA 1, a specific evaluation team was set up, as shown in Table 7.

Table 7: Stream 2 CIP, CWA 1 Evaluation team.

Full Name	Affiliation	Role / Required expertise
Selby Knudsen	Trilateral	Lead Evaluator
Stergios Mosialos	Hellenic Police	Evaluator
Gabriel Pestana	INOV	Evaluator
Shenae Lee	SINTEF	Evaluator

3.2.1.2 Results from Experts Discussions and Interviews

The presentation of the contents of the CWA was interactive with the feedback from the attendees. As the semantic layer alone contains more than 700 concepts, it would have been impractical to discuss all of them in detail during the meeting. Instead of discussing all the concepts, its organization and rationale was discussed during the meeting, but the whole list of the concepts was circulated in an excerpt of the document, so the attendees could provide their feedback offline in the form, which contained a specific question for any missing concept in the document. The presentation also included a discussion on every single operational need. The attendees produced their own comments there, but the excerpt circulated to them also contained both the operational needs and the conclusions, so they could have more time to consider their responses offline.

In general terms, all of attendees agreed with the necessity for the solutions proposed and the rationale and structure of the document. They also agreed with the operational needs and the conclusions of the evaluation of EDXL-CAP and EDXL-SitRep contained in the CWA. They also provided feedback about some missing concepts, that were finally collected in the filled forms.

3.2.1.3 Questionnaire Results

Here is a resume of the answers from the participants:

Q.4. Did you miss any important concept that you think should be included in the list of concepts supported? The aim of this question was to identify any missing concept in the semantic layer. It included a box for providing any number of concepts and additional explanations. Hence, in this case, an empty box was considered as no concept found to be missing. When the box was not empty, its content was analysed for the suitability of all concepts and its integration in a single and non-redundant set of concepts to be proposed to the workshop for its addition to the semantic layer.

<table>
<tr><th>Question 4</th><th>answer</th><th>Number</th></tr>
<tr><td rowspan="4">Q.4. Did you miss any important concept that you think should be included in the list of concepts supported? If so, please indicate them:</td><td>No missing concept</td><td>3</td></tr>
<tr><td>1. availability of hospital before a crisis starts;
2. Whether a building has collapsed or partially collapsed;
3. Facility type road/bridge/airport</td><td>3</td></tr>
<tr><td>Within Facility type concepts these might be important:
• Industrial plant or factory
• Transportation facility (e.g., Airport, train or metro station)
• Highway
• State Building (some of these might be critical, such as National Assemblies, Secret Services Headquarters, etc.)
• International Organisation’s Buildings (some of these might be critical, such as NATO or EU buildings within the country)
• Harbour or port
• Lighthouse (not quite sure on this one)
• Wind turbine and photovoltaic power plant (these might be considered within Electric power plant?)</td><td>10</td></tr>
<tr><td>For identifying CBRN-E events, I don’t see where it distinguishes between the different type of event. Or even for other events, I may have missed it, but I don’t see type of event anywhere</td><td>1</td></tr>
</table>

Q5. Did you miss any important operational need that you think that should be included in the list of operational needs to be supported for the evaluation of EDCL-CAP and EDXL-SitRep? The aim of this question was to identify any operational need missing in the semantic layer. It included a box for providing any number of concepts and additional explanations. Hence, in this case, an empty box was considered as no operational need found to be missing. When the box was not empty, its content was analysed for the suitability of all operational needs and its integration in a single and non-redundant set of concepts to be proposed to the workshop for its addition to the operational needs considered for the evaluation of EDXL-CAP and EDXL-SitRep.

Question	Answer	Number
Q5. Did you miss any important operational need that you think that should be included in the list of operational needs to be supported for the evaluation of EDCL-CAP and EDXL-SitRep? If so, please indicate them:	Yes	0
	It is not very clear which phases of crisis management EDCL-CAP and EDXL-SitRep are designed for. Are they suited for early detection of the unwanted situation (e.g. earthquake)? Or rather, are they suited for creating alarms during abnormal situations (after an accident or an incident has occurred)	1
	NO	6

Q6. Do you want to amend any of the existing operational needs? The aim of this question was to validate the operational needs included in the document. It included a box for providing a full explanation for any amendment. Hence, in this case, an empty box was considered as no operational need to be amended. Actually, as this field could have contained any number of operational needs to be amended, the response is conceptually not binary. The response could conceptually have included any number of operational needs potentially amended. However, as finally no operational need was required to be amended by any stakeholders, we provide a condensed summary as a Yes/no question to foster readability.

Question	Answer	Number
Q6. Do you want to amend any of the existing operational needs? If so, please indicate the amendment:	Yes	0
	NO	7

Q7. Do you disagree with any of the of the conclusions of the evaluation of EDXL-CAP and EDXL-SitRep? The aim of this question was to validate the opinion of the stakeholders regarding the conclusions produced in the CWA regarding the suitability of the standards EDXL-SitRep and EDXL-CAP to support the transmission of data in crises involving CIS. It provided a free text box for providing a full description of any disagreement. Hence, in this case, an empty box was considered as no disagreement. Actually, as this field could have contained any number conclusions in disagreement, the response is conceptually not binary. The response could conceptually have included any number disagreements. However, as finally no disagreement was stated by any stakeholders, we provide a condensed summary as a Yes/no question to foster readability.

Question	answer	Number
Q7. Do you disagree with any of the of the conclusions of the evaluation of EDXL-CAP and EDXL-SitRep? If so, please indicate the disagreement and the reason:.	Yes	0
	NO	7

None of the four questions were yes/no questions or even questions with a level of agreement. Instead, the answers could have potentially had as many possible responses as concepts included in the semantic layer or operational needs included in the evaluation of EDXL-CAP and EDXL-SitRep. And what is even more complex, a result of 10 missing concepts is not necessarily worse than a response indicating a single concept. For that reason, the document provides tables that already interprets the answers according to the nature of each question and calculates the KPIs based on that interpretation. The results of the calculated KPIs are shown in Figure 5 below, but it does not make much sense to provide a graphic showing the level of agreement for each question, as the decision of the number of levels to be shown would be subjective.

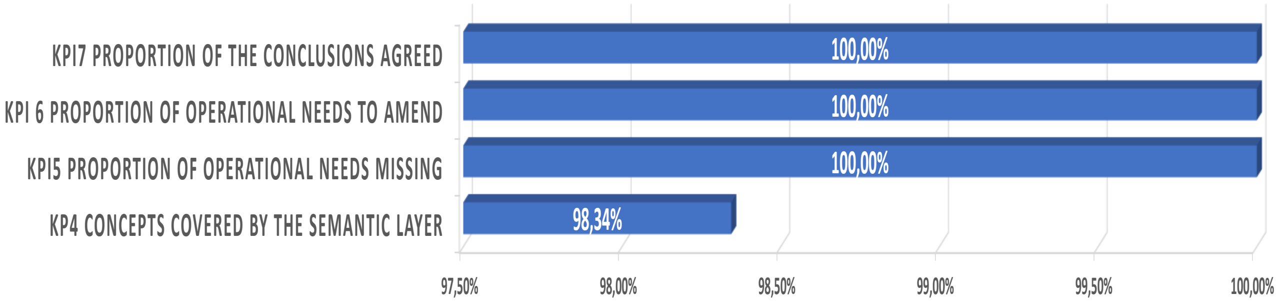


Figure 5: Summary of the assessment obtained for each KPI related to the classification of the FSX.

(1) In particular, it should be noted that KPI 5 to 7 have a very successful 100% evaluation, but this is not surprising taking into account that the operational needs were extracted first from a scenario designed by a firefighter responder and later refined by the input of three types of responders more.

(2) KPI4 is a little less optimistic. In fact, the answers from the responders not only found some missing concepts, but in some cases opened the way to add more concepts that were related with the ones identified by the attendees. Taking into account the total number of concepts identified from the ones provided by the attendees were a total of 34 new concepts, this would lower the rate of concepts successfully included in the semantic layer to only 95,84%.

3.2.1.4 Results from Formal Aspects Evaluation

The evaluation regarding formal aspects was conducted by Janne Kalli (SFSO with the help of the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3 [35], chapter 2.1.9) and gave the following results:

A high Likert scale score was reached throughout: As an average 4.67 out of 5. No value was below 4, except for the question concerning the headings of a depth bigger than 3 levels. The index of the final published document was generated including only three levels of depth. The overall impression of the reviewer was: ‘Usually the bibliography is the final thing written on a CEN deliverable (including CWAs). Overall looks very good and solid deliverable!’. The reason why the bibliography is not at the end of the document is because some of the Annexes are normative and it is important to leave them at the end of the document as the content of the last one constitutes the semantic layer proposed by the CWA.

3.2.1.5 Results from RAF Tool Analysis

During the STRATEGY project the stream 2 standardisation activities were assessed for both CWAs also with the help of the ResiStand Assessment Framework (RAF), see D2.2 [5] Section 6.2 RAF intake sheet for gap CIP-10 [4] . For the final evaluation the RAF supported analysis was reconsidered and updated. More details about the RAF in general can be found in chapter 2.4 of the present deliverable or in STRATEGY deliverable D1.2 [2] “Project Handbook”.

However, in the case of the CWA1 of stream2, it contains a chapter for an evaluation of the EDXL-CAP and EDXL-SitRep. This evaluation is not a standardization proposal, but an evaluation of two already existing standards. As the RAF tool is used for evaluating the feasibility of new standardization proposals, the RAF tool is not applicable for this part of the document but only for the part that deals with the definition of a Semantic Layer. The RAF Tool Intake sheets for that part of the CWA1 presented in this section, are an update of the RAF Tool Intake sheet presented in D2.2 [5] section 6.2 [4]. An overview of the completed RAF intake sheets for both TS are presented respectively in:

- Figure 6: Intake sheet: General overview
- Figure 7: Intake sheet: Urgency and development
- Figure 8: Assessment sheet
- Figure 9: Impact sheet

The final assessment shows a medium result in terms of feasibility (3.3 out of 5), a considerable value in terms of impact (4 out of 5) and a very high value (ASAP) in terms of urgency (5 out of 5). The consortium is working with OASIS in the adoption of the ideas expressed in the CWA in the EDXL standards and has also participated in a meeting of ISO/TR 22351:2015 (Message structure for exchange of information) for bringing to it the suggestions on the use of the Semantic Layer. More details on the pre-standardization activities carried out to promote the CWA can be found in deliverable document D5.2 [37].

It is worth to mention the reasons why the feasibility has got only a moderate value. Though the workshop released a complete document, as explained in deliverable document D5.2 [37], the fact that ISO/TC 292/WG 3 Emergency Management (STRATEGY C liaison approved), which was evaluating the viability of revision of on ISO/TR 22351:2015 is being reorganized, has not given the chance to STRATEGY to continue the initiated contacts in March 2022 and propose the use of the semantic layer in the very probably review of this reference as ISO/Technical Specification. In practice, this has lowered the follow-up perspectives only to moderate, because despite the technical groups have shown interest in the CWA, the lack of financing due to the finalisation of the project won’t allow some partners to guarantee the continuation of the work when the WG is finally reorganized and the work on ISO/TR 22351:2015 is resumed. The consortium has prepared their proposals based on the CWA before the completion of the project to send them to the WG when their work is resumed, but without attending the meetings to defend the proposal and not being actively involved in the revision of the CWA, the probability of adoption by that WG is hindered. However, the consortium is also working actively with OASIS and the OGC to their support to the ideas proposed in the CWA and their possible adaption.

Intake

ResiStand

Title of the proposed standard: Semantic namespace for concepts applicable to critical infrastructures

Identification number:

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
STRATEGY	Hybrid
	-
	-
	-
	-

Type of standard: Technical Specification

Scope of the standard: Definition of a namespace that includes a set of names for concepts applicable to the protection of crisis infrastructures, together with their meanings.The names included in that namespace, together with their description, will be used as a basis for the proper interpretation of such data by the software modules aiming to exploit them, allowing for the exchange and interoperability of both sensors and intelligence modules.

Example or illustration 1:

Example or illustration 2:

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Unknown	As far as we know, there are no common legislation for the whole set of critical infrastructures that deal with the format of data exchanged
with national and regional legislation:	Unknown	As far as we know, there are no common legislation for the whole set of critical infrastructures that deal with the format of data exchanged

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	The compliance with this standard will contribute to the ability of freely exchange
Governmental organisations:	Yes	The compliance with this standard will contribute to the ability of freely exchange
NGOs:	Unknown	
Industry/SMEs:	Yes	The compliance with this standard will contribute to the interoperability of any sensor
Consultancy organisations:	Unknown	
Research institutes:	Unknown	
Standardisation bodies:	No	Although standardization bodies will need to include the ability for referring namespaces
Others:	Unknown	

Figure 6: Intake sheet for CWA1 of stream 2: General overview

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	It is essential for end users to have the ability to adopt new sensors and intelligence modules that are ready to cope with new threats without having to adapt their systems to the new sensors and modules.
Industry:	The adoption of this standard should reduce significantly the development costs because it will help to avoid the need to create specific software for interpreting each way of expressing the information for every sensor or the
Research:	The research community will benefit from the same advantages as the industry as long as they develop software from their sensors. And more specifically, the development of new, experimental sensors to cope with new
Policy makers:	Policy makers will benefit from a economy of costs caused by the interoperability of the components of the system resulting from breaking the silos of each individual provider
Citizens:	The adoption of this standard should reduce the time needed to incorporate new sensors compliant with it, resulting in higher security. Additionally, the reduction of costs resulting from increasing the interoperability of the

Urgency of the standard: (when is it needed for implementation?)	Very high (ASAP)	The lack of interoperability of system components due to the inexistence of a common representation for the data is endemic and is resulting in delays in adopting new technologies for a very long period of time, continuously jeopardizing unnecessarilly

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	Check the completeness of the set of concepts from the point of view of responders
Governmental organisations:	Yes	Check the completeness of the set of concepts from the point of view of policy makers
NGOs:	No	Though the NGOs might be interested in receiving and sending data to and from
Industry/SMEs:	Yes	Agree on the list of concepts to be used and the way to represent them
Consultancy organisations:	No	
Research institutes:	No	
Others:	No	
Preferred leading type of stakeholder:	-	

Expected barriers and constraints:	Main providers of command & control centers take advantage of current lack to turn the market of crisis management software into a de-facto monopolium and therefore may not be interested in supporting this initiative

Other information

Additional remarks:	

Figure 7: Intake sheet for CWA1 of stream 2: Urgency and development.

Assessment

ResiStand

Title of the proposed standard: Semantic namespace for concepts applicable to critical infrastructures

Identification number:

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
STRATEGY	Hybrid
	-
	-
	-
	-

Scope of the standard: Definition of a namespace that includes a set of names for concepts applicable to the protection of crisis infrastructures, together with their meanings.The names included in that namespace, together with their description, will be used as a basis for the proper interpretation of such data by the software modules aiming to exploit them, allowing for the exchange and interoperability of both sensors and intelligence modules

Type of standard: Technical Specificatio

Urgency (when needed?): Very high (ASAP)

Overall impact: Considerable

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high

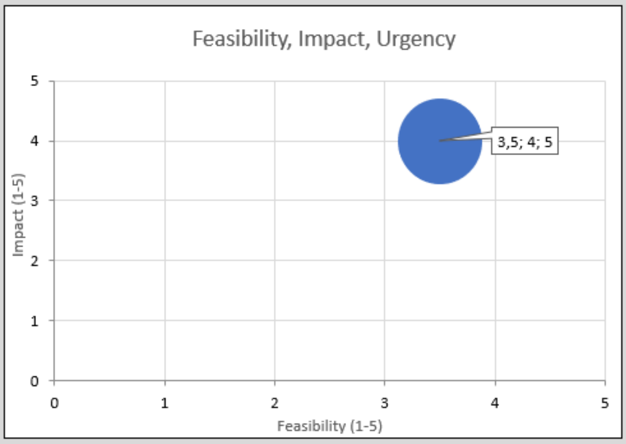
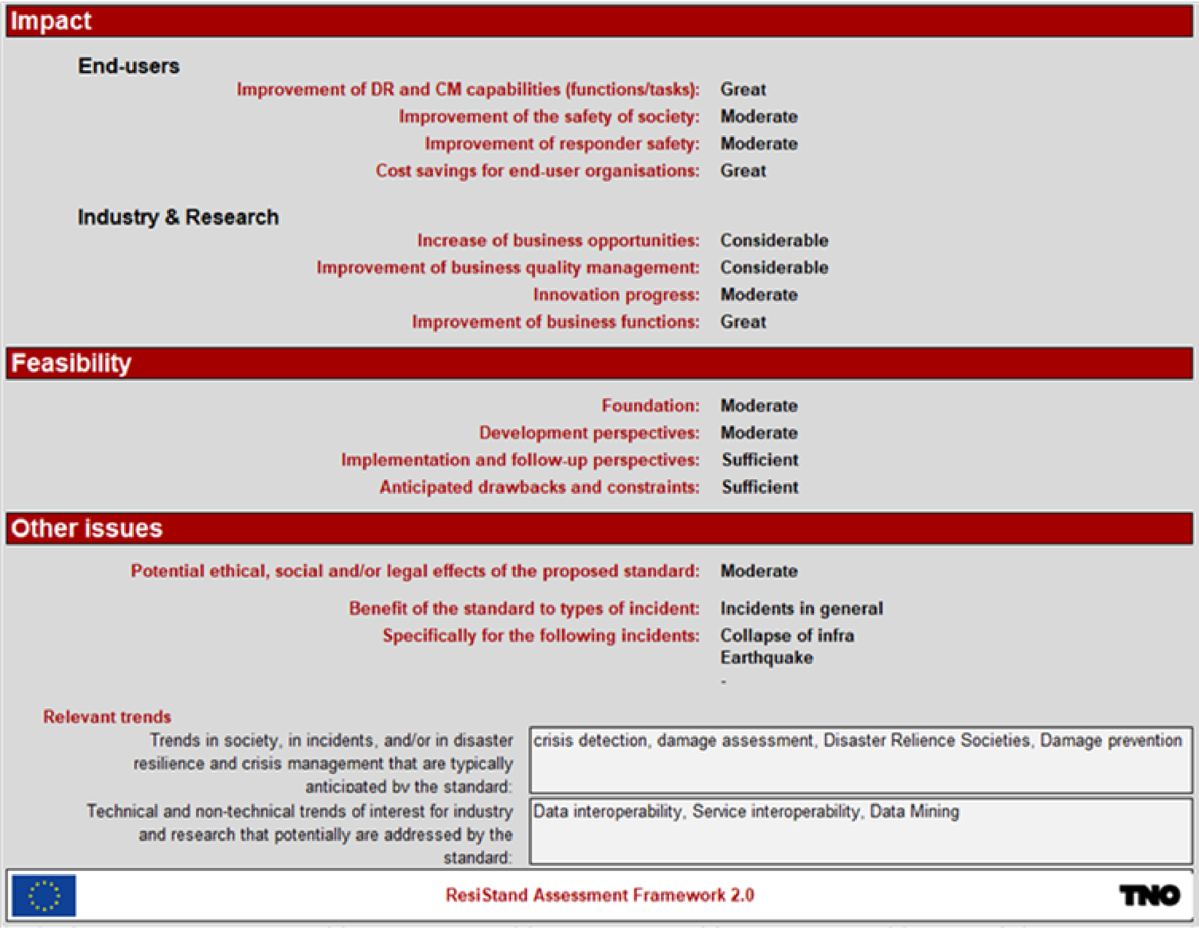


Figure 8: Assessment sheet of CWA1 of stream 2



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Moderate
Improvement of responder safety:	Moderate
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Considerable
Innovation progress:	Moderate
Improvement of business functions:	Great

Feasibility

Foundation:	Moderate
Development perspectives:	Moderate
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Moderate
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	Collapse of infra Earthquake -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	crisis detection, damage assessment, Disaster Relience Societies, Damage prevention
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Data interoperability, Service interoperability, Data Mining

ResiStand Assessment Framework 2.0

TNO

Figure 9: Impact sheet of CWA1 of stream2

3.2.1.6 Conclusions

Though some of the attendants asked for the inclusion of terms for the number of beds available in the hospitals in the semantic layer, that concept is already present in standard EDXL-HAVE, and in fact has already been reported periodically in countries such as Spain. As the semantic layer is designed to not collide with any existing standards by not including the concepts that are included in them, there is not really any point in adding this concept.

Apart from that, the attendees provided a very valuable feedback of missing concepts in the sematic layer, for a total of 14 concepts missing or in need to be amended, though one of them was the number of beds available of hospitals that was not really missing, which makes a total of 12 missing concepts to be included in the semantic layer.

There are two concepts that deserve special attention:

- there was no way to express in the semantic layer whether a building had collapsed or not. Though the semantic layer already included a concept for that, it did not include the proper values for this. This caused the revision of this concept and its format.
- The semantic layer does not contain any values for the type of a CBRNE event. Though EDXL-CAP includes a field for the type of event, it does not include any type for the types of events

Chemical, Biological, Radiological, Nuclear or Explosive. For this reason, its omission makes necessary to include them in the semantic layer.

Finally, the answer to Q5 regarding which phases of crisis management EDCL-CAP and EDXL-SitRep are designed for, do not really refer to a missing operational need, but a demand for a general clarification of what EDXL-CAP and EDXL-SitRep are meant to be used for. Hence, we have considered that response as 'no' but included the demanded clarification in the CWA document.

Apart from that, the fact that none of the attendees identified any operational need missing or to be amended and they agreed with all the conclusions let being very optimistic on the quality of the evaluation of the suitability of EDXL-CAP and EDXL-SitRep carried out in CWA 1.

3.2.2 Stream 2 Critical Infrastructure Protection CWA 2

3.2.2.1 Basis of the Final Evaluation

The final evaluation of the

CWA "Emergency management – Incident situational reporting for Critical Infrastructures"

was performed on the basis of the Stream 2 workshop and the FSX in March 2022 in Gualdo Tadino, Italy. Details on the FSX scenarios and the workshop discussion that took place prior to the FSX can be found on deliverable D2.18 [20] "*FSX Scenarios planning report*" and D4.1 [34] "*Full scale exercise demonstration report*".

The first day of the FSX was devoted to setting up the technical implementation of the Incident Reporting CWA as well as testing the infrastructure and system in order to check its full functionality. As in the TTX, in the FSX an online form was setup based on the content of the CWA, in order to fit the CWA to a more realistic, operational and technical environment. The specificities of the technical implementation are described in detail in deliverable D3.3 [27] "*Standards selection, implementation, testing and interoperability assessment for CIP*".

The CWA was discussed in detail, mainly with local stakeholders (fire officers, critical infrastructure operators) and STRATEGY partners during the second day of the FSX.

During the fourth day of the FSX, the actual day the field scale exercises took place, the CWA was inserted in the operational environment through the scenario (described in deliverable D4.1 [34] "*Full scale exercise demonstration report*") and demonstrated to the players and observers of the FSX that were located in the main command post of the exercise.

The CWA version tested was version no 5, which incorporated all the recommendations that came up during the TTX as well as additional improvements pointed out by the CEN workshop participants.

For the final evaluation two questionnaires were made in order to gather the necessary data: (a) for the evaluators and (b) for the players. In addition, the CWA leader kept specific notes of the comments of the participants.

At this point, it must be noted that the most significant input for the final evaluation was gathered during the workshop day, exactly prior to the FSX day. The day of the implementation of the FSX the CWA was demonstrated in the players of the exercise during the evolution of the scenario. Due to the multiple and consecutive events no detailed discussion took place on the CWA content. This had been discussed the previous day.

The evaluation team was set up by the following three persons presented in Table .

Table 8. The evaluation team.

Full Name	Affiliation	Role / Required expertise
Katerina Valouma	SATWAYS	Lead Evaluator
Ioannis Sekadakis	HP	Evaluator
Georg Neubauer	AIT	Evaluator

Table 9. List of organizations participating in the small group.

Organisation Name	Type/Description
Hellenic Police (Greece)	End user (LEA)
CNVVF (Italy)	End user (Firefighter)
KEMEA (Greece)	RTO
Regional Civil Protection Agency (Umbria Marche-Italy)	End user (LEA)
Dam infrastructure (Italy)	SME
SATWAYS (Greece)	SME
Austria Institute of Technology (Austria)	RTO
Local police (Italy)	End user (LEA)
LEA - Law Enforcement Agency; RTO - Research and Technology Organisation; SME - Small and medium-sized enterprise	

Overall, the input data for the final evaluation came from:

- The Expert discussions
- Plenary discussion that tool place the day of the FSX
- Notes taken by the CWA leader
- Questionnaires for the evaluators and the players
- Videos and photos

3.2.2.2 Results from Expert Discussions and Testing during the FSX

The Expert discussions that took place during the FSX were extremely important for the final evaluation as, during this process, the information and content of the CWA were checked one by one. Furthermore, there was the necessary time for the participants to use the online form (https://survey123.arcgis.com/share/35a360a8d90249a998ea59997867e0db) in order to test it in action before the FSX.

The focus group was well balanced in terms of first responders, critical entities representatives, industry and researchers.

The main outcome was that the CWA is on the right track and accomplishes its scope. It became clear that:

- incident notification and reporting is well supported by the CWA.
- Common operational picture and situation awareness are also well supported.
- Information either in a brief or in a more detailed format can be supported.
- First responders acquire a good situational awareness.
- Restoration times of disrupted services can be reduced.

3.2.2.3 Questionnaire Results

In order to create a more qualitative schema for the final evaluation of the CWA, two questionnaires were developed, one for the evaluators and one for the participants, in both an online and paper format.

The data gathered from the participants cannot be considered as statistically valid, as the sample was small. Many participants of the focus group did not answer the questionnaire, most probably due to the language barrier. During the discussion, this was overcome through a translation between the participants but for the answers to the questionnaires this was not feasible.

Thus, quantitative aspects cannot be considered statistically valid. Nevertheless, some important points came up. Below, the questions through which quality aspects can be deduced are presented.

Briefly the main outcomes from the questionnaires are presented below:

Question 1: Please specify, the extent, to which the proposed CWA facilitates the information sharing between organisations (especially between affected Cis and competent authorities or affected Cis and first responders) and the provision of a common situational picture.

The proposed CWA facilitates to a great extent the information sharing between organisations and the provision of a common operational picture.

Question 2: Is the identification of first responder's operational needs for deployment of resources, to confront the emergency, facilitated by the contents of the proposed CWA.

The operational needs of first responders for the deployment of resources is considerably facilitated.

Question 3: To what extent the proposed CWA supports the process of restoration time and recovery of disrupted services?

Restoration and the recovery time of the disrupted services is supported considerably. Nevertheless, one answer pointed out that in this domain the proposed CWA offers limited support.

Question 4: To what extent the proposed CWA assists to the enhancement of the resilience of critical infrastructures?

The proposed CWA assists the enhancement of the resilience of critical infrastructures to a considerable extent.

Question 5: According to your opinion, the proposed CWA provides:

The proposed CWA provides a good situation awareness.

Question 6: To what extent the proposed CWA facilitates the development of novel technological tools, related to the reporting and notification of incidents and emergencies?

The CWA can support considerably the development of new innovative technological tools.

Question 7: To what extent the proposed CWA could be compatible to existing technical standards and specifications (if aware of any)?

The CWA can be compatible with existing technical standards to a great extent.

Question 8: Should all the following four types of incident report (bold fonts) exist in the incident report?

All the four types of incident reports were found to be useful.

Question 10: Would you prefer a shorter version of the incident report?

The length of the incident report is adequate.

Question 11: Was it difficult for you to understand the content of the proposed CWA? (Question for CI operators)

The content of the CWA is easy for CI operators.

Question 12: Do you think that it required too much time in order to fill in and submit the incident report? (Question for CI operators)

The time needed to complete a report is not too much.

Question 14: To what extent the proposed CWA covers the purposes of incident notification and reporting?

The proposed CWA covers to a considerable and great extent incident notification and reporting purposes.

Question 15: Does the proposed CWA bring added value to the management and response of disruptive incidents that occur in critical infrastructures/entities?

The proposed CWA bring added value to the management of and response to the incident.

Question 16: Would you consider using the proposed CWA as a means of conforming to the recently published Directive EU 2557/2022 on the resilience of critical entities?

The use of the CWA as a tool for conforming with the EC Directive 2557/2022 is seen in a positive way by the participants.

Question 17: Do you think that the printed template is useful?

The printed template was found useful.

Question 18: Are the colours in the printed template helpful?

The suggested colouring in specific set of information was also found useful.

Question 19: Please identify/mention, briefly, the strengths of the proposed CWA.

The strengths of the CWA are the provision of common operational picture, situation awareness and incident reporting.

Question 20: Please identify/mention, briefly, any weaknesses of the proposed CWA.

As a weakness it was mentioned that it seems difficult to be adapted by all types of critical infrastructures.

Question 21: How likely would you recommend the current CWA to become a standard, taking into consideration the completeness of its content and the existing standardization gaps it addresses?

It is very likely that the CWA would be proposed to become a new standard.

The overall results are presented in Table 10 and Figure 10. The data regarding the overall evaluation of the CWA shows that the CWA was viewed positively by the exercise participants.

Table 10. Overall results from the participants feedback form in terms of percentages (only quantifialble questions are presented).

Question	Strongly disagree	Disagree	Neither agree/disagree	Agree	Strongly agree	No answer	AVERAGE SCORE
Q1	0%	17%	17%	0%	67%	0%	4,2
Q2	0%	17%	17%	50%	17%	0%	3,7
Q3	0%	17%	17%	50%	17%	0%	3,7
Q4	0%	17%	17%	50%	17%	0%	3,7
Q5	0%	0%	33%	33%	33%	0%	4,0
Q6	0%	17%	17%	33%	33%	0%	3,8
Q7	0%	0%	17%	17%	33%	33%	4,3
Q14	0%	17%	0%	33%	33%	17%	4,0
Q15	0%	0%	17%	17%	50%	17%	4,4
Q16	0%	0%	17%	0%	50%	33%	4,5
Q17	0%	0%	0%	17%	50%	33%	4,8
Q18	0%	0%	0%	0%	83%	17%	5,0
Q21	0%	0%	17%	0%	67%	17%	4,6

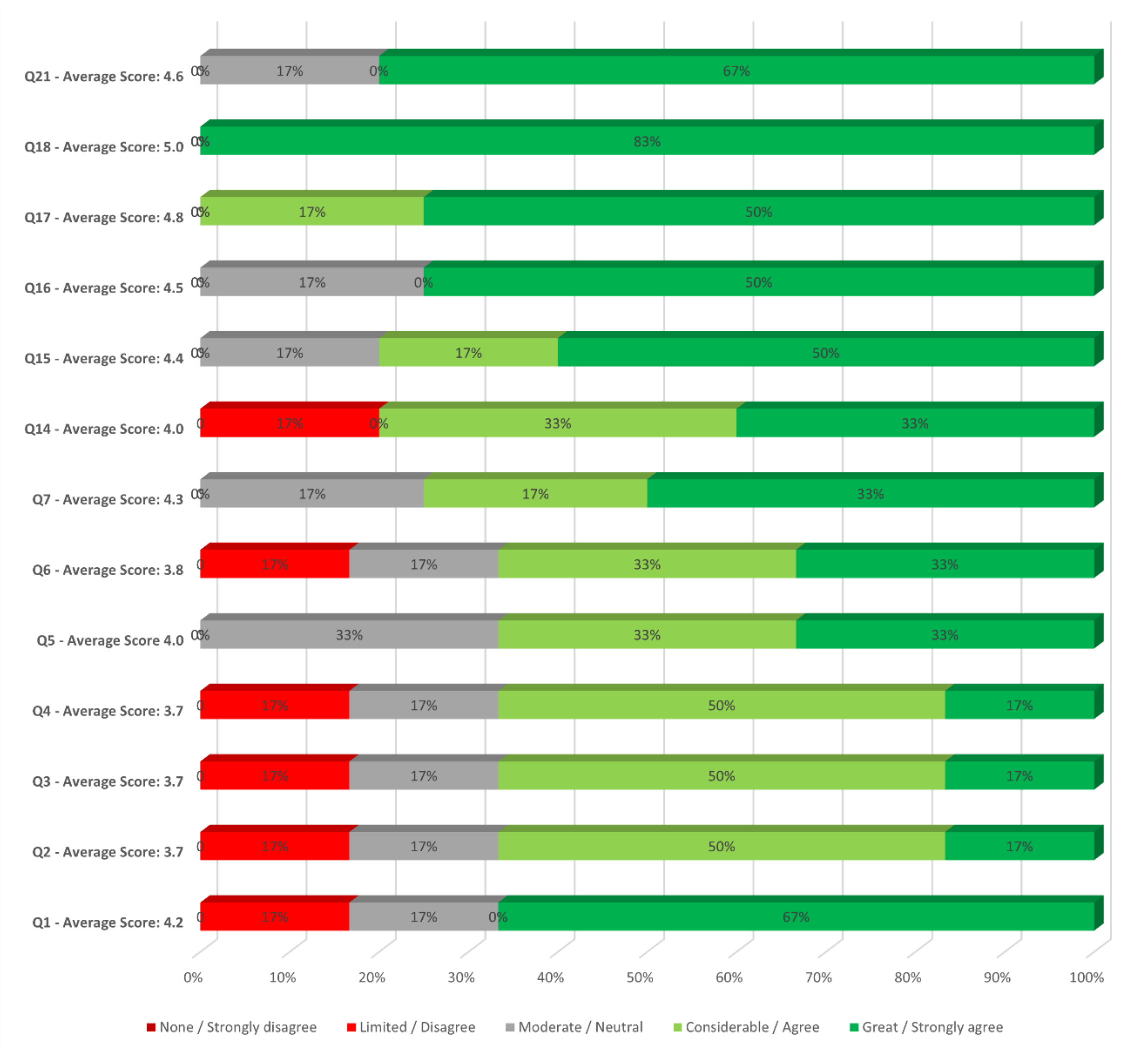


Figure 10. Score distribution obtained for the statements of CWA 2 of Stream 2 that could be quantified.

The criteria and KPIs for CWA 2 of Stream 2 are the same with the ones that were studied in the TTX. These are mentioned in detail in deliverable D4.3 [35] “Intermediate evaluation and validation of crisis management solutions and pre-standardization items (from all user groups)” and briefly mentioned here in.

The criteria for CWA 2 are the following:

1. Overall resilience enhancement
2. Enhanced situation awareness and improved collaboration between stakeholders
3. Maintenance and restoration
4. Applicability with other solutions

The KPIs for CWA 2 are the following:

- KPI 1 Enhanced common operational picture and information exchange.
- KPI 2 Facilitation of identification of first responders' needs.
- KPI 3 Reduction of recovery and restoration time and CI resilience enhancement.
- KPI 4 Facilitation of the development of relevant technological tools.
- KPI 5 Compatibility with technical standards.

In Table 11 below the matching of the criteria and KPIs with the questions of the evaluator's form and participants feedback form.

Table 11. Stream 2 – CWA 2 (incident reporting) criteria, KPIs and question/statement for evaluation (Cx stands for capability in the evaluators' form). Bold denotes the question being more related to the KPI.

	Criteria	KPI		Questions Evaluators' Questionnaire	Questions Players' Questionnaire
1	Overall resilience enhancement	1	Enhanced common operational picture and information exchange	C1-T1, C1-T2	**Q1**, **Q5**, Q8, Q9, Q14, Q15, Q17, Q18
		3	Reduction of recovery and restoration time and CI resilience enhancement	C3-T1	**Q3**, **Q4**, Q8, Q15, Q16
2	Enhanced situation awareness and improved collaboration between stakeholders	1	Enhanced common operational picture and information exchange	C1-T1, C1-T2	**Q1**, **Q5**, Q8, Q9, Q14, Q15, Q17, Q18
		2	Facilitation of identification of first responders' needs	C2-T1	Q2
3	Maintenance and restoration	3	Reduction of recovery and restoration time and CI resilience enhancement	C3-T1	**Q3**, **Q4**, Q8, Q15, Q16
4	Applicability with other solutions	4	Facilitation of the development of relevant technological tools	C4	Q6, Q17, Q18
		5	Compatibility with technical standards	C5-T1	**Q7**, Q18

The overall assessment of the KPIs grade is presented in Figure 11. The grading of the KPIs is considered very positive taking into account the low number of answers from the participants feedback form and the positive evaluation from the evaluators. The assessment of the KPIs has been based on the most relevant questions from the participants' feedback form. For KPI 1 and KPI 3 the average of Q1, Q5 and Q3, Q4 have been considered, respectively. Comparatively, to the other KPIs, the grading of KPI is somewhat expected as Q5 is an extremely difficult question to be answered and contains a subjectivity from the user perspective.

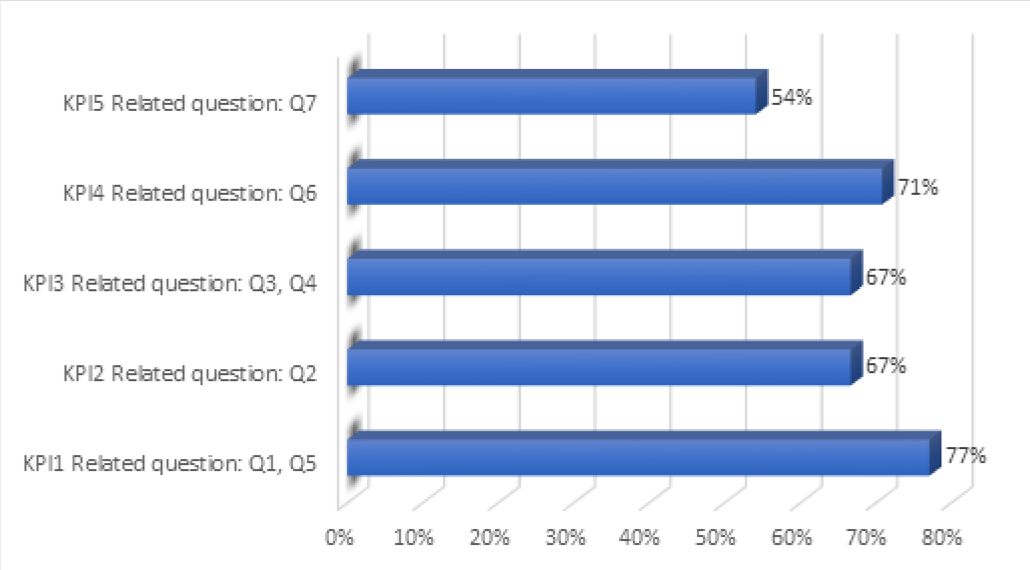


Figure 11. Summary of the assessment obtained for each KPI. For KPI 1 and KPI 3 the average of Q1,Q5 and Q3,Q4 have been considered, respectively.

3.2.2.4 Results from Formal Aspects Evaluation

The evaluation regarding the formal aspects was conducted by Janne Kalli (SFS) with the help of the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3 [35], chapter 2.1.9) and gave the following results:

The average score of the questions from the formal evaluation is equal to 4.8. In question Q.9 (participants in the KoM) the score received was equal to 1 because the document evaluated was one step before the submission to CEN – CENELEC. These have been included in the final document. Similarly in Q.19 (Are the normatively referenced documents ISO/IEC or CEN/CENELEC documents) is equal to 1, as the version examined is not the final submitted to CEN/CENELEC. This has been corrected and it was decided to keep some normatively references documents that are not documents of ISO/IEC or CEN/CENELEC as these are used as normative reference in the document.

Consequently, the CWA passed the formal evaluation successfully. In addition, the CWA passes through a public commenting process.

3.2.2.5 Results from RAF tool Analysis

The proposed CWA addresses issues directly referring to critical entities services and their impact to society. Any type of incident can be handled through the CWA. Overall and in simple words, the CWA is compatible with the newly published EC Directive 2557/2022 on the resilience of critical entities.

Among the benefits of the CWA are:

- Incident reporting and notification in a standardised way
- The creation of a common and updated operational picture
- Common understanding of the situation

- Information in many forms, from a simple description of what has happened to more detailed information such as the severity, the number of persons affected and the restoration time
- Applicability to all types of critical entities
- Benefits for industry through the creation of new technological tools in existing or new software
- Cross-organization interoperability, enhanced coordination and collaboration between Critical entities and emergency services
- Improved restoration times

The assessment score using the RAF tool is equal to Considerable impact, High feasibility and High Urgency (within 1 year, although the option ASAP is valid) (Figure 12 and Figure 13).

Intake

ResiStand

Title of the proposed standard:	Incident situational reporting for Critical Infrastructures
Identification number:	

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):	Organisation or project consortium	Stakeholder category
	KEMEA	Research
		-
		-
		-
		-

Type of standard:	Workshop Agreement
Scope of the standard:	Definition of the specific information, the content, that needs to be communicated by CI operators to the national competent authority (e.g. a National Coordination Centre), responsible for supervising CIs or first responders Coordination Centres, in case a potentially damaging and/or disruptive incident occurs, which could also be capable of affecting one or more interconnected CIs.
Example or illustration 1:	An earthqauke occurs. The ground motion affects a Critical Infrastructure (e.g. an oil refinery) and the oil production needs to be paused. The CI operator/security liaison officer according to data from the SCADA system decides to
Example or illustration 2:	A CI (powr plant) is under a terrorist attack. A sudden telephone alerts for potential bomb explosion in the next 30 minutes. The CI operator/security liaison officer fills in the deidcated form providing the relevant information and

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Yes	Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities and repealing Council Directive 2008/114/EC.
with national and regional legislation:	Yes	For example, in Greece Presidentail Decree of 39/2011 adjusts the legislation in a way that is harmonised to the European one described above.

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Common understanding of the situation, accurate and timely operations, common
Governmental organisations:	Yes	Governmental organisations, responsible for supervising CIs will be directly informed by
NGOs:	Yes	For the case of volunteer FRs
Industry/SMEs:	Yes	Other CIs and technology solution providers that could develop a tool for this reason
Consultancy organisations:	Yes	Organisation that are policy makers or provide similar services
Research institutes:	Unknown	
Standardisation bodies:	Yes	Enhancing existing standards related to the use and protection of Critical Infrastructures
Others:	Yes	CI operators of interconnected infrastuctures will be updated

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	CI operators will have a standard format to fill in and provide Public Bodies and first responders with all the crucial data relevant to the operation status of the CI and evaluation of the evolving situation. Operators of interconnected
Industry:	Industry may contribute to the development of cut-edge tools allowing for sophisticated incident reporting with standardised content.
Research:	
Policy makers:	The demonstration of a successful operation of a coordination center connected with multiple CIs may lead to institutionalization of the center.
Citizens:	Public Bodies will have the capability, through the standardisation of the reports, to timely warn and alert citizens, in case damages of CIs affect the population.

Urgency of the standard: (when is it needed for implementation?)	High (within 1 year)	Standardisation of incident reporting of the status of CIs is essential in order, to achieve a common way of information interpretation, thus raising situational awareness, to facilitate response operations and to timely restore operational continuity.

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	First responders will participate in the testing and validation of the proposed standard,
Governmental organisations:	Yes	Governmental organisations operating a national control centre for CI will assess the
NGOs:	No	Not required but cannot excluded if they want to participate.
Industry/SMEs:	Yes	Agree on the feasibility on the collection of the data required by the report.
Consultancy organisations:	Yes	Consultancy organisations will be the policy makers, who will provide consultation upon
Research institutes:	No	
Others:	Unknown	

Preferred leading type of stakeholder:	First Responders	End users, both the CI operators and the first responders.(governmental organisations to
Expected barriers and constraints:	The difficulty and/or willingness by industry providers of command & control centers to adapt their systems to standardised format.	

Other information

Additional remarks:	

ResiStand Assessment Framework 2.0 — TNO

Figure 12. Screenshot of the “INTAKE“ sheet of the RAF tool for CWA-2 (Stream 2).

Assessment

ResiStand

Title of the proposed standard: Incident situational reporting for Critical Infrastructures

Identification number:

Proposed standardisation activity and its perspectives

	Organisation or project consortium	Stakeholder category
Proposing organisations or projects:	KEMEA	Research
		-
		-
		-
		-

Scope of the standard: Definition of the specific information, the content, that needs to be communicated by CI operators to the national competent authority (e.g. a National Coordination Centre), responsible for supervising CIs or first responders Coordination Centres, in case a potentially damaging and/or disruptive incident occurs, which could also be capable of affecting one or more interconnected CIs.

Type of standard: Workshop Agreement

Urgency (when needed?): High (within 1 year)

Overall impact: Considerable

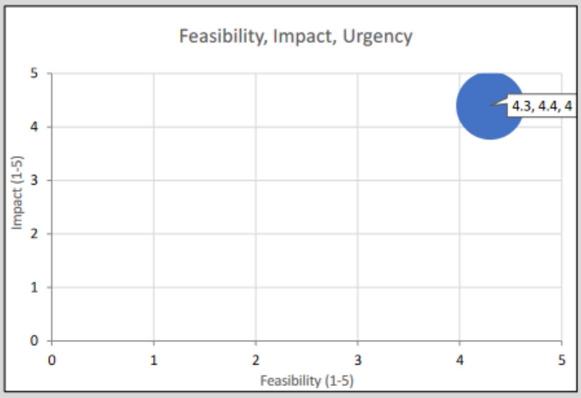
Feasibility: High

Legend (scores presented in the rectangle)

1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high

2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great

3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Great
Cost savings for end-user organisations:	Considerable

Industry & Research

Increase of business opportunities:	Moderate
Improvement of business quality management:	Considerable
Innovation progress:	Considerable
Improvement of business functions:	Great

Feasibility

Foundation:	Sufficient
Development perspectives:	Amply sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	CBRN attack Earthquake -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Developments in disaster resilience towards a more holistic approach for safety and security. Improved exchange of information and networking. Engaging of society as a whole.
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Changes in crisis management, developments in crisis management and disaster resilience, command-control and communication technology, protection of the public in general

ResiStand Assessment Framework 2.0 TNO

Figure 13. Screenshot of the “ASSESSMENT“ sheet of the RAF tool for CWA-2 (Stream 2).

3.2.2.6 Conclusion

As stated previously, three evaluators were assigned for this CWA. Due to the number of the evaluators, their answers cannot be used for direct statistical outcomes. For the evaluators' form the guidelines provided in the CWA: *Evaluation of exercises – Implementation Guidelines* were used, thus identifying capabilities and tasks.

The capabilities and tasks identified, along with the answers from the evaluators, are the following:

- Common operational picture and situational awareness

  Task 1. The information shared and proposed by the CWA provides a good situational awareness (Answers: 2 Strongly Agree, 1 Agree)
  Task 2. The CWA provides enhanced situational awareness (Answers: 2 Agree, 1 no answer)
  Task 3. Use and acceptance from the point of view of CIS. Are Cis ready to make use of such a CWA? (Answers: 2 Neutral, 1 Agree)
  Task 4. Based on the observations you made, do you think that the CI operator devoted too much time in order to provide all the necessary information (answers: 1 Agree, 1 Disagree, 1 Neutral)
- Adequately covering first responders' needs for deployment

  Task 1. Level of satisfaction of first responders for providing the necessary resources (field forces) (Answers: 2 Agree, 1 Neutral).
- Resilience and restoration

  Task 1. Support to the reduction of recovery time and stop of disruption (Answers: 2 Agree, 1 Strongly Agree).
- Promoting use of technological tools

  The concept/information is capable of supporting the development of new incident reporting tools? The concept/information asked is capable to alter existing incident reporting legacy systems? (Answers: 3 Yes).
- Compatibility with other existing standards and regulations

  Task 1. Is the concept/information in line/compatible with other existing standards (OASIS, M/ETHANE, other). (Answers: 1 Partially, 2 No answer).

The overall outcome from all three evaluators is positive. Not even one major drawback has been mentioned.

Some indicative comments from the evaluators are provided below:

- Comment on the resilience and restoration capability: "*Timely and adequate provision of information related to infrastructure needs and the incident will reduce recovery time. Also, very useful to provide info on affected services to prioritize actions*"
- Comment on the promoting use of technological tools: "*Yes, is capable to apply new incident reporting tools either automatic systems through webservices or through records by users*".
- Comment on the common operational picture and situational awareness: "*Some of the information could be optional e.g. UTC time*".
- Comment on the common operational picture and situational awareness: "*when asking for risks make clear if you ask for risks occurring in sequence of the event that occurred or if you mean the risk of the occurrence of an incident*".

- Comment on the common operational picture and situational awareness: "*for the incident description provide a standardized taxonomy and free text*".

The FSX proved that the CWA "*Emergency management – Incident situational reporting for Critical Infrastructures*" is on right path. The structure of the document and its content are more than sufficient in order to fit the purposes of incident notification and reporting, as well as to enhance the situation awareness and common operational picture.

Compared to the intermediate evaluation more fields were added, the content was described in a clear way and the potential to foster technological development became clear. This was displayed through the successful implementation of an online form as well as of the colour rules for printed template and the successful demonstration in the FSX of a command and control system that receives incident notification and reporting. The compatibility of the CWA with other standards was better understood by the participants. This latter case is an issue of subjectivity of the respondent as the extent to which the respondent to the questionnaire is aware of other existing technical standards is crucial for the overall answer.

As suggested by the title of the CWA "Emergency management – Incident situational reporting for Critical Infrastructures" the preexisting standards of EDXL-CAP and EDXL-SitRep hold pertinence within the context of the CWA and are duly addressed therein.

3.3 Final Evaluation of Stream 3 Response Planning

3.3.1 Basis of the Final Evaluation

Considering the progress of the CEN Workshop on the

CWA "Structuring an emergency response plan for crisis management stakeholders"

that has taken place following the intermediate evaluation of the document (see D4.3 [35] for further details), the updated concepts /outcomes of the said pre-standardization activity were again assessed primarily as part of the FSX (see chapter 2 of the present deliverable). Specifically, with respect to the thematical stream 3 of STRATEGY on Response planning, the final evaluation was realised in two sequential steps. These were:

a) Dedicated training session held on the 28th of March 2023.

During this session, the concepts, scope, objectives, and methodology of the CWA in consideration were presented and discussed as part of a small-scale training / tabletop exercise that was participated by representatives of operational stakeholders that included the organizations listed in the following table. Participants had the opportunity to evaluate the validity and completeness of the proposed response plane template versus their current planning approaches using a realistic scenario for supporting relevant discussions. In addition, practical examples of digital plans and benefits in line with the CWA concepts along with the operational plan that is currently being used by CNVVF for the case of waste disposal plant emergencies (which also follows the principles proposed by the CWA in discussion), was presented. Feedback was collected though a specifically designed questionnaire, the analysis of which is presented subsequently.

Table 12: List of organizations with attendants in the dedicated workshop for stream 3 – Response Planning.

Organisation Name	Type/Description
KEMEA (Greece)	RTO - Research and Technology Organisation;
CNVVF (Italy)	End-user (Firefighter)
Valabre (France)	End-user (Firefighter)
Hellenic Police (Greece)	End-user (Law Enforcement)
Ministry of National Defence (Greece)	End-user (Armed Forces)
ARPA UMBRIA (Italy)	End-user (Regional Agency for the protection of the Environment)
Prefettura – Ufficio Territoriale del Governo di Perugia (Italy)	End-user (Local Government Agency Prefecture)
STWS (Greece)	SME - Small and medium-sized enterprise

During the final stage of the training session in discussion and in consideration of any outcomes and conclusions that resulted, participants were engaged in providing feedback through a structured questionnaire that was prepared specifically for the occasion (also available online through a QR Code).

b) Full Scale Exercise held on the 30th of March 2023.

On the 30th of March, the actual Full-Scale Exercise was held. In the context of the Response Planning stream of the project (stream 3), the proposed concept of homogeneous response planning approach for establishing an efficient information exchange capacity during crisis response, was facilitated through a technical set-up that was based on the ENGAGE C2 system (provided by STWS) and is described in detail in D3.4 [28].

The feedback collection process was further supported through a team of specifically appointed evaluators that focused on collecting additional data / feedback as part of a parallel "note keeping" process targeting discussions of the session or short interviews of participants. Results are presented in the subsequent sections. The evaluation team is provided in the following table.

Table 13: KPIs team for Stream 3 – "Response planning".

Full Name	Affiliation	Role / Required expertise
Katerina Valouma	STWS	Lead Evaluator
George Sakkas	KEMEA	Evaluator
Stergios Mosialos	Hellenic Police	Evaluator

Through the aforementioned mechanism, data collected during the FSX was used as a basis for the final evaluation and included:

- Input from the live discussion between the participants of the exercise
- Notes collected by evaluators and CWA chairman
- Registration forms with information on the respondents
- Evaluation questionnaires filled in by the participants at the end of the TTX.

Further to the above, additional discussions took place and feedback was collected during the 2 Interoperability Events and Workshops on Crisis Management and Civil Protection held in Rome (Italy) and Berlin (Germany) in February (15, 16) and April (26, 27) 2023, respectively. The aim of said events has been to analyze the expected interoperability benefits though engaging a wide spectrum of both European and Non-European Crisis management stakeholders. Participants included representatives from Fire Services, Emergency Medical Services, Police, Armed forces, Academia, Industry, RTOs etc. In addition, representatives from the Advisory as well as the End-User advisory board of the project were present, providing feedback as per their own field of expertise (EMS, Fire Service, Emergency Communication, etc.) as well as to the project as a whole. The discussions held and the data collection approach that included the holding of group discussions, is described in detail in D4.2.

3.3.2 Results from Experts Discussions and Testing during the FSX

As explained in the previous section, discussions were held primarily during the training session on the 28th of March 2023 (as part of the Full-Scale Exercise). In this respect, upon a detailed presentation of the CWA concept as well as the Emergency Response Plan Structure, participants were engaged in a small-scale tabletop exercise that was based on a realistic hypothetical scenario. This assumed the onset of a series of multiple safety and security emergencies that included among others, a terrorist attack in natural gas installations, a CBRN incident, and a fire in a plastic recycling centre. In this respect, at specific points during the unfolding of the scenario, moderated discussions were held that corresponded to specific sections of the plan template proposed by the CWA vis a vis the current planning practices of participating stakeholders.

During the exchange of opinions among participants (with different operational background), the general benefit behind the homogenization of the structure of response plan documents was acknowledged. In this context, the facilitation of the exchange of mission critical information during a crisis, when plans follow a specific structure and content guidelines, was highlighted specifically for cases if digital plans. This capacity will largely contribute to the establishment of a common operational picture, especially for cases where multi-stakeholder intervention is required.

Further to the above, discussions highlighted the alignment of the response plan structure proposed by the CWA on "Structuring an emergency response plan for crisis management stakeholders" (please refer to D5.3 [38] for further details) to the plans already practiced by end users. In general, the proposed plan template was found to include all those critical sections for covering the provisions that would govern the operational mandate of end-users (first responders) with respect to crisis management. In addition, the expected content guidelines were deemed as being sufficient and helpful to future response planners for elaborating new plans or modifying existing ones.

Further to the above, the concept of the CWA was presented and discussed during the events / workshops of the project specifically designed to investigate the impact on interoperability with respect to STRATEGY's pre-standardization activity (see 5.3.1 and D4.2). In discussion held during said events (participated by a wide spectrum of European and non-European Stakeholders) representatives of the Food and Agriculture Organization of the United Nations (also participating the project's Advisory Board) highlighted the need to consider crises among those falling in scope of the template in consideration, that relate to animals. In other words, it was clarified that in numerous hazardous events, animals are affected in which cases a specific type of response that is undertaken by specialized responders that will have to exchange information with first responders. As such following a relevant document structure will facilitate interoperability with organization focusing on minimizing the impact to livestock.

3.3.3 Questionnaire Results

The scoring analysis stemming from the questionnaire as answered by the participants during the FSX - targeting the evaluation of the CWA ""Structuring an emergency response plan for crisis management stakeholders"", is subsequently provided. As followed for the case of the intermediate evaluation (described in D4.3 [35]) the average score calculated per question (using a Likert Scale) was used for quantifying the impact of the CWA along the considered evaluation dimensions and was elaborated using the equation explicated in D4.3 [35], chapter 2.

The figure below graphically depicts the scoring analysis. The questions from the respective questionnaire) correspond on a 1 to 1 basis to the KPIs listed in section 3.3 of D4.3 [35] (i.e., Q1 is related to KPI 1.a, as also approached for the case of the intermediate evaluation). In this respect, the average score calculated per question was used for quantifying the impact of the CWA.

As evident from the following figure, almost all questions that were evaluated using the Likert Scale were highly rated.

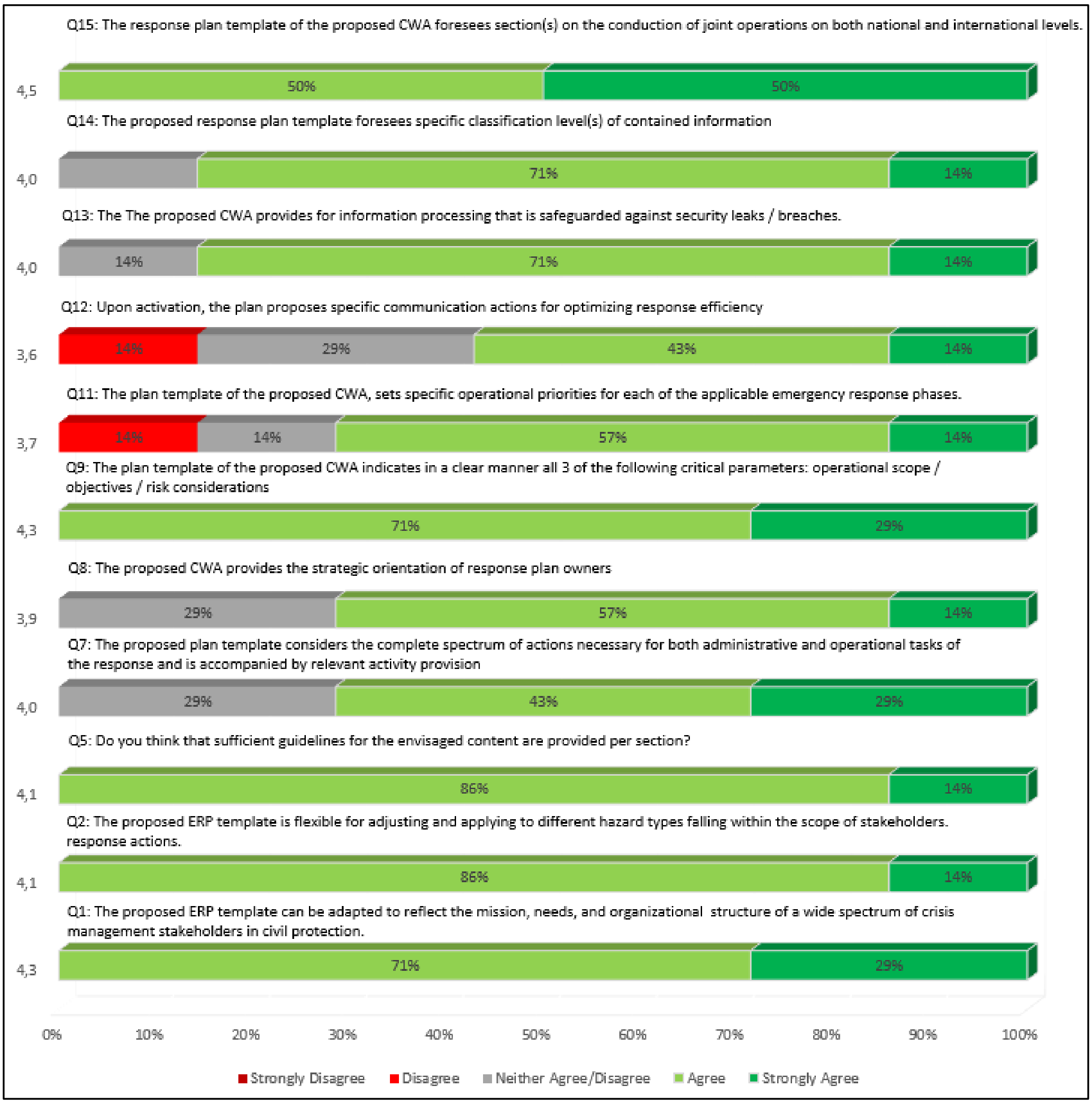


Figure 14: Scores' distribution obtained for all the statements in Stream 3 – Response Planning. questionnaires

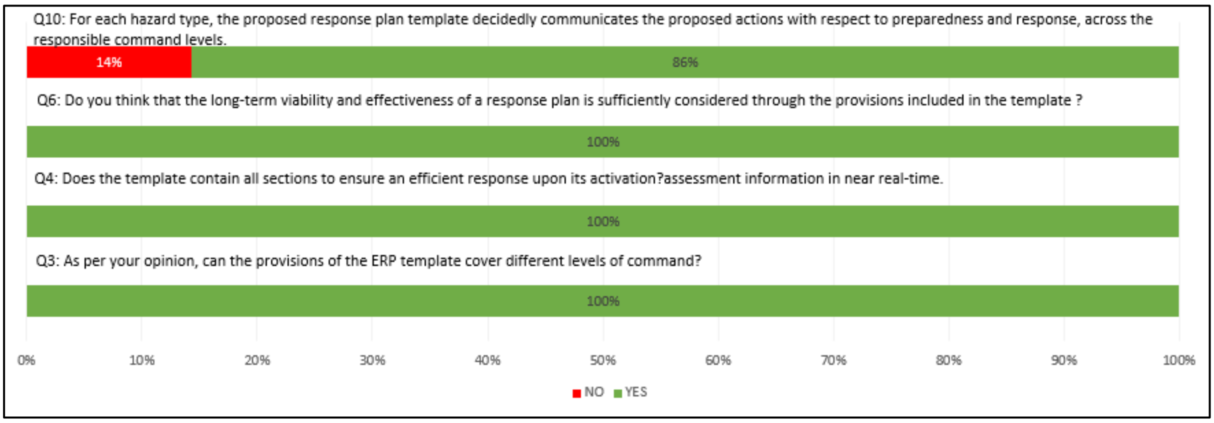


Figure 15: Answers' distribution for YES / NO questions in Stream 3 – Response Planning questionnaires.

Taking into account also the free text comments, the following conclusions can be drawn:

Q1: The proposed ERP template can be adapted to reflect the mission, needs, and organizational structure of a wide spectrum of crisis management stakeholders in civil protection. (Average Score 4.3)

This high rating received by the exercise participants has been perceived as a recognition of the proposed template structure's inherent ability to handle the response requirements for a broad range of crisis management stakeholders. In this regard, it was found that the core structure of the suggested template does contain all those thematic sections for ensuring availability of all necessary operational information required for the efficient response of stakeholders, and which are also required to provide a uniform format of the relevant documents in the future.

Q2: The proposed ERP template is flexible for adjusting and applying to different hazard types falling within the scope of stakeholders (Average Score 4.1)

As for the case of the previous question, this high scoring was interpreted as a measurement of the response plan template's inherent flexibility for accommodating various hazard types that fall within the scope of operational of crisis management stakeholders, while also ensuring the capacity for being applied and tailored to meet specific needs.

Q3: As per your opinion, can the provisions of the ERP template cover different levels of command? (Average Score 4.1)

As for the case of the previous question, this high scoring was interpreted as a measurement of the response plan template's inherent flexibility for accommodating various levels of command that fall within the scope of operational of crisis management stakeholders, while also ensuring the capacity for being applied and tailored to meet specific needs.

Q5: Do you think that sufficient guidelines for the envisaged content are provided per section? (Average Score 4.1).

Evaluating crisis management stakeholders acknowledged that the instructions supplied with regards to the content that needed to be filled in each section of the template were sufficient. This is a crucial component for crisis management stakeholders to consider when deciding whether to use the template and how to make the transition from current response planning procedures seamless.

Q7: The proposed plan template considers the complete spectrum of actions necessary for both administrative and operational tasks of the response and is accompanied by relevant activity provisions (Average Score 4.0).

The statement's high score highlights the fact that the suggested template for developing an emergency response plan, would be applicable to an extended spectrum of stakeholders in crisis management who would be involved in managing a civil protection crisis (including both the underlying operational as well as administrative processes). The answers referring to a more neutral stance (neither agree nor disagree) has been communicated as a willingness for further investigation (against existing plan structure) and provision of further feedback should any significant observations arise.

Q8: The proposed CWA provides the strategic orientation of response plan owners (Average Score 3.9).

Considering the feedback received for a part of this question, it has been highlighted that the response plan template contains provisions pertinent to the strategic orientation of stakeholder utilizing the response plan in discussion. This is interpreted as a support for the plan's application at all relevant levels of command (while providing the flexibility of being further adapted to optimize such a potential as per any specificities).

Q9: The plan template of the proposed CWA indicates in a clear manner all 3 of the following critical parameters: operational scope / objectives / risk considerations (Average Score 4.3).

Taking into account the high scoring assigned to the respective question, Stakeholders have noted the plan's clarity with regard to the scope, objectives, and risk factors. The large majority of end users who evaluated the template clearly believe that it includes all the necessary provisions to outline its applicability across a broad spectrum of end users and for a variety of hazard types. In this regard, responses were largely in line with the aforementioned assertion with all of them agreeing or strongly agreeing to the said position.

Q11: The plan template of the proposed CWA, sets specific operational priorities for each of the applicable emergency response phases. (Average Score 3.7).

The response plan template has been acknowledged to encompass the necessary flexibility / provisions for setting the respective operational priorities (that are stakeholder & response phase-specific), considering its intent to support stakeholders across all phases of their engagement in emergency response. Most respondents appeared to agree and strongly agree with the remark. One "Neither Agree nor Disagree "answer was received which was seen as "having arguments for being both in favour and against" and has therefore been viewed as a neutral position. As far as the “Disagree” answer is concerned, additional clarifications were provided on the fact that the template only aims at remaining sufficiently generic for providing the general framework for incorporating the said operational priorities (and not to set them). In any case, the comment has been considered for the finalisation of the CWA.

Q12: Upon activation, the plan proposes specific communication actions for optimizing response efficiency (Average Score 3.6).

This question reflects the opinion of the end-users that the template under consideration does include sections for integrating the specific communication actions to be taken by the responsible stakeholders during response to ensure efficiency. The majority of replies that were received, agreed/strongly agreed with the aforementioned assertion. The two that were left were 1 "Neither Agree nor Disagree option," which was seen as "having arguments for being both in Favour and against" and has therefore been perceived as a neutral position, and 1 “Disagree” answer that mainly refereed to the original expectation of the respondent that the template would include specific actions. It was clarified that that the objective of the template is to merely provide the capacity for potential planners to include such information in the plan documents on the basis of the said template and not to propose specific communication actions, which are however stakeholder specific.

Q13: The proposed CWA provides for information processing that is safeguarded against security leaks / breaches. (Average Score 4.0).

All the responses emphasized the availability of provisions in the template that would protect the confidentiality of data contained in a response plan against breaches and malicious attacks that aim at undermining the effectiveness of the response when deemed necessary. In this respect, the provisions of sections A.1.3 *Classification level* and A.1.5 *Record of distribution* will introduce a control system for guaranteeing that only responsible parties have access rights.

Q14: The proposed response plan template foresees specific classification level(s) of contained information. (Average Score 4.0).

The vast majority of responses appreciated the availability of clauses that set specific classification levels with regards to the information included in the response plan. More specifically, section A.1.3 defines the classification level of the document (according, among others, to Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p.53).

Q15: The response plan template of the proposed CWA foresees section(s) on the conduction of joint operations on both national and international levels. (Average Score 4.5).

The suggested template does have explicit elements for facilitating collaboration among different crises management stakeholders in situations where multi-stakeholder involvement is necessary. As acknowledged by FSX participating respondents, in order to enable the performance of joint operations on a national and international scale, respectively, sections like A.7 “Liaisons network – interoperability” and A.8 “International cooperation – help” have been added.

Besides the questions assessed using the Likert scale, additional evaluation dimensions were approached from a YES/NO perspective. This is because they are perceived to have a binary interpretation with respect to their impact on the response plan template in consideration. **Error! Reference source not found.** summarizes the respective results. For cases of not receiving any answers (and in the absence of further justification), it was considered as not having experience and as such these were not considered for further analysis.

Other evaluation variables were tackled from a YES/NO perspective in addition to the aforementioned questions evaluated using the Likert scale. This is due to the perception that they can only have one of two effects on the response plan template under review. The corresponding results, shown in the figure above, reveal a full alignment with respect to the KPIs set by the Questions Q3, Q4 and Q6 of the respective questionnaire (see D4.3 [35] – section 3.3 for further details). A “No” response was received as part of Q10 with regards to provisions for communicating the proposed actions with respect to preparedness and response, across the responsible command levels, with no additional rationale. This was considered as a need to highlight further the fact that the proposed plan template aims at providing the core set of provisions/clauses of plans that could be further adapted by response planers for building their specific plans and not setting organisation specific details as part of the plan template.

3.3.4 Results from Formal Aspects Evaluation

The formal aspects evaluation of the pre-standardization documents, that was conducted based on a specific questionnaire, was opted to be supported by National Standardization bodies provided their experience in the relevant domain. Specifically, with respect to the CWA on Response Planning the evaluation was conducted by a representative of ASRO that was not engaged in the elaboration process of the document in consideration for ensuring a non-biased outcome. The process resulted in the acknowledgement of an outcome that fully respected the quality aspects set out by the respective KPIs as shown in the figure below. Remarks were limited to minor editorial recommendations regarding the Tables / Figures used that were explained and addressed as part of the final document.

In a nutshell, based on the evaluation outcomes the total average score is 4.82 out of 5 which indicates that the CWA of Stream 3 has fulfilled the expected quality standards and with respect to the content and structure and is suitable for (a) creating the necessary impact among the stakeholders community, (b) supporting the work of the Technical Committees of CEN/TC 391 toward investigating its potential to work towards continuing the work of this pre-standardization activity upon submission of a New Work Item Proposal.

3.3.5 Results from RAF Tool Analysis

As described generally in chapter 2, for each CWA/TS a ResiStand Assessment Framework (RAF) supported evaluation was performed in WP2 and reconsidered and updated in the framework of Task T4.3.

Regarding stream 3 the output from the RAF Excel tabs labelled "INTAKE" (m) and "ASSESSMENT" (Figure 17) is shown in the figures below. The "Intake" sheet is used as the first data entry point and provides information about the need for and potential effects of the CWA on various stakeholder groups. Further analysis was based on these data. The description of the CWA's scope, coupled with an illustration of its actual use and its target audiences, laid the framework for the upcoming study.

The RAF analysis for the case of the pre-standardisation activity of stream 3, predicts that the CWA on "Exchanging building and infrastructure damage information with Common Alerting Protocol" will significantly improve crisis management efficiency through enhanced interoperability among stakeholders, specifically in situations where response takes place in a multi-stakeholder environment. It is anticipated that by homogenising the structure based on which Emergency Response Plan (ERP) documents are being elaborated, will greatly facilitate the exchange of operational information across all phases of crisis management, thus largely increasing the communication efficiency, specifically during crises. As a result, the situational awareness of crisis management stakeholders will be enhanced and will expectedly lead to an improved management of available resources. Improving the situational awareness because of the enhanced inflow of information will also enable first responders to prioritize their actions resulting in a significant reduction of the impact brought on by disasters.

Intake

ResiStand

Title of the proposed standard:	Standards for cross-border cooperation between organisations
Identification number:	ResPI-3

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):	Organisation or project consortium	Stakeholder category
	Satways	Industry
	CNVVF	End-users
	Hellenic Police	End-users
	HMOD	End-users
	CINAMIL	End-users

Type of standard:	Unknown
Scope of the standard:	To develop process standards to facilitate interoperability between organisations and cross-border cooperation, to enhance synergies and develop interagency frameworks.
Example or illustration 1:	A generic response plan template is planned to be proposed (in order to cover different types of situations). However, our activities (use cases, TTXs etc) will be focused on two cases: the CBRN incidents and incidents
Example or illustration 2:	

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Yes	1. Work Programme 2018-2020: Secure societies - Protecting freedom and security of Europe and its citizens
with national and regional legislation:	Yes	SEVESO Directive implemented in Greece by the J.M.D. 172058/2016 (G.G. 354/B/2016) (point 2 above)

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Firefighters, Police, Crisis Management Operation Centers etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	Unknown	
Industry/SMEs:	Yes	Technology companies developing incident management and decision support
Consultancy organisations:	Unknown	
Research institutes:	Yes	Research Centers when developing new solutions/applications for public safety
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protection Mechanism

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	End users will benefit from the collaborative response planning process and the exchange of digitized response plans.
Industry:	Technology providers will be able to desing interoperable systems and take into consideration unified and collaborative response planning processes
Research:	Research projects in the field of security, civil protection and public safety will be able to use the proposed standard during implementation of solutions and trials/pilots
Policy makers:	Strengthening civil protection capacities in the partner countries; Strengthening cooperation between the partner countries, and bringing them closer to the EU Civil Protection Mechanism
Citizens:	Create safer societies

Urgency of the standard: (when is it needed for implementation?)	High (within 1 year)	Nowadays, the existence of emergency situations that spread across borders, requires the involvement of different agencies. The basis for an effective multi-agency coordination and collaboration process is the clear identification of agencies'

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	To adopt the standard proposed
NGOs:	Unknown	
Industry/SMEs:	Yes	To provide the technical systems/solutions/applications
Consultancy organisations:	No	
Research institutes:	Yes	To desing new ideas, projects and implement when testing/piloting projects in the
Others:	Yes	Standardisation organisations to promote standardisation and benefits of the standard
Preferred leading type of stakeholder:	First Responders	Firefighters, Police, Crisis Management Operation Centers etc.

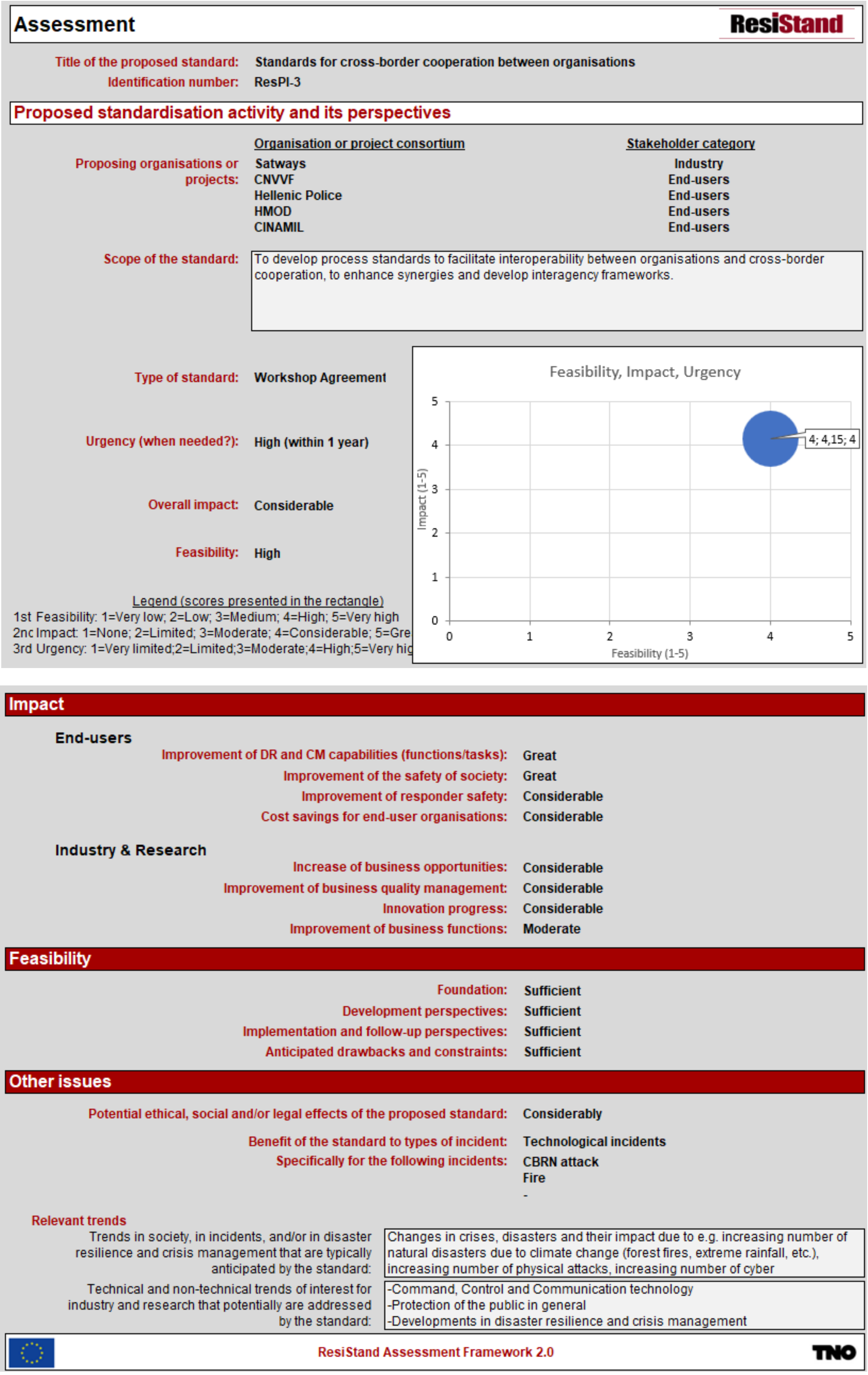
Expected barriers and constraints:	

Other information

Additional remarks:	

ResiStand Assessment Framework 2.0 TNO

Figure 16: RAF 2.0 – Intake tab filled for the CWA on Response Planning (Stream 3)



Assessment ResiStand

Title of the proposed standard: Standards for cross-border cooperation between organisations
Identification number: ResPI-3

Proposed standardisation activity and its perspectives

Proposing organisations or projects: Organisation or project consortium	Stakeholder category
Satways	Industry
CNVVF	End-users
Hellenic Police	End-users
HMOD	End-users
CINAMIL	End-users

Scope of the standard: To develop process standards to facilitate interoperability between organisations and cross-border cooperation, to enhance synergies and develop interagency frameworks.

Type of standard: Workshop Agreement

Urgency (when needed?): High (within 1 year)

Overall impact: Considerable

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nc Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Gre
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very hig

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Considerable
Cost savings for end-user organisations:	Considerable

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Considerable
Innovation progress:	Considerable
Improvement of business functions:	Moderate

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Technological incidents
Specifically for the following incidents:	CBRN attack Fire -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Changes in crises, disasters and their impact due to e.g. increasing number of natural disasters due to climate change (forest fires, extreme rainfall, etc.), increasing number of physical attacks, increasing number of cyber
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	-Command, Control and Communication technology -Protection of the public in general -Developments in disaster resilience and crisis management

ResiStand Assessment Framework 2.0 TNO

Figure 17: RAF 2.0 – Assessment tab filled for the CWA on Response Planning (Stream 3)

Provided the above, taking as input all the respective knowledge gained as part of the validation steps taken during the period in consideration, outputs of the RAF tool assessment, reveal the potential and the benefit that the specific pre-standardization activity is expected to have. Considering the urgency and impact that covering this need is expected to have, the CWA is also considered to be used as a basis for the submission of a New Work Item Proposal (NWIP) (please refer to D5.3 [38] for more details).

3.3.6 Conclusions

The feedback received during the validation steps that were foreseen for the period after the tabletop Exercises (see D2.12 [14]- and D4.3 [35] for relevant details) has largely underlined the necessity of homogenization of the response planning approach with respect to structuring the plan documents.

As became evident form the comments collected through surveys and conversations the KPIs have received favourable assessments from the FSX participants and attendees of relevant project events (interoperability events – described in D4.2). The evaluation, comments and observations made during the exercises served as helpful steering tool for improving the CWA's content. In this context, participants of the respective CEN workshop have appreciated their value in enhancing the CWA towards establishing a useful tool for organisations to structure (either existing and/or new) plans.

As a result, the document has been enriched and updated accordingly. In general, the proposed structure of the CWA received positive feedback for its effectiveness in facilitating coordination and collaboration among relevant authorities during emergencies. It was also commended for enhancing the decision-making process and enabling a more efficient and timely response following a disaster. With the CWA a common approach has been developed, that can help practitioners homogenize their planning processes according to an existing pre-standardization deliverable.

In general, the document effectiveness in promoting coordination and collaboration between crisis management stakeholders (First Responders) during emergencies was appreciated. In addition, the document potential in improving the decision-making process towards optimising the efficiency of the response operations was identified.

3.4 Final Evaluation of Stream 4 Command & Control

3.4.1 Final Evaluation of Stream 4, CWA 1

3.4.1.1 Basis of the Final Evaluation

Taking into account the advancements made within the two CEN Workshops of Stream 4 focused on

> CWA "Collaborative emergency response - Common addressing format and emergency identification protocol" and,
>
> "Management of forest fire incidents – SITAC-based symbology"

which have occurred subsequent to the intermediate assessment of the document (please consult D4.3 for more in-depth information), the updated concepts and results of this preliminary standardization effort were once again appraised. This evaluation primarily occurred as an integral component of the Full Scale Exercise (FSX) conducted in Gualdo Tadino, located in the northeastern Umbria region of Italy's Perugia province, spanning from March 27th to March 31st, 2023. This event was organized by the Italian National Fire and Rescue Service and encompassed the practical assessment of all pre-standardization undertakings within the STRATEGY project under authentic operational circumstances. Notably, concerning the thematic stream 4 of the STRATEGY initiative which pertains to Command and Control, the ultimate assessment took place in a dual-step progression. These steps were as follows:

a) Two dedicated training sessions (one per CWA) held on the 28th of March 2023.
   In these sessions, the concepts, extent, aims, and procedure of the considered CWA were introduced and deliberated upon. This occurred within the framework of a compact training session or tabletop exercise, attended by operational stakeholders' representatives. Among the participants were various organizations listed in the table below. During this engagement, attendees had the occasion to assess the effectiveness and comprehensiveness of the suggested agencies' communication protocol and the symbol set for the forest fires in comparison to their existing planning methodologies. This evaluation was conducted using a realistic scenario that facilitated pertinent discussions.
   Feedback was gathered through a specifically tailored questionnaire, the subsequent analysis of which is presented.

Table 14 List of organisations with attendants in the dedicated workshop

Organisation Name	Type/Description
KEMEA (Greece)	RTO - Research and Technology Organisation;
CNVVF (Italy)	End-user (Firefighter)
Hellenic Police (Greece)	End-user (Law Enforcement)
Ministry of National Defence (Greece)	End-user (Armed Forces)
ARPA UMBRIA (Italy)	End-user (Regional Agency for the protection of the Environment)

Organisation Name	Type/Description
Prefettura – Ufficio Territoriale del Governo di Perugia (Italy)	End-user (Local Government Agency Prefecture)
STWS (Greece)	SME - Small and medium-sized enterprise

In the concluding phase of the discussed training sessions, participants were involved in sharing their feedback. This was facilitated through the utilization of a well-structured questionnaire, meticulously crafted for this purpose. To enhance the efficiency of the feedback collection process, a group of designated evaluators was assigned. These evaluators were tasked with gathering supplementary data and feedback in tandem with a parallel "note-keeping" process. This process was aimed at capturing discussions within the session and conducting brief interviews with the attendees. The outcomes of this evaluation are detailed in the subsequent sections. The composition of this evaluation team can be found in the following table.

Table 15: KPIs team for Stream 4 – “Command and Control".

Full Name	Affiliation	Role
Giannis Chasiotis	STWS	Member of Stream-4-KPI-Group
Leonidas Perlepes	STWS	Member of Stream-4-KPI-Group
Ilias Gkotsis	STWS	Member of Stream-4-KPI-Group
Alex Pagkozidis	STWS	Member of Stream-4-KPI-Group
Juliane Schlierkamp	FhG-INT	Member of Stream-8-KPI-Group
Wilson Antunes Maj	CINAMIL	Member of Stream-8-KPI-Group
Pertti Woitsch	WCO	Member of Stream-8-KPI-Group

b) Full Scale Exercise held on the 30th of March 2023.

On March 30th, the official execution of the Full-Scale Exercise took place, with broad participation from various operational stakeholders. This inclusive group encompassed not only high-ranking representatives from national-level political and operational entities associated with the Italian hosts but also a diverse array of contributors. The event was orchestrated through the enactment of two distinct operational scenarios, the comprehensive depiction of which can be found in deliverable D4.1. Within the context of Stream 4 Command and Control and more specifically, using the approaches that are proposed by both Stream 4 CWA, multiple agencies were able to communicate, exchanging operational information and depicting information that was related to a forest fire in a common way. This was made possible through the implementation of a technical arrangement centred around the ENGAGE C2 system, provided by Satways Ltd. For a thorough understanding of this arrangement, please refer to the detailed explanation provided in deliverable D3.5.

Through the aforementioned mechanism, data collected during the FSX was used as a basis for the final evaluation and included:

- Input from the live discussion between the participants of the exercise
- Notes collected by evaluators and CWA chairman.
- Registration forms with information on the respondents
- Evaluation questionnaires were filled in by the participants at the end of the TTX.

Further to the above, additional discussions took place and feedback was collected during the 2 Interoperability Events and Workshops on Crisis Management and Civil Protection held in Rome (Italy) and Berlin (Germany) in February (15,16) and April (26,27) 2023 respectively. The aim of the said event has been to analyze the expected interoperability benefits through engaging a wide spectrum of both European and Non-European Crisis management stakeholders. Participants included representatives from Fire Services, Emergency Medical Services, Police, Armed forces, Academia, Industry, RTOs etc. In addition, representatives from the Advisory as well as the End-User advisory board of the project were present, providing feedback as per their own field of expertise (EMS, Fire Service, Emergency Communication, etc.) as well as to the project as a whole. The discussions held and the data collection approach that included the holding of group discussions is described in detail in D4.2.

3.4.1.2 Results from Expert Discussions and Testing during the FSX

In the previous days of the FSX, stakeholders were grouped per CWA topic and expertise, presenting them with the current version of the CWA, discussing technical and operational details, and discussing possible extensions and improvements.

During the FSX exercise and specifically on the 30th of March, STRATEGY partners and external experts engaged in the exercise as players represented stakeholders and provided key feedback with regard to all the streams. Besides the players, who were engaged in the exercise scenarios, there were also people who attended the FSX as observers. These persons contributed to the FSX by providing comments in the context of an open discussion session and evaluation through a dedicated questionnaire.

In general, all participants were informed of the plan of the meeting, the objectives of the FSX and the STRATEGY project approach through a quick presentation. It was highlighted that two scenarios will support the exercise implementation:

- Scenario A: Earthquake causes the partial collapse of a dam element.
- Scenario B: Wildland-Urban-Interface fire triggers waste-disposal and treatment plants.

Within scenario B the content of both Stream 4 CWAs was promoted. More specifically, the use of a common identification protocol for information sharing among the agencies and the use of a common symbology in collaborative response activities were promoted. These were also showcased through a mini preparatory workshop on the 28th of March. The importance of a common understanding of all partners involved in response is well established, considering that disasters are agnostic to borders and an incident can affect different countries. During cross-border emergencies, joint efforts of involved services could be facilitated using common symbology. Participants were divided into three small groups corresponding to civil protection stakeholders of different types of agencies and of different origins involved in a multidisciplinary disaster. They were provided with an indicative scenario of a WUI fire, including several steps that tried to include several types of operational tasks and different needs, offering them the possibility to depict them on the map using the proposed symbol set, take decisions and share information among the three teams (crisis communication).

After the demonstration, two dedicated sessions (one per Stream's CWA) were designed as an open dialogue session for the participants to summarize their experiences/comments. Additionally, online questionnaires have been shared with the participants to collect feedback for validating and evaluating the final results produced in the context of the CWA. After the FSX all responses have been analysed.

Based on the results and comments collected, the following changes or improvements (or generic comments) were provided:

- Regarding CWA "Collaborative emergency response - Common addressing format and emergency identification protocol":
    - The proposed identification protocol should be in line with the directives and policies that are used by the EU authorities.
    - The acronym of the countries that are used by the protocol should be in line with the updated ISO (published in 2022) standard.
    - The proposed identification protocol should be expandable, being able to facilitate information sharing among more types of agencies and authorities, than the usual ones.
    - The proposed protocol should be agnostic to the internal hierarchy of agencies, in order to be able to support different types of agencies from different countries, as different hierarchy schemes are used by the agencies around the EU.

3.4.1.3 Questionnaire Results

The subsequent section presents the analysis of scores derived from the questionnaire responses given by participants during the FSX. This analysis is directed towards the assessment of the CWA titled "Collaborative emergency response - Common addressing format and emergency identification protocol." Just as it was done for the intermediate evaluation, as described in D4.3, the average score per question was employed to measure the impact of the CWA across the evaluated dimensions. This process was detailed using the equation provided below. The approach taken for this scoring analysis adheres to the assumption that the assigned scoring scale (including options like Strongly Disagree, Disagree, Neither agree nor disagree, etc.) is in accordance with the principles of the Likert scale.

As evident from the following figure, almost all questions that were evaluated using the Likert Scale were highly rated.

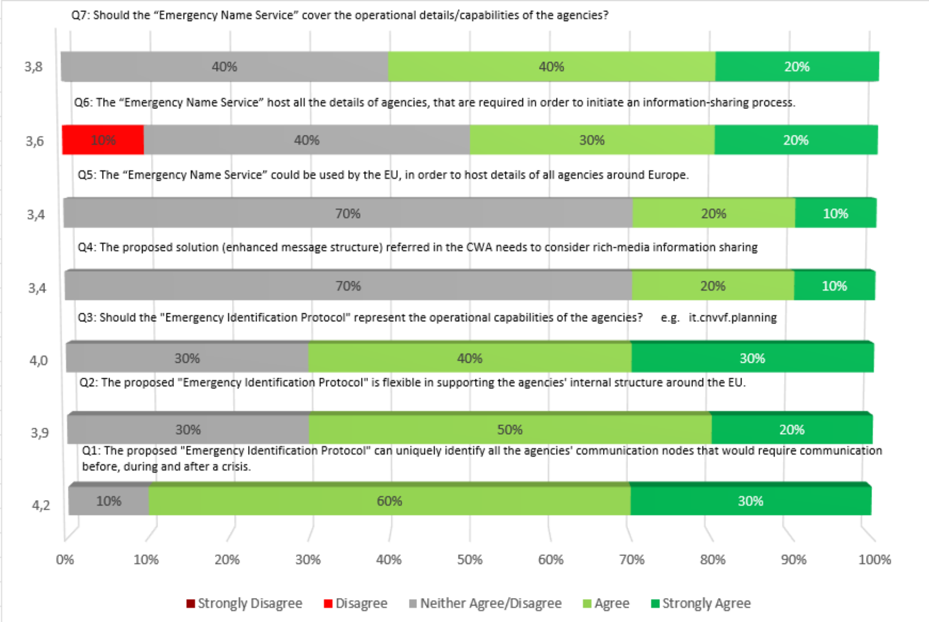


Figure 18 Answers for each question of the evaluation questionnaire for Stream 4 CWA "Collaborative emergency response - Common addressing format and emergency identification protocol"

<u>Q1: The proposed "Emergency Identification Protocol" can uniquely identify all the agencies' communication nodes that would require communication before, during and after a crisis. (Average Score 4.2)</u>

The protocol that is proposed describes an approach that could be adopted by the agencies, in order to uniquely identify all their communication nodes. This would facilitate communication among the agencies before during and after a crisis. However, some doubts were expressed as currently not all agencies around the EU are equipped with C2 systems, being able to exchange information.

<u>Q2: The proposed "Emergency Identification Protocol" is flexible in supporting the agencies' internal structure around the EU. (Average Score 3.9)</u>

The proposed protocol has been designed to be agnostic to agencies' internal structure, being flexible to be adapted to all agencies' needs. However, the special operations/responsibilities of a number of agencies around the EU, would require the enhancement of the list of the operations that are covered by the protocol.

<u>Q3: Should the "Emergency Identification Protocol" represent the operational capabilities of the agencies? e.g. it.cnvvf.planning (Average Score 4.0)</u>

The proposed protocol should represent the operational capabilities of the agencies. However, a small number of participants mentioned that this approach should be avoided in order for the communication nodes that will be identified, to be generic and capable of operating in any type of incidents.

<u>Q4: The proposed solution (enhanced message structure) referred in the CWA needs to consider rich-media information sharing (Average Score 3.4).</u>

The proposed protocol has been designed in order to support the sharing of rich-media information. However, some participants were wary of this approach, as the sharing of rich media information should always be consistent with agency sharing policies (especially as this information is typically classified as confidential).

<u>*Q5: The "Emergency Name Service" could be used by the EU, in order to host details of all agencies around Europe. (Average Score 3.4).*</u>

According to the results, the proposed "Emergency Name Service" could be used by the EU, in order to host details of all agencies around Europe. However, a number of the participants were quite reluctant to this approach as it is highly dependent on political decisions around the EU (being difficult to adapt at the central level).

<u>*Q6: The "Emergency Name Service" host all the details of agencies, that are required in order to initiate an information-sharing process. (Average Score 3.6).*</u>

The "Emergency Name Service" is capable of hosting all the details of the agencies, that are required in order to initiate an information-sharing process. This covers the hosting of either technical (e.g. IP, supporting protocols etc.) or operational (e.g. responsibilities of the node) details. However, a number of participants would prefer this service to host only information that is related to the operational capabilities of the nodes, avoiding any technical information.

<u>*Q7: Should the "Emergency Name Service" cover the operational details/capabilities of the agencies (Average Score 3.8).*</u>

This question is quite relevant to the previous one. A number of participants would like to have all types of details in the service catalogue while others would like to have only the operational details of the agencies stored.

3.4.1.4 Results from Formal Aspects Evaluation

A questionnaire for the evaluation of formal aspects of the pre-standardisation documents and technical specifications provided for by the different streams has been created and was filled by Speranta Stomff from ASRO (National Standardization Body) for this CWA. More details per category-criteria (each one contains several KPIs and each one is liked with one question) used to evaluate the CWA are depicted in the following figure. As presented in the following Figure, 5 of 12 criteria have been completely achieved, while 6 out of 13 have achieved a score of 4 out of 5 and above on a Likert scale.

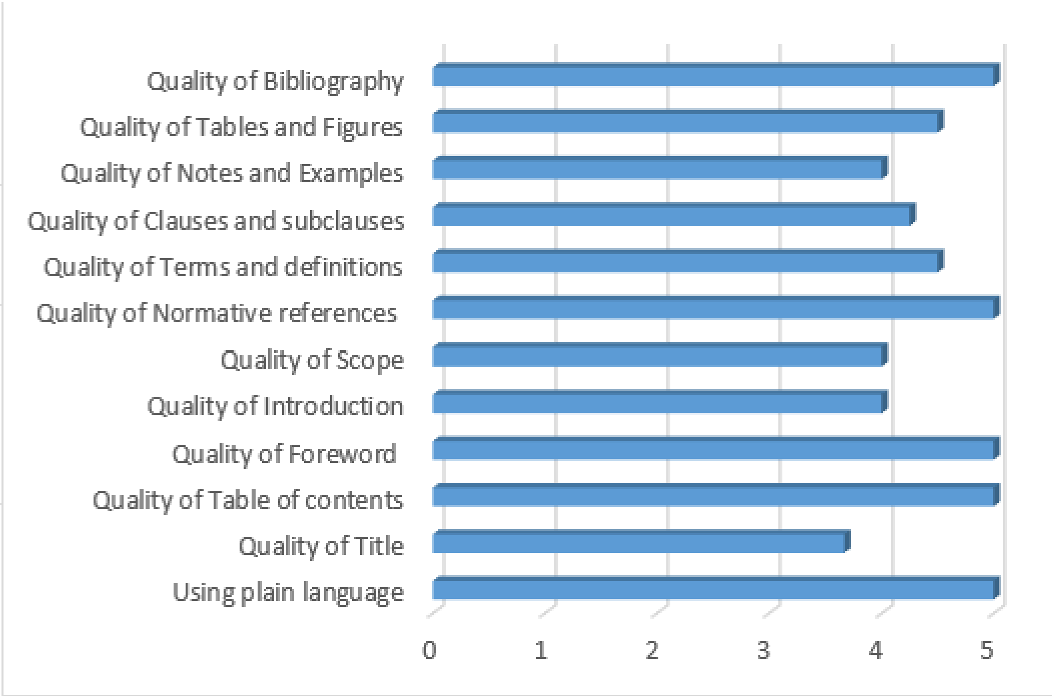


Figure 19: Average values of questions per each formal aspect evaluation criterion.

In a nutshell, based on the evaluation outcomes the total average score is 4.46 which indicates that the CWA in discussion of this chapter is appropriate and has a comprehensible structure.

3.4.1.5 Results from RAF tool Analysis

In addition to the aforementioned questionnaires, discussions, and other evaluation methods employed to assess the CWA, the ResiStand Assessment Framework (RAF) has been utilized. Developed as part of the ResiStand project, the RAF serves as a tool to aid organizations or individuals in gauging the potential effects of a prospective standardization initiative and its practical feasibility within the field of disaster resilience and crisis management.

Accordingly, the RAF was applied to analyze several aspects related to the CWA. This encompassed mapping out the potential advantages it could bring, verifying its alignment with essential ethical, legal, and societal considerations, and considering the organizational prerequisites necessary for the development, implementation, and potential progression of the CWA into a full-fledged standard.

To initiate this process, we began by completing the initial phase, which involved filling out the intake sheet. This intake sheet offers insights into various elements, including the urgency associated with establishing the proposed standard and initial textual descriptions outlining the potential impacts for different categories of stakeholders.

Intake

ResiStand

Title of the proposed standard: Standards for Situtational Awareness and Common operational Picture

Identification number: CC-01

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
Satways	Industry
CNVVF	End-users
Hellenic Police	End-users
HMOD	End-users
	-

Type of standard: Workshop Agreement

Scope of the standard: The standardization of the collaboration and exchange of operational information among the agencies that are involved in an emergency situation.
The proposed standard covers:
Standardisation of interfaces, to promote data interoperability to get the same picture of the disaster.

Example or illustration 1: The onset of emergency situations that spreads either across borders or into a specific country, requires involvement of different civil protection agencies, often with different C2 systems. The collaboration among the

Example or illustration 2:

Compliance with legislation

	Compliant (Y/N)?	Explanation
with European legislation:	Yes	1. Work Programme 2018-2020: Secure societies - Protecting freedom and security of Europe and its citizens
with national and regional legislation:	Yes	1. National Special Plan on the "Human Loss Management" M.D. 3384/2006 (G.G. 776/28-6-2006)

Target groups for applying the standard

	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Crisis Management Operation Centers, Firefighters, Police etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	Unknown	
Industry/SMEs:	Yes	Technology companies developing incident management and decision support solutions,
Consultancy organisations:	Unknown	
Research Institutes:	Yes	Research Centers when developing new solutions/applications for public safety
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protection Mechanism

Potential impact and urgency

Benefits for stakeholder categories

	Description of who will benefit and in what way
End-users:	During emergency situations that requires involvement of different civil protection agencies raise several issues with regards to the common understanding, communication and collaboration among the involved agencies.
Industry:	Technology providers will be able to ensure interoperability among different Command and Control Systems and their associated functions
Research:	Research projects in the field of security, civil protection and public safety will be able to use it during implementation
Policy makers:	Strengthening civil protection capacities in the partner countries; Strengthening cooperation between the partner countries, and bringing them closer to the EU Civil Protection
Citizens:	Create safer societies

Urgency of the standard: (when is it needed for implementation?) Moderate (< 2 yrs)

There is a clear need for standardisation of the collaboration and exchange of operational information among agencies (both military and civil protection). In modern societies there is an imperative need of common understanding between agencies, either on national or

Development

Required types of stakeholders

	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	To adopt the proposed standard
NGOs:	Unknown	
Industry/SMEs:	Yes	To provide technical systems/applications
Consultancy organisations:	Yes	
Research Institutes:	Yes	To desing new ideas, projects and implement when testing/piloting projects in the domain
Others:	Yes	Standardisation organisations

Preferred leading type of stakeholder: First Responders — Crisis Management Operation Centers should be vital members during the development

Expected barriers and constraints:

Other information

Additional remarks:

ResiStand Assessment Framework 2.0 TNO

Figure 20: RAF 2.0 – Intake tab filled for the CWA on collaborative emergency response (Stream 4)

Impact - Industry & Research

ResiStand

Potential impact of the proposed standard

Score: Considerable

Applicability to trends

Relevant trends

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	-Command, Control and Communication technology -Developments in disaster resilience and crisis management

Increase of business opportunities

Score: Considerable

	Potential increase	Explanation
Increasing sales on existing market: *(increased market share)*	Considerable	
Access to new markets: *(e.g. geographical or sectoral, dual use, new product)*	Considerable	
New partnerships: *(between companies, public-private, etc.)*	Moderate	
Improved profit margin: *(e.g. economies of scale)*	Considerable	
Cost reduction: *(as a result of cheaper procurement, production and sales)*	Limited	

Additional remarks

Improvement of business quality management

Score: Moderate

	Potential improvement	Explanation
Customer satisfaction/image/reputation:	Moderate	
Consistent quality:	Moderate	

Additional remarks

Innovation progress

Score: Great

	Potential progress	Explanation
New identified market needs:	Great	
Knowledge development and transfer:	Considerable	
Applies scientific knowledge to practice:	Considerable	

Additional remarks

Improvement of business functions

Score: Moderate

	Performance improvement	Explanation
Inbound logistics: *(receiving and storing incoming goods or material for use)*	None	
Production / Operations: *(processing, quality assurance, health, safety and environment)*	None	
Outbound logistics: *(storing, transporting, and distributing goods to customers)*	None	
Marketing and Sales: *(e.g. market analysis, marketing, contracting, sales)*	Moderate	
Service: *(customer care and technical support)*	None	
Management & Administration: *(general mgt., finance, control, legal, facility mgt., IT, HR)*	None	
Engineering / Construction: *(efficient engineering, design, construction)*	Moderate	
Research & Development: *(efficient R&D, knowledge mgt., research, product development)*	Limited	
Procurement: (procurement activities, screening and selection of suppliers, negotiating and contracting)	None	

Additional remarks:

Other information

Additional remarks:

ResiStand Assessment Framework 2.0

TNO

Figure 21: RAF 2.0 – Impact tab filled for the CWA on collaborative emergency response (Stream 4)

Assessment

ResiStand

Title of the proposed standard: Standards for Situational Awareness and Common operational Picture
Identification number: CC-01

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
Satways	Industry
CNVVF	End-users
Hellenic Police	End-users
HMOD	End-users
	-

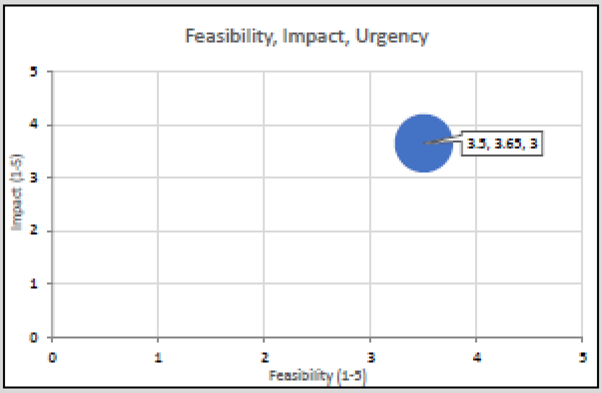
Scope of the standard: The standardization of the collaboration and exchange of operational information among the agencies that are involved in an emergency situation.
The proposed standard covers:
Standardisation of interfaces, to promote data interoperability to get the same picture of the disaster.
Standardisation of information and information management exchange across borders and organisations.

Type of standard: Workshop Agreement

Urgency (when needed?): Moderate (< 2 yrs)

Overall impact: Considerable

Feasibility: High



Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks): Great
Improvement of the safety of society: Considerable
Improvement of responder safety: Moderate
Cost savings for end-user organisations: Limited

Industry & Research

Increase of business opportunities: Considerable
Improvement of business quality management: Moderate
Innovation progress: Great
Improvement of business functions: Moderate

Feasibility

Foundation: Moderate
Development perspectives: Sufficient
Implementation and follow-up perspectives: Insufficient
Anticipated drawbacks and constraints: Amply sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard: Considerably

Benefit of the standard to types of incident: Technological incidents
Specifically for the following incidents: CBRN attack
Wildfire
-

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard: Changes in crises, disasters and their impact due to e.g. increasing number of natural disasters due to climate change (forest fires, extreme rainfall, etc.), increasing number of physical attacks, increasing number of cyber incidents/attacks, increase of cascading

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard: -Command, Control and Communication technology
-Developments in disaster resilience and crisis management

ResiStand Assessment Framework 2.0 TNO

Figure 22: RAF 2.0 – Assessment tab filled for the CWA on collaborative emergency response (Stream 4)

Based on the assessment carried out, the CWA emerges as a matter of great importance, primarily owing to the pressing requirement for effective communication between relevant stakeholders and response authorities. Delving into specifics, the CWA presents a moderate yet comprehensive overall impact, with the potential to deliver benefits to end-users and the industry at large. This is paired with a substantial likelihood of the CWA's successful development and implementation. Furthermore, the potential ethical, social, and legal ramifications of the proposed standard are notable and warrant consideration.

3.4.1.6 Conclusion

In summary, the input collected from both the questionnaires distributed among FSX participants and the open discussion can be encapsulated as follows: the proposed common identification protocol and naming service could facilitate information sharing among the agencies that are involved in a situation. The proposed approach is filling gaps that currently exist and blocks effective information sharing among the agencies C2 in case of an emergency. Hence, the primary takeaway indicates that the CWA should be brought to its conclusion and formally submitted, accompanied by making necessary final adjustments. This entails fine-tuning the structure and the information that the protocol will host and the type of operations that will be supported, taking into consideration the collaboration practises that currently are used by other organisations (e.g. INSARAG). These considerations can be integrated and deliberated upon by the members of the CWA.

3.4.2 Final Evaluation of Stream 4, CWA 2

3.4.2.1 Basis of the Final Evaluation

The final evaluation was performed for

CWA “Management of forest fire incidents – SITAC-based symbology”

on the base of Stream 4 – Command and Control FSX, through realistic disaster scenarios involving First Responders and practitioners, and gathering user feedback, on June 27th – 31st 2023, at Gualdo Tadino (Italy) organized by CNVVF. During the FSX, both CWAs of Stream 4 were supported by the ENGAGE C2 system that is provided by Satways ltd, and optoacoustic material.

Table 16: List of organisations attending the CWA’s session on the 28th of June (as identified in questionnaires)

Organization Name	Type/Description
CNVVF (The National Fire and Rescue Service - Italy)	End user (Firefighter)
ENTENTE VALABRE (France)	End user (Firefighter)
Agenzia Forestale Regionale (Regional Forestry Agency - Italy)	End user (Forestry)
Preffetura provinciale di Perugia (Italy)	End user (Prefecture)

Several data were collected the days before and the same day of the FSX as a base for the final evaluation:

- Input from the live discussion between the participants of the exercise
- Notes collected by evaluators and CWA chairman
- Registration forms with information on the respondents
- Evaluation questionnaires were filled in by the participants at the end of the TTX.

To support the data collection a specific evaluation team was set up (Table 15). The data collection team has been composed of STWS personnel, being responsible for the CWA development.

Table 17: Data collection team

Full Name	Affiliation	Role / Required expertise
Giannis Chasiotis	STWS	Lead Evaluator
Leonidas Perlepes	STWS	Evaluator
Ilias Gkotsis	STWS	Evaluator – CWA chairman

3.4.2.2 Results from Expert Discussions and Testing during the FSX

During the previous days of the FSX, stakeholders were grouped per CWA topic and expertise. They were presented with the current version of the CWA and were asked to discuss the technical and operational details, as well as suggest possible improvements.

During the FSX exercise and specifically on 30th of March, STRATEGY partners and external experts engaged in the exercise as players. These stakeholders provided key feedback with regards to all the streams. Besides the players, who were engaged in the exercise scenarios, there were also people who attended the FSX as observers. These persons contributed to the FSX by providing comments in the context of an open discussion session and evaluation through a dedicated questionnaire.

In general, all participants were informed on the plan of the meeting, the objectives of the FSX and the STRATEGY project approach through a quick presentation. It was highlighted that two scenarios will support the exercise implementation:

- Scenario A: Earthquake causes the partial collapse of a dam element.
- Scenario B: Wildland-Urban-Interface fire triggers waste-disposal and treatment plants.

Within scenario B the use of a common symbology in collaborative response activities, was promoted. This was also showcased through a mini preparatory workshop on the 28th of March. The importance of a common understanding of all partners involved in response is well established, considering that disasters are agnostic to borders and an incident can affect different countries. During cross-border emergencies, joint efforts of involved services could be facilitated using common symbology. Participants were divided in three small groups corresponding to civil protection stakeholders of different type of agencies and of different origin involved in a multidisciplinary disaster. They were provided with an indicative scenario of a WUI fire, including several steps that tried to include several types of operational tasks and of different needs, offering them the possibility to depict them on the map using the proposed symbol set, take decisions and share information among the three teams (crisis communication).

After the demonstration, a dedicated session was designed as an open dialogue session for the participants to summarize their experiences/comments. Additionally, an online questionnaire has been shared to the participants to collect feedback for validating and evaluating the final results produced in the context of the CWA. After the FSX all responses have been analysed.

Based on the results and comments collected, mainly on the section of the CWA that the proposed symbolset was included and presented, the following changes or improvements (or generic comments) were provided:

- The symbols used is simple and effective. The colour code is well detailed. The actions in progress and those planned, represent all the actions expected during a forest fire intervention. The next significand and possibly complicated thing now is the training component. Teams must initially become the “owners” of this new tool, working in the end in the same way. This may be the most difficult point, the acceptability and application of the graphic charter, while respecting the colour codes and symbolism.
- Secondary flanks symbols should be elaborated and use different filling pattern to indicate the speed rather than the arrow size
- Sensitive point symbols, especially the generic one (e.g. gas-electric-nuclear) should be checked and described with more details

3.4.2.3 Questionnaire Results

The figure below graphically depicts the scoring analysis stemming from the questionnaire as answered by the participants during the FSX - targeting the evaluation of the CWA “Management of forest fire incidents – SITAC-based symbology”. The average score calculated per question was used for quantifying the impact of the CWA along the considered evaluation dimensions and was elaborated using the equation below. The considered scoring analysis approach has assumed the assigned scoring scale (i.e. Strongly Disagree, Disagree, Neither agree nor disagree etc.) as being in line with the Likert scale.

As evident from the following figure, almost all questions that were evaluated using the Likert Scale (Q1, Q2, Q3, Q4, Q5) were highly rated.

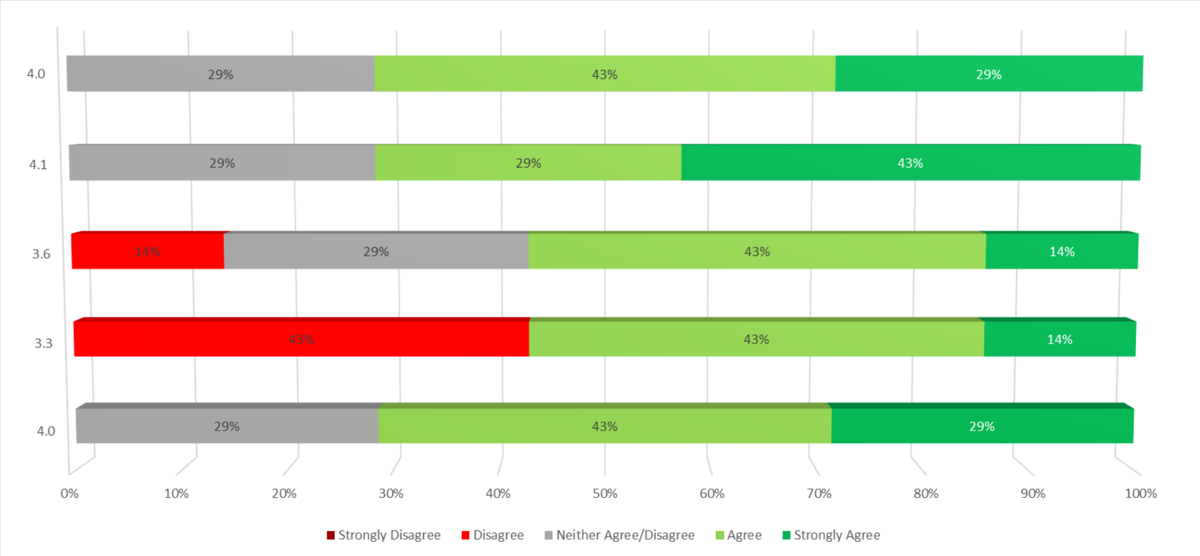


Figure 23 Answers for each question of the evaluation questionnaire and the KPIs values

Q1: The shape and size of the icons in the proposed symbols' set can properly reflect the identity and the meaning they represent and they can fit to both printed and digital use according to the considered scope of operations . (Average Score 4.0)

The shape and size are agreed to reflect the category, meaning and information of a specific symbol, even if it is a symbol that is combined with a container (explanatory text). Most (if not all) of the proposed symbols were easy to draw and read by the participants, affiliated with different entities and of different disciplines. This was totally encouraging both for the CWA working group, but also for them to easily consider of adopting and using it.

Q2: The colour pallet used for the visualization of the information through the set of symbols icons is considered adequate and representative for the purpose it is defined. (Average Score 3.3)

The proposed colours are adequate, which makes not only the symbols easy to distinguish but also understand. A specific type of organization though, among the responders, use a color for another purpose, so previous training should be considered when using the proposed symbol set and maybe national adoption of the CWA.

Q3: The meaning of the symbols is clear and easily understandable in both digital and printed format. (Average Score 3.6)

Most of the responders agreed that the proposed set of symbols can be easily understood and do not lead to misunderstandings, independent to the environment/medium used with (digital or paper maps). Nevertheless, a comment was related to check again the capability to draw all symbols also in black and white versions easily.

Q4: The operational requirements of wildfire response coordination are adequately reflected in the proposed categories and symbols. (Average Score 4.1)

The majority of participants agreed that the key needs and requirements with regards to wildfire management are depicted in the proposed categories and symbols, leading to enhanced decision making during the pressure and peculiarities of the incident.

<u>Q5: The requirements of fire response mapping as tested in the FSX can be adequately solved by the proposed CWA-SITAC. (Average Score 4.0)</u>

In the last question, the needs that follow the visualization of response activities onto a map, are efficiently addressed by the proposed CWA, as stated by the FSX participants. Moreover, the operational requirements are also considered to be addressed though using the proposed symbol set, at least those related to the incident commander and the personnel operating in the field.

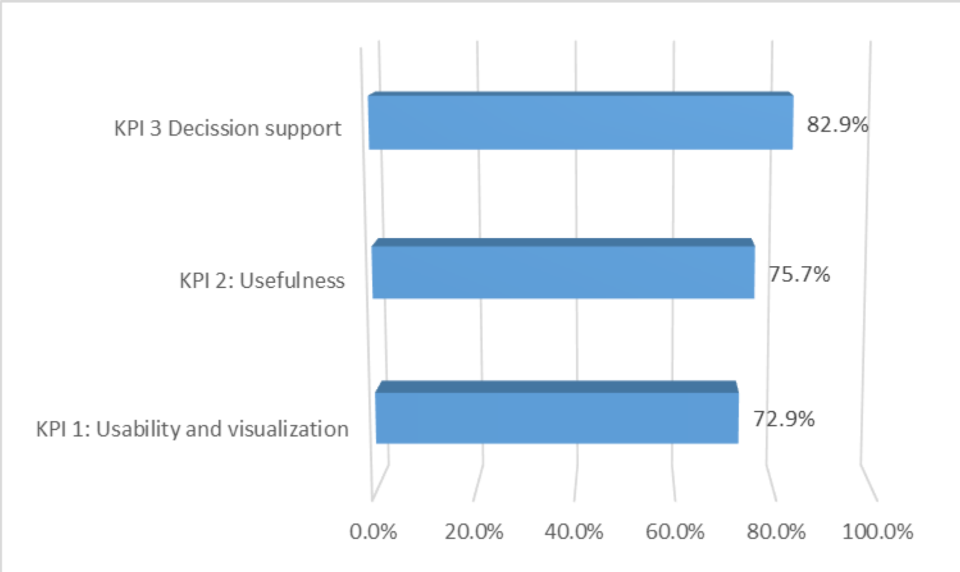


Figure 24 Criteria values as average of the questions part of the KPIs

The above questions were linked with specific KPIs, as depicted in the figure above, related to the usefulness of the proposed symbol-set, its usability and visualization appropriateness, and the support offered to the users in order to take decisions.

3.4.2.4 Results from Formal Aspects Evaluation

A questionnaire for the evaluation of formal aspects of the pre-standardisation documents and technical specifications provided for by the different streams, have been created and was filled by Speranta Stomff from ASRO (National Standardization Body) for this CWA. More details per category-criteria (each one contains several KPIs and each one is liked with one question) used to evaluate the CWA are depicted in the following figure. As presented in the figure 9 of 14 criteria have been completely achieved, while 3 out of 14 have achieved a score of 4 out of 5 and above in a Likert scale.

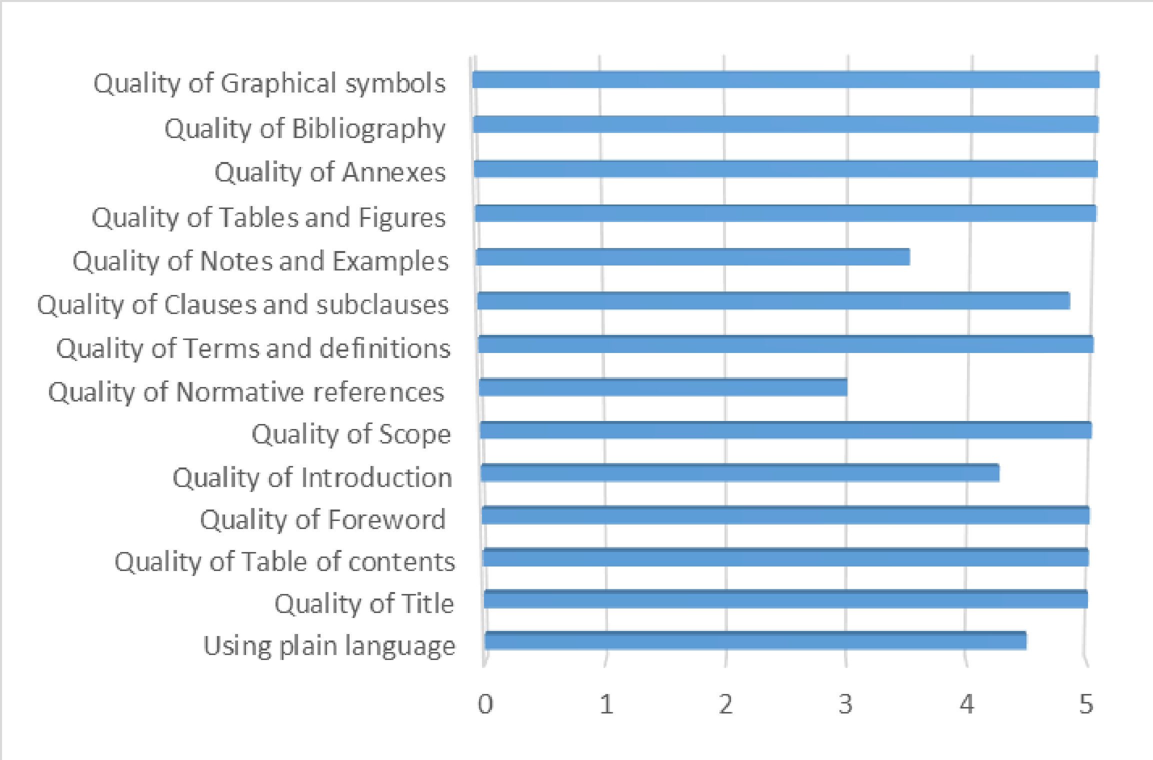


Figure 25 Criteria values on formal aspects fulfilled as average values of questions per criterion

Overall, based on the evaluation outcomes the total average score is 4.67 which indicates that the CWA discussed in this chapter is appropriate and has a comprehensible structure.

3.4.2.5 Results from RAF tool Analysis

Further to the above questionnaires, discussions and in general the evaluation means used for assessing the CWA, the ResiStand Assessment Framework (RAF) has been used. RAF has been developed within the ResiStand project to support organisations or individuals in assessing the impact of a possible standardisation project and the feasibility of developing and implementing it in the domain of disaster resilience and crisis management.

Thus, we used RAF to map the potential benefits of the CWA, to check whether it is compliant with essential ethical, legal, social issues, and to consider the organisational conditions under which the CWA will be developed and implemented, but also go ahead to become a standard.

Initially we have completed the first stage by filling the intake sheet in Figure 21, which provides, among other things, insight into the urgency aspect of having the proposed standard available, and some first textual information about the potential impact for various stakeholder categories.

Intake

ResiStand

Title of the proposed standard: Management of forest fire incidents – SITAC-based symbology

Identification number: CC-02

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
Satways	Industry
CNVVF	End-users
Hellenic Police	End-users
HMOD	End-users
CINAMIL	End-users

Type of standard: Workshop Agreement

Scope of the standard: The standardisation of the way the operational information can be depicted on paper based (and digital) maps by the response teams in the field and shared among involved agencies.

Example or illustration 1: In order to establish a standard way for displaying operational information on paper based maps or in CC systems, it is necessary to define a common and agreed set of rules for symbol development and use,

Example or illustration 2:

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Unknown	No (to the best of our knowledge) specific legislation related to developing symbols for management of forest fire incidents.
with national and regional legislation:	Unknown	No (to the best of our knowledge) specific legislation related to developing symbols for management of forest fire incidents.

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Firefighters, Civil Protection, Crisis Management Operation Centers, Police etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	Unknown	
Industry/SMEs:	Yes	Technology providers, developing incident management and decision support
Consultancy organisations:	Unknown	
Research institutes:	Yes	Research Centers when developing new solutions/methodological frameworks for
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	DG ECHO, DG HOME, EU Civil Protection Mechanism

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Establishing a standardized symbology that can be used by the forest fire management services and public safety agencies across the EU, enhancing collaboration among cross-border agencies, EUCPM modules
Industry:	Using a common symbol set to depict operational information, will enhance the Interoperability among different Command and Control Systems and their associated functions, leading to wider market
Research:	
Policy makers:	Strengthening civil protection and wildfire management capacities in the EU countries, strengthening cooperation among their agencies, and providing the ability to develop legislations/regulations based on a
Citizens:	Indirect benefit, as enhanced management and response to forest fire incidents will lead to environmental impact mitigation and decrease of casualties

Urgency of the standard: (when is it needed for implementation?) Very high (ASAP) — The standardisation of a common way the operational information is displayed to a map and C2 systems is crucial, since symbology is a powerful tool in human communication that surmount language barriers and provide a visual and quick

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	To test and validate the proposed standard
Governmental organisations:	Yes	To adopt the proposed standard
NGOs:	No	
Industry/SMEs:	Yes	To consider when developing technical systems/applications
Consultancy organisations:	No	
Research institutes:	Yes	To desing new ideas, projects and implement when testing/piloting projects in the
Others:	Yes	Standardisation organisations

Preferred leading type of stakeholder: First Responders — Fire Services, Civil Protection Agencies, Volunteers, Forest services etc.

Expected barriers and constraints: Adoption of the standard (common symbol set) and operational use may take time and needs training of personnel (especially in those countries that had a different approach)

Other information

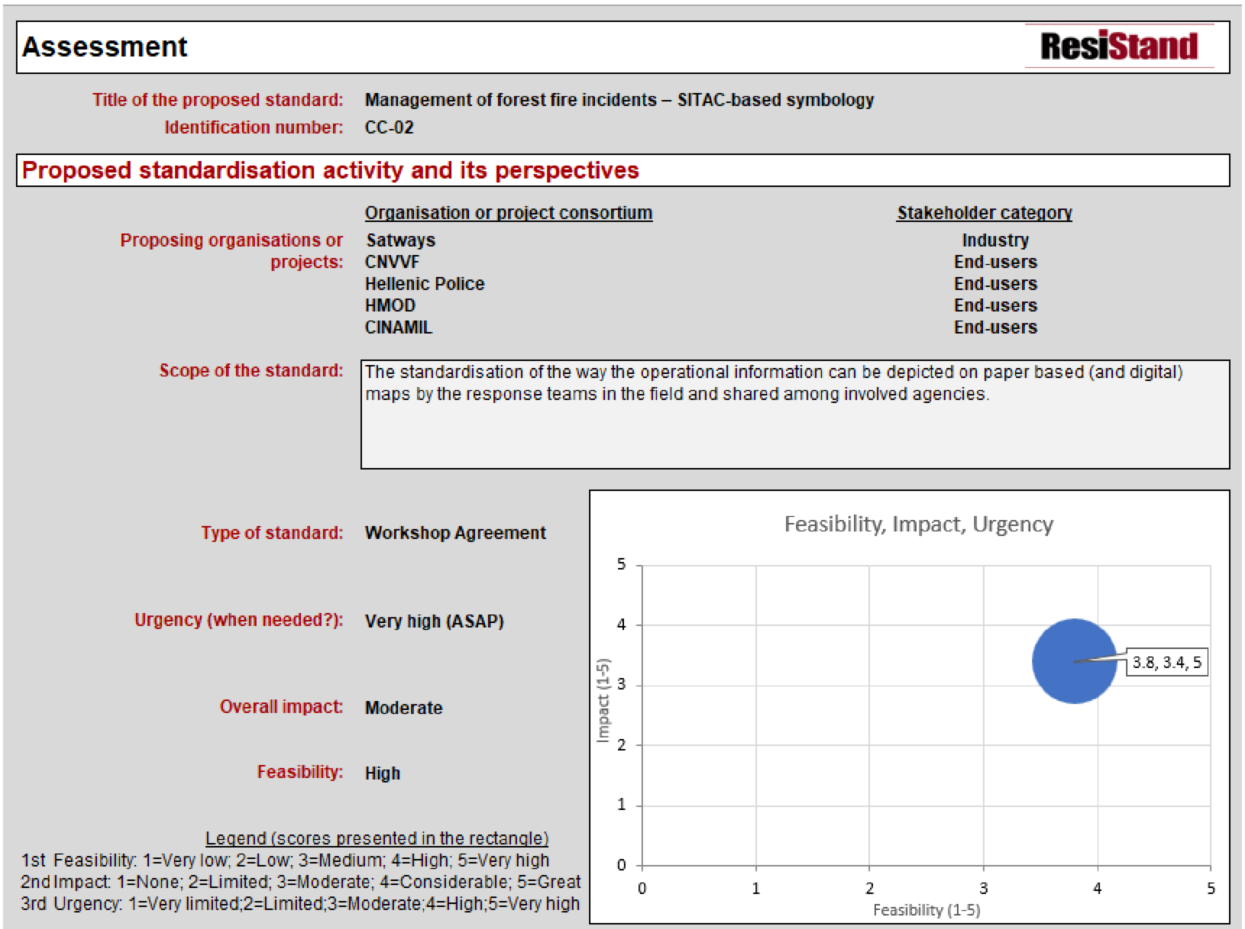
Additional remarks:

ResiStand Assessment Framework 2.0 — TNO

Figure 26: RAF intake sheet filled for “Management of forest fire incidents – SITAC-based symbology” CWA

In the next stage (stage 2) we completed the Impact tabs (Practitioners, Industry & Research) which provide detailed insight into the impact aspect of having the standard established.

In the final stage (stage 3) we replied to questions on relevant ethical, societal and legal issues in the concerned worksheet, and finally the Feasibility tab assessing the feasibility of successfully developing and implementing the proposed standard (Figure 22).



Assessment ResiStand

Title of the proposed standard: Management of forest fire incidents – SITAC-based symbology
Identification number: CC-02

Proposed standardisation activity and its perspectives

Proposing organisations or projects: Organisation or project consortium	Stakeholder category
Satways	Industry
CNVVF	End-users
Hellenic Police	End-users
HMOD	End-users
CINAMIL	End-users

Scope of the standard: The standardisation of the way the operational information can be depicted on paper based (and digital) maps by the response teams in the field and shared among involved agencies.

Type of standard: Workshop Agreement

Urgency (when needed?): Very high (ASAP)

Overall impact: Moderate

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Considerable
Improvement of responder safety:	Moderate
Cost savings for end-user organisations:	Limited

Industry & Research

Increase of business opportunities:	Moderate
Improvement of business quality management:	Considerable
Innovation progress:	Moderate
Improvement of business functions:	Moderate

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Insufficient
Anticipated drawbacks and constraints:	Amply sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Natural incidents
Specifically for the following incidents:	Wildfire Fire -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Changes in crises, disasters and their impact due to e.g. increasing number of natural disasters due to climate change (forest fires, extreme rainfall, etc.), increasing number of physical attacks (arsons), increasing number of incidents by
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Command, Control and Communication technology Protection of the public in general Developments in disaster resilience and crisis management

ResiStand Assessment Framework 2.0

Figure 27 RAF assessment sheet filled for "Management of forest fire incidents – SITAC-based symbology" CWA

Given the conducted assessment, the CWA seems to be of high priority due to the urgent need for communication among involved stakeholders and responding authorities to forest fire incidents, aiming to offer benefits to natural incidents, more specifically to wildfires/forest fires. In more details, the CWA offers a moderate overall impact (potential benefits for end-users and industry), with a high feasibility of a successful development and implementation of the CWA. Moreover, the potential ethical, social and/or legal effects of the proposed standard are considerable.

3.4.2.6 Conclusion

In conclusion, the feedback gathered both by the questionnaires distributed to the FSX participants and during the open discussion, expressed the view that the proposed wildfire management related symbol set is really useful and important for the stakeholders involved during such an incident. It was stated that the main purpose and priority is to be utilize the symbol set on paper maps in the field, in order to quickly depict the current situation and communicate it among stakeholders. The emphasis on paper maps is due to the circumstances that operating personnel face, making digital maps and respective equipment difficult to use. Thus, the main conclusion was that the CWA should be finalized and submitted, making some final modifications, fine tuning its categories and symbols in some case, always taking under consideration the similarities of currently used symbols at national level, standard operating procedures and best practices (that can be inserted and discussed by the CWA members).

The symbol set of the CWA titled "Management of Forest Fire Incidents – SITAC-based Symbology" is presently adopted by a considerable number of first responders, although not universally. However, some practitioners adhere to alternative protocols and recommendations. In order to achieve the objective of standardized symbols across Europe, embracing change becomes an imperative necessity.

3.5 Final Evaluation of Stream 5 Early Warning and Rapid Damage Assessment

3.5.1 Final Evaluation of Stream 5, CWA 1

3.5.1.1 Basis of the Final Evaluation

As reported in D4.3 [35], the result from the TTX concerning stream 5 were analysed, presented and discussed as part of T5.5 activities (see D5.5 [40]), contributing to improve some of the

> CWA 1 "Requirements and recommendations for social media early warning messages in crisis and disaster management"

clauses, and in making the CWA more respondent to the needs of first responders and practitioners. The text was simplified and made it clearer to read, and less vague in some points, for an improved applicability of the document.

The final evaluation was performed on the base of activities conducted during the FSX. Details about the event can be found in D4.1 "Full scale exercise demonstration report" [34] and especially focused on stream 5 in D3.6 [30]

The activities regarding stream 5 CWA 1 were: (i) a workshop conducted on the 29th of March 2023 with professional journalists and CNVVF personnel, dedicated to the application of the CWA to draft alerting messages for the FSX scenarios, and (ii) an evaluation during the FSX on the 30th of March 2023, where the prepared messages were published using the specific digital tool developed during the project ("Social Media Alerting Tool" – SMAT - for further details see D3.6 [30]) and discussed by the FSX participants as part of the operational scenario.

The affiliations of the participants and their role in the FSX event can be seen in the table below.

Table 18: List of organisations with attendants in the FSX related to Stream 5 CWA 1.

Organisation Name	Type/Description
CNVVF	End-user (Firefighter)
Journalist	Communication expert
Freelance journalist	Communication expert
Journalist/Journalism lecturer	Communication expert
RTO - Research and Technology Organisation; SME - Small and medium-sized enterprise	

Several data were collected previous to and during the FSX as a base for the final evaluation:

- Input from the live discussion between the participants of the exercise
- Expert session and other discussions
- Notes collected by evaluators
- Registration forms with information on the respondents

- Questionnaires filled in by the participants

The evaluation questionnaire presented in D4.3 [35] was used in the FSX for collecting feedback from workshop's participants. The questionnaire was reviewed after the TTX and adapted for the FSX, also considering the use of the developed web tool (see D3.6 [30]), with two questions added to collect feedback about the use of the web tool. The following table presents the Criteria, KPIs and how they were matched in the questionnaires of TTX and FSX.

Table 19: Mapping of criteria, KPIs and questions for TTX and FSX (in bold the new questions).

	Criteria	KPI		TTX Questions	FSX Questions
1	Usefulness	1	Future use of CWA	- Q1) "I will use the CWA in my operations in future" [Likert scale 1-5] - Q2) "I will implement the CWA in our processes in future" [Likert scale 1-5]	- Q1) "I will use the CWA in my operations in future" [Likert scale 1-5] - Q2) "I will implement the CWA in our processes in future" [Likert scale 1-5]
		4	Emergency Communication support	- Q7) "The proposed CWA support the communication during emergencies" [Likert scale 1-5] - Q8) "The proposed CWA cover all the important topics related to the design of messages for social media during emergencies" [Likert scale 1-5]	- Q7) "The proposed CWA support the communication during emergencies" [Likert scale 1-5] - Q8) "The proposed CWA cover all the important topics related to the design of messages for social media during emergencies" [Likert scale 1-5]
2	Usability	2	Facilitation	- Q3) "The proposed CWA facilitates the preparation of social media messages during emergencies" [Likert scale 1-5] - Q4) "The proposed CWA facilitates the preparation of texts to share in social media during emergencies" [Likert scale 1-5] - Q5) "The proposed CWA facilitates the preparation of graphic material to share in social media during emergencies" [Likert scale 1-5]	- Q3) "The proposed CWA facilitates the preparation of social media messages during emergencies" [Likert scale 1-5] - Q4) "The proposed CWA facilitates the preparation of texts to share in social media during emergencies" [Likert scale 1-5] - Q5) "The proposed CWA facilitates the preparation of graphic material to share in social media during emergencies" [Likert scale 1-5]
		3	Direct Usage	- Q6) "I could use the CWA tomorrow to write my own Social Media Early Warnings messages." [Likert scale 1-5]	- Q6) "I could use the CWA tomorrow to write my own Social Media Early Warnings messages." [Likert scale 1-5]

	Criteria	KPI		TTX Questions	FSX Questions
				- Q9) “The CWA is usable within the current framework of your institution” [Likert scale 1-5] - Q10) “What hinders you from using the CWA?” [free text]	- Q9) “The CWA is usable within the current framework of your institution” [Likert scale 1-5] - Q10) “What hinders you from using the CWA?” [free text] - Q11 “Using a semi-automated digital tool to support the preparation of social media messages would help in using the CWA [Likert scale 1-5] - Q12 “I thought the digital tool used during the FSX was easy to use” [Likert scale 1-5]
3	Impact	5	Perceived Potential of Social Media	- Q11) Social Media Messages have the potential to save lives in case of emergency [Likert scale 1-5] - Q12) Social Media Messages have the potential to harm the health of citizens in case of emergency [Likert scale 1-5]	- Q13) Social Media Messages have the potential to save lives in case of emergency [Likert scale 1-5] - Q14) Social Media Messages have the potential to harm the health of citizens in case of emergency [Likert scale 1-5]
		6	Impact of Social Media	- Q13) The impact of a Social Media message to a crisis or disaster depends on the timing it is sent [Likert Scale 1-5] - Q14) The impact of a Social Media message to a crisis or disaster depends on the formulation of a message [Likert Scale 1-5] - Q15) The impact of a Social Media message to a crisis or disaster depends on the design of the message [Likert Scale 1-5] - Q16) The proposed CWA will change the impact of Social Media messages to crisis and disasters [Likert Scale 1-5]	- Q15) The impact of a Social Media message to a crisis or disaster depends on the timing it is sent [Likert Scale 1-5] - Q16) The impact of a Social Media message to a crisis or disaster depends on the formulation of a message [Likert Scale 1-5] - Q17) The impact of a Social Media message to a crisis or disaster depends on the design of the message [Likert Scale 1-5] - Q18) The proposed CWA will change the impact of Social Media messages to crisis and disasters [Likert Scale 1-5]

	Criteria		KPI	TTX Questions	FSX Questions
				- Q17) How you think does the proposed CWA contribute to crisis and disaster management? [free text]	

To support the data collection a specific evaluation team was set up (Table 20).

Table 20: Stream 5 CWA 1 Evaluators team.

Full Name	Affiliation	Role / Required expertise
Andrea Capaccioli	DBL	Lead
Natascia Erriu	CNVVF	Member

3.5.1.2 Results from Expert Discussions and Testing during the FSX

The two activities regarding stream 5 CWA 1 introduced above had different scopes. The workshop was used to apply the CWA for drafting messages related to FSX scenarios, but with a different angle from the application that was done during the TTX, where participants were mostly practitioners, i.e. first responders with a more technical focus. The three journalists participating in the workshop analysed how the CWA can be applied from a communication expert point of view, with a focus on how easy it is to draft a clear and effective message using the recommendations and guidelines from the CWA.

The workshop resulted in a fruitful discussion, with the drafting of 14 messages (7 for Facebook and 7 for Twitter) related to the FSX scenario A. In total 4 persons participated to the workshop: 3 professional journalists and 1 member of CNVFF. During the discussion two main aspects emerged: the complexity of drafting a message is in combining a clear, simple language that every citizen can understand, with the complexity of providing all the information needed for an alert. This can be further amplified in the case of social media that have a fixed number of characters for the messages (like Twitter). The communication experts' view on this was essential to clarify how an expert approaches the writing of such messages, and how the recommended message structure, proposed in the CWA, helps in organizing the message and in focusing more on the right words to use. A second aspect emerged during the discussion was related to the impact of disinformation and the need to work on increasing awareness for the citizens about knowing the official channels for emergency information and recognize the fake ones or the not reliable.

This second point has been touched also in the discussion in the plenary session of the FSX, where participants from LEAs and from the prefecture recognised the need and also the opportunity provided by the CWA 1 in terms of supporting the use of social media that municipalities and local authorities are doing at the moment. The use of social media has been recognized as important for the emergency communication toward the citizens, but strong limitations are posed by the lack of skills and competencies especially in smaller local authorities, such as small municipalities, where it is difficult to have dedicated staff to the communication. A standardisation of the preparation of early warning messages could help in this situation.

3.5.1.3 Questionnaire Results

The evaluation questionnaire was administered at the end of the workshop to the participants.

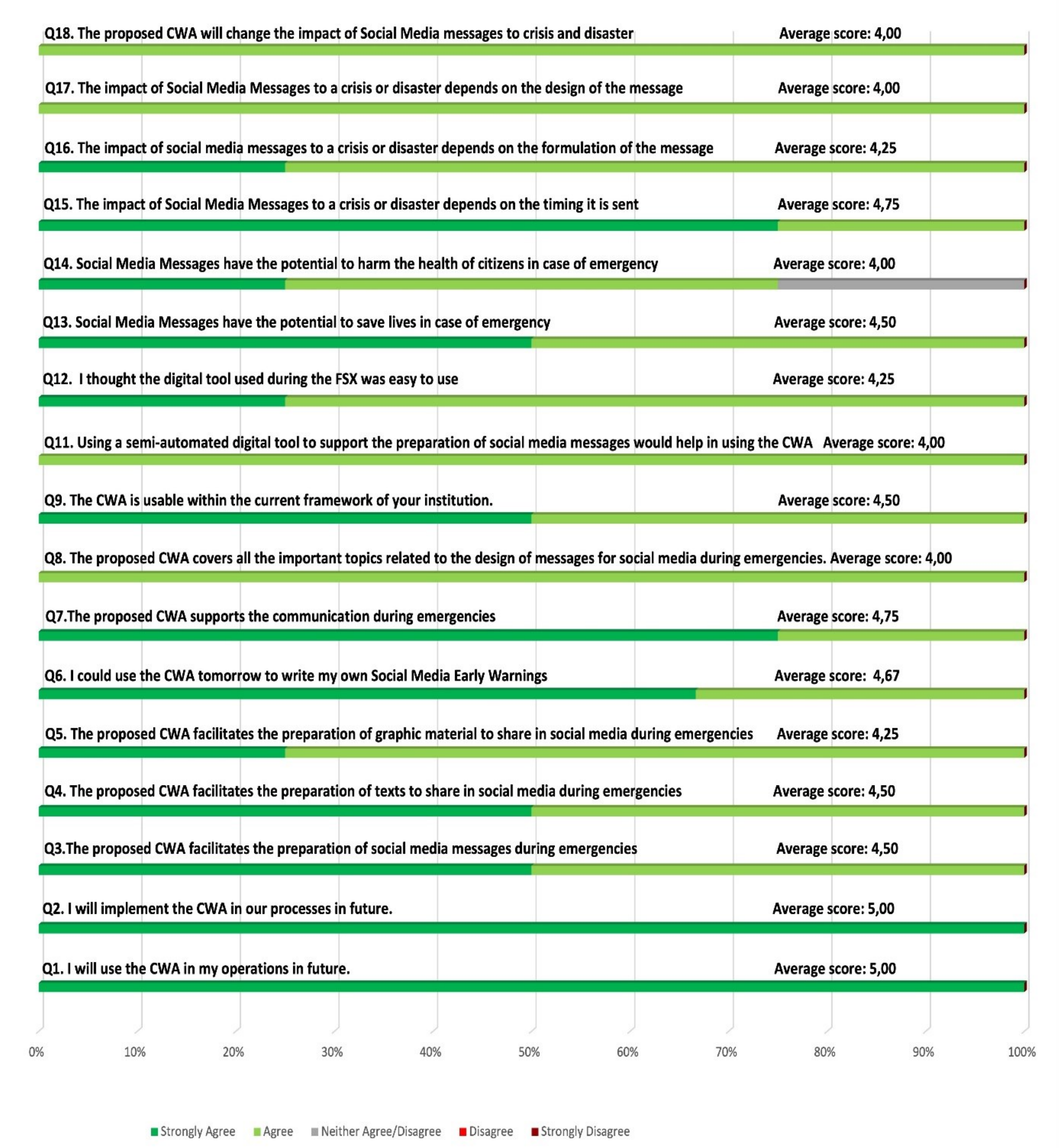


Figure 28 presents all the answers received from participants to the 17 Likert scale questions of the evaluation questionnaire.

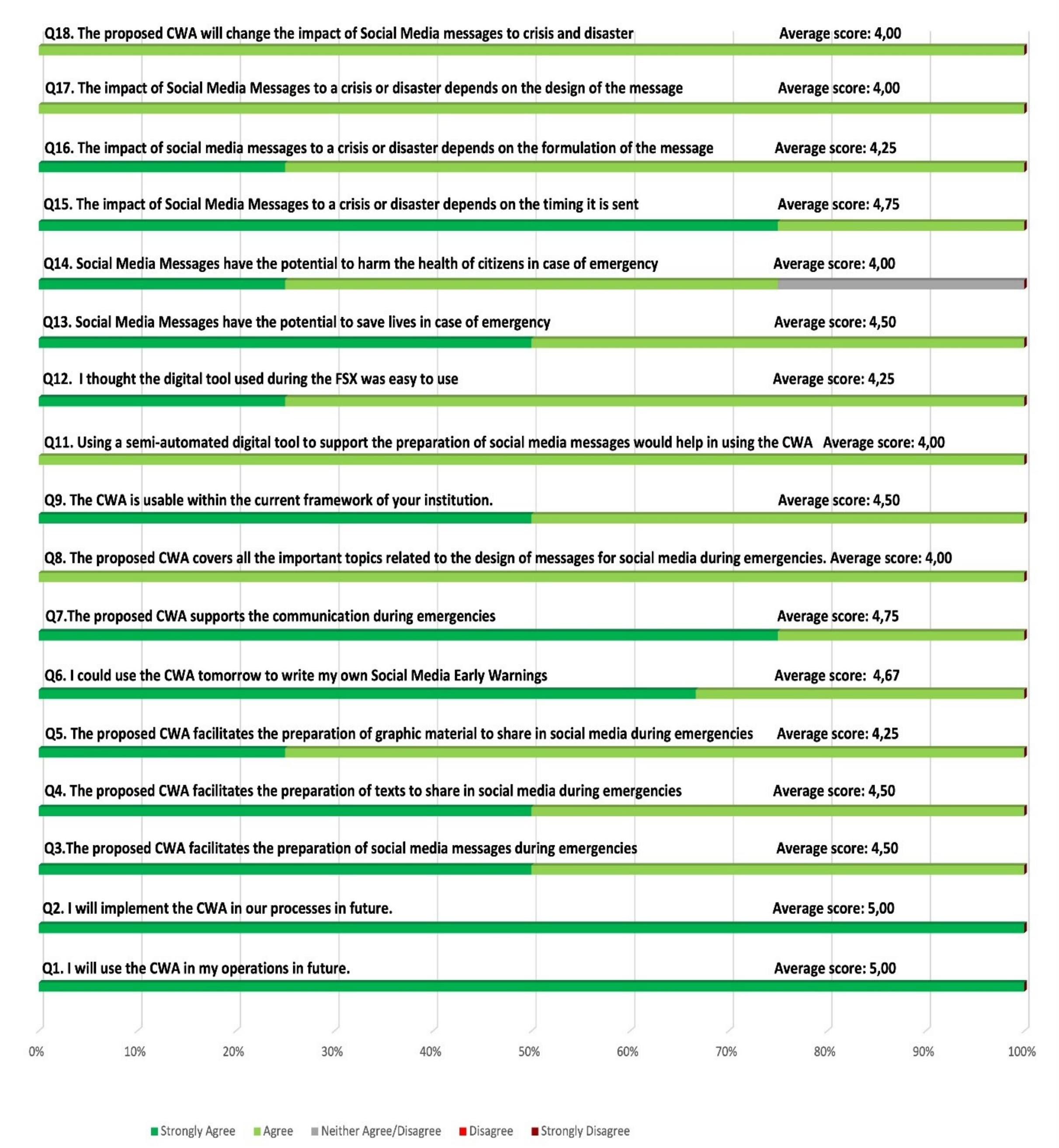


Figure 28: Scores' distribution obtained for all the statements in Stream 5, CWA 1 questionnaires.

It must be noted that the questionnaire was administered only to the 4 participants in the workshop. Also, the comparison with the intermediate evaluation is not properly possible since respondents are different: in the TTX they were first responders, practitioners and experts, whilst in the workshop they were journalists. But, as discussed in the previous section, the results from the workshop are interesting, since they provide and integrate with a different point of view the results of the TTX.

The different KPI results were calculated using the same methodology as for the intermediate evaluation described in D4.3. [35] Figure 29 shows the results for each KPI.

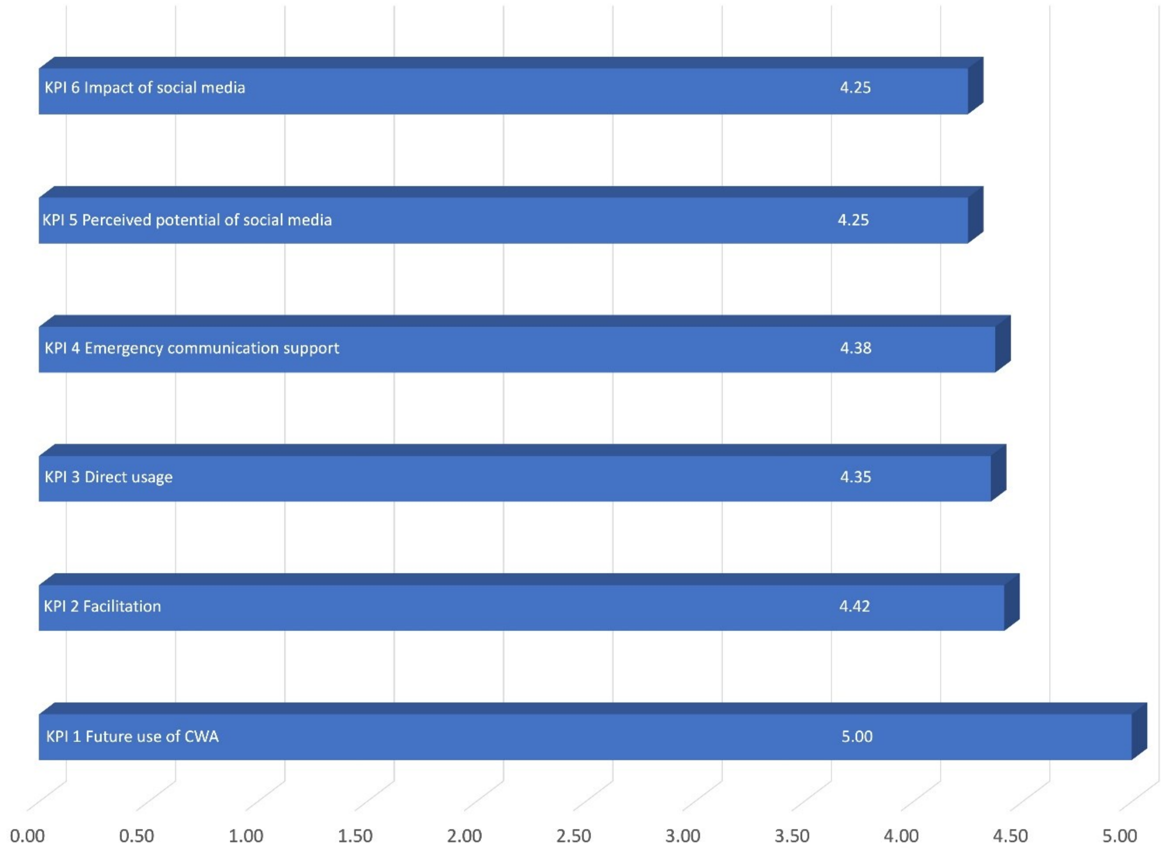


Figure 29: KPIs values as average of the questions part of the KPI.

In particular, looking at each KPI:

KPI 1: Future use of CWA

This KPI indicates the willingness of using the CWA 1 in future operations both individually and from an organisation point of view. The result (KPI1 = 5) cannot be considered relevant in this case, since the two questions (Q1 and Q2) were answered only by one respondent, the member of CNVVF (note: these questions were not relevant for the journalists).

KPI 2: Facilitation

This KPI indicates if the CWA 1 facilitates the work of end users in preparing social media early warning messages, and content of the messages in case of an emergency. As in the case of the intermediate evaluation, this KPI obtained the highest value (KPI2 = 4.42) showing a general consideration for this CWA 1 to be effective in facilitating the preparation of social media early warning messages.

KPI 3: Direct usage

This KPI explores how the proposed CWA 1 is directly usable from the end users in their organisations, considering the actual internal processes and organisational aspects. This KPI was integrated with two new questions related to the possible use of specific digital tool for the application of the CWA, and in particular also one question related to the use of the SMAT tool used during the FSX (see D3.6 [30]).

For KPI3 the result (KPI3 = 4.35) is high and in line with the overall positive evaluation made by the participants. In particular both the two new questions received a high evaluation (Q11 = 4 and Q12 = 4.25), participants considered the possible use of digital tool relevant for the direct application of the CWA 1 and positive the experience with the SMAT tool.

KPI 4: Emergency communication support

This KPI refers to the fact if the proposed CWA 1 is an added value to the communication during emergencies, covering all the necessary aspects. KPI4 result (KPI4 = 4.38) is very positive, showing that also from a communication point of view the CWA is effective in supporting emergency communication in general (Q7 = 4.75), but also in terms of topics covered by the guidelines (Q8 = 4.00).

KPI 5: Perceived potential of social media

This KPI explores the impact of social media messages in the emergency communication scenario, in order to understand the potential of designing effective and clear social media messages. KPI result (KPI5 = 4.25) highlights how participants consider the potential impact of social media. As also discussed in previous section, this result makes clear that as for other channels of communication social media must be approached considering the potential impact that they can have, and thus this CWA 1 can be a way to approach them in a more effective way.

KPI 6: Impact of social media

This KPI considers how the impact of a social media messages is affected by the designing of the message and by the proposed CWA 1. The result of the KPI (KPI6 = 4.25) is in line with all the other KPI results, showing how also from a communication perspective how a social media early warning messages is designed affect the impact of the message. Especially, timing was considered very relevant (Q15 = 4.75).

3.5.1.4 Results from Formal Aspects Evaluation

The CWA went through the formal aspects evaluation made by ASRO. The evaluation results were all positive (all questions answered with values of 4 or 5) and the CWA 1 was recognised having a good structure and a user-friendly language. Some minor comments were made about formatting issues due to the file export from the ISO OSD platform, which was used for drafting the document, and about referencing of standards mentioned in the text.

3.5.1.5 Results from RAF tool Analysis

The ResiStand Assessment Framework (RAF) was already used in WP2 to make an initial assessment of the standardization opportunities, for this CWA 1 see D2.5 [8]. A final assessment has been made after the completion of the CWA, and here the results are presented. The Intake sheet (Figure 30) shows the main scope and the potential impact of the CWA. The Assessment sheet (Figure 31) presents the final results of the assessment. It can be seen that the final assessment shows very good results in terms of feasibility (3.8 out of 5) and impact (3.5 out of 5), while for urgency (3 out of 5) the result is sufficient. Implementation and follow-up perspectives was considered amply sufficient since the CWA has been published, and the support of the project made it possible to develop, test it and collect multiple positive feedback as described in D4.3 [35], D4.4, D5.5 [40]. The results and the comments from CWA participants and from the STRATEGY evaluation were positive and thus follow-up perspectives of this CWA were considered amply sufficient.

In general, the assessment results are positive, and the opportunity provided by this CWA 1 in terms of improvement of the safety of society is considerable, as well as a considerable business opportunity.

Intake — ResiStand

Title of the proposed standard:	Requirements and recommendations for SM EW messages in crisis and DM
Identification number:	

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):	Organisation or project consortium	Stakeholder category
	Deep Blue	Research
		-
		-
		-
		-

Type of standard:	Workshop Agreement
Scope of the standard:	This document provides requirements and recommendations for the content and structure of social media early warning messages and notifications in crisis and disaster management towards the public. This document is applicable to all stakeholders in crisis and disaster management that seek to disseminate social media early warning messages and notifications to the public.This document does not cover how and when to
Example or illustration 1:	the design of a common format (visualisation and text) to frame and present alert messages by providing consistent and coherent information across different social media channels.
Example or illustration 2:	

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Yes	At the moment there are no rules on
with national and regional legislation:	Yes	

Target groups for applying the standard	Target group	Free space for additional comments
First Responders:	Yes	Firefighters, police, civil protection… they are the users of the standard
Governmental organisations:	Yes	Users of the standards: municipalities, regions, government
NGOs:	Yes	NGOs working on supporting first responders (e.g. civil protection association, online
Industry/SMEs:	Yes	Companies who want to develop tools for applying the proposed guidelines in an
Consultancy organisations:	Unknown	
Research institutes:	Unknown	
Standardisation bodies:	No	
Others:	No	

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Create and issue clear and unified messages among first responding authorities and understandable by the local population as well as vulnerable groups such as disabled, elderly people, or foreigners in order to
Industry:	IT companies could develop specific tool to automate and create messages applying the requirements and recommendations of the document
Research:	
Policy makers:	More informed citizens through a better communication
Citizens:	Clear messages and information, more awareness and reduction of risks

Urgency of the standard: (when is it needed for implementation?)	Moderate (< 2 yrs)	

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	main stakeholdrs and user of the standard: firefighters, civil protection, emergency
Governmental organisations:	Yes	political support for adoption and spread: ministries, regions, municipalities. They
NGOs:	Unknown	
Industry/SMEs:	Yes	Development of automated tools: software development companies
Consultancy organisations:	Unknown	
Research institutes:	Yes	development of standard
Others:	Yes	Citizens: citizens organisations and representatives would help in testing and provide

Preferred leading type of stakeholder:	First Responders	Main actor for the standard
Expected barriers and constraints:	Adoption could be slow also due to fragmentation and difference among countries on who is in charge for alerting and deciding which messages should be sent	

Other information

Additional remarks:	

ResiStand Assessment Framework 2.0 — TNO

Figure 30: RAF final assessment for CWA 1 - Intake sheet.

Assessment

ResiStand

Title of the proposed standard: Requirements and recommendations for SM EW messages in crisis and DM

Identification number:

Proposed standardisation activity and its perspectives

Proposing organisations or projects:	Organisation or project consortium	Stakeholder category
	Deep Blue	Research
		-
		-
		-
		-

Scope of the standard: This document provides requirements and recommendations for the content and structure of social media early warning messages and notifications in crisis and disaster management towards the public. This document is applicable to all stakeholders in crisis and disaster management that seek to disseminate social media early warning messages and notifications to the public. This document does not cover how and when to use social media.

Type of standard: Workshop Agreement

Urgency (when needed?): Moderate (< 2 yrs)

Overall impact: Considerable

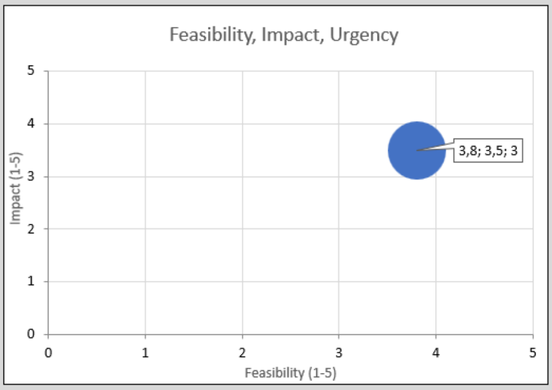
Feasibility: High

Legend (scores presented in the rectangle)

1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high

2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great

3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Considerable
Improvement of responder safety:	Considerable
Cost savings for end-user organisations:	None

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Considerable
Innovation progress:	Great
Improvement of business functions:	None

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Amply sufficient
Anticipated drawbacks and constraints:	Insufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	-
	-
	-

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Developments in disaster resilience and crisis management due to increasing use of social network for public and emergency communication, to alert citizens.
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Technical trends related to the development of tools for emergency communication and alerting, especially for social network integration within emergency communication and alerting - Non-technical trends related with the increase usage of

ResiStand Assessment Framework 2.0

TNO

Figure 31: RAF final assessment for CWA 1 - Assessment sheet.

3.5.1.6 Conclusion

To conclude, the results of this evaluation together with the previous intermediate evaluation results show how the work done on CWA 1 “Requirements and recommendations for social media early warning messages in crisis and disaster management” has been proven effective and the initial need (see D2.5 [8]) considered by the work of Stream 5 has been addressed. The CWA 1 which has been tested during the TTX first and the FSX afterwards, has been considered valuable and received good feedback, making a document that could be a first step toward a more systematic contribution to standardization of social media early warning messages. Existing standards for warnings have been considered and the CWA is conformed with existing standards for warning – on Social Media and general.

The thorough evaluation process with its two steps (first the TTX and then the FSX) at different CWA maturity levels, made it possible to approach the CWA 1 integrating different perspectives. For the TTX First Responders, practitioners and emergency management experts gave a strong input for the CWA improvement, testing its applicability and effectiveness. In the FSX workshop the CWA 1 was first evaluated by communication experts highlighting the relevance for their role and the importance of a clear, structured and inclusive communication approach for early warning social media messages. In a second activity, during the FSX field exercise the CWA 1 was tested in a simulated operative situation, showcased in a final demonstration to the actors in the front-line of the emergency management, both from the First Responders' side and from the institutional decision-making side, validating its application and usefulness in such a scenario. This integration together with the extensive work done in D5.5 [40] has proven to be effective for producing a CWA 1 receiving very positive comments and validation by different actors.

Also, as shown by the RAF assessment there are businesses opportunities that could be tackled by market players that could support the work of first responders and local authorities. With the publication of the CWA 1, it will open another opportunity, with the possibility for interested actors to further develop the CWA 1 into a proper standard.

3.5.2 Final Evaluation of Stream 5, CWA 2

3.5.2.1 Basis of the Final Evaluation

Also, for

> CWA 2 “Exchanging of building and infrastructure damage information with Common Alerting Protocol”,

the final evaluation was performed on the base of activities conducted during the FSX. As previously mentioned, details about the event regarding stream 5 can be found in D3.6 [30] “Standards selection, implementation, testing and interoperability assessment for Early Warning and Rapid Damage Assessment”.

The evaluation was conducted as part of the activities of the FSX event, in two phases: (i) a dedicated workshop on the 29th of March 2023 with the participation of a small group of experts (First Responders representatives and infrastructure operators) where the timely communication of damage information was discussed, while also the proposed communication protocol developed within the CWA was tested and evaluated, and (ii) during the FSX, where the CWA 2 was used to disseminate to the stakeholders the estimated damages and the status of a dam after the occurrence of an earthquake (according to the operational scenario).

The affiliations of the participants related to stream 5, CWA 2 and their role in the FSX event can be seen in the table below.

Table 21: List of organisations with attendants in the dedicated workshop on stream 5, CWA 2.

Organisation Name	Type/Description
KEMEA (Greece)	RTO
CNVVF (Italy)	End-user (Firefighter)
Valabre (France)	End-user (Firefighter)
Dam of Valfabbrica (Italy)	End-user (Infrastructure operator)
STWS (Greece)	SME
RTO - Research and Technology Organisation; SME - Small and medium-sized enterprise	

Several data were collected previous to and during the FSX as a base for the final evaluation:

- Input from the live discussion between the participants of the exercise
- Expert session and other discussions
- Notes collected by evaluators
- Registration forms with information on the respondents
- Questionnaires filled in by the participants

The evaluation questionnaire was developed using as a basis the one that was initially used for the evaluation of the TTX, following also the guidelines and template of the Participant Feedback Form (PFF), as provided by the Task Leader. More specifically, the questionnaire was adapted by modifying question #3 "*The current version of CAP protocol (v1.2) is enriched by introducing as parameters at least 2 types of buildings and infrastructures, which will subsequently be associated with the CAP message and disseminated along with their damage assessment information in near real-time.*" Since the elaborated version of the CWA is developed to be applicable by any type of building or engineering structure, question #3 was adapted as "*The current version of CAP protocol (v1.2) is enriched by introducing at least 4 parameters providing information on the affected buildings and infrastructures, which will subsequently be associated with the CAP message and disseminated along with their damage assessment information in near real-time.*". The new question is part of KPI #1 "Decision-making support".

To support the data collection a specific evaluation team was set up (Table 22).

Table 22: Stream 5, CWA 2 Evaluation team.

Full Name	Affiliation	Role / Required expertise
Ilias Gkotsis	STWS	Lead
Danai Kazantzidou	KEMEA	Member

3.5.2.2 Results from Experts Discussions and Testing during the FSX

The mentioned expert session on March 29th, 2023, brought together the CWA Leader, a group of experts, and participants from the CEN Workshop to discuss the proposed approach. In essence, the CWA specifies a protocol aimed at communicating the status of buildings or infrastructure in the form of an alert after a disaster. The results are communicated to the command-and-control centers of emergency management agencies. As part of the CWA 2, an extensive list of parameters deemed important for stakeholders was identified and included in the distributed message. These parameters are categorized into three groups: (i) Building and infrastructure descriptors, (ii) Damage descriptors, and (iii) Usability descriptors.

During the discussion, the experts proposed the inclusion of two new parameters in the communication protocol.

1. Dangerous Substances in a Building/Infrastructure

The first parameter falls under the category of Building and Infrastructure descriptors. According to the experts, it is crucial to provide information about the presence of dangerous substances in a building or infrastructure. Additionally, if known, the specific type of substance should also be transmitted. After a disaster, damaged buildings or infrastructure may release hazardous chemicals, asbestos, lead, and other harmful substances, posing significant risks to the health and safety of individuals, including emergency responders, workers, and nearby residents. As a result, the new parameter "dangerous substances in a building/infrastructure" has been added to the CWA 2.

2. Induced Risk

The second parameter pertains to the damage descriptors. It refers to the risk posed to a building or infrastructure that is not directly damaged by the disaster. The significance of such risks is largely dependent on external hazards that may endanger a building, such as a heavily inclined tree nearby. Even if the building itself remains undamaged, its use may become unsafe. Therefore, communicating this information to the operator responsible is crucial to ensure appropriate measures are taken to assess and address any potential risks.

By incorporating these two new parameters into the communication protocol, the CWA aims to enhance its effectiveness in post-disaster alert dissemination and enable emergency management agencies to make more informed decisions to mitigate risks and safeguard public safety.

This second point has been touched also in the discussion in the plenary session of the FSX, where participants from LEAs and from the prefecture recognised the need and also the opportunity provided by the CWA 1 in terms of supporting the use of social media that municipalities and local authorities are doing at the moment. The use of social media has been recognized as important for the emergency communication toward the citizens, but strong limitations are posed by the lack of skills and competencies especially in smaller local authorities, such as small municipalities, where it is difficult to have dedicated staff to the communication. A standardisation of the preparation of early warning messages could help in this situation.

During this FSX exercise, several stakeholders from different disciplines participated. Two scenarios were prepared, one was related to a dam collapse and the second was related to a fire incident. The alerting protocol proposed by the CWA was related to the first scenario. According to this scenario, an earthquake causes the partial collapse of a dike element. At the first stages of response, the CWA was used, a damage alarm and the related details were communicated to the command-and-control centre of the responsible authority, informing them for a potential damage on the dike, caused by the earthquake. During the scenario execution, the alarm messages that were related to the damages of the dam infrastructure were presented to the participants automatically using the alerting protocol that is proposed by the CWA. The details of the damages that were estimated to the dam were presented to the users. According to the received feedback by the participants, it is considered that the alerting protocol, in such cases, plays a crucial role in ensuring that accurate and timely information reaches the relevant stakeholders. This facilitates effective decision-making, response coordination, and resource allocation during an emergency situation. It helps prevent confusion and supports a coordinated effort to address the incident's impact and aftermath.

3.5.2.3 Questionnaire Results

The figure below graphically depicts the scoring analysis stemming from the questionnaire as answered by the participants during the 1st phase of the evaluation (as described in subchapter 3.5.2.1) targeting the evaluation of the CWA 2 “Exchanging of building and infrastructure damage information with Common Alerting Protocol”. The average score calculated per question was used for quantifying the impact of the CWA along the considered evaluation criteria (described in D4.3 [35] in Section 3.5.2 and was elaborated following the same approach as within the Intermediate Evaluation, using the equation presented in Chapter 2 of Deliverable D4.3 [35].

As evident from the following figure, all questions that were evaluated using the Likert Scale (Q1, Q4, Q6, Q7, Q8) were highly rated with scores ranging from 3.7 to 4.7 (out of 5).

Figure 32: 24 Scores' distribution obtained for all the statements in Stream 5, CWA 2 questionnaires (1st part).

Moreover, it should be highlighted that the evaluation dimensions were approached from a YES/NO perspective for some questions. The figure below summarizes the respective results.

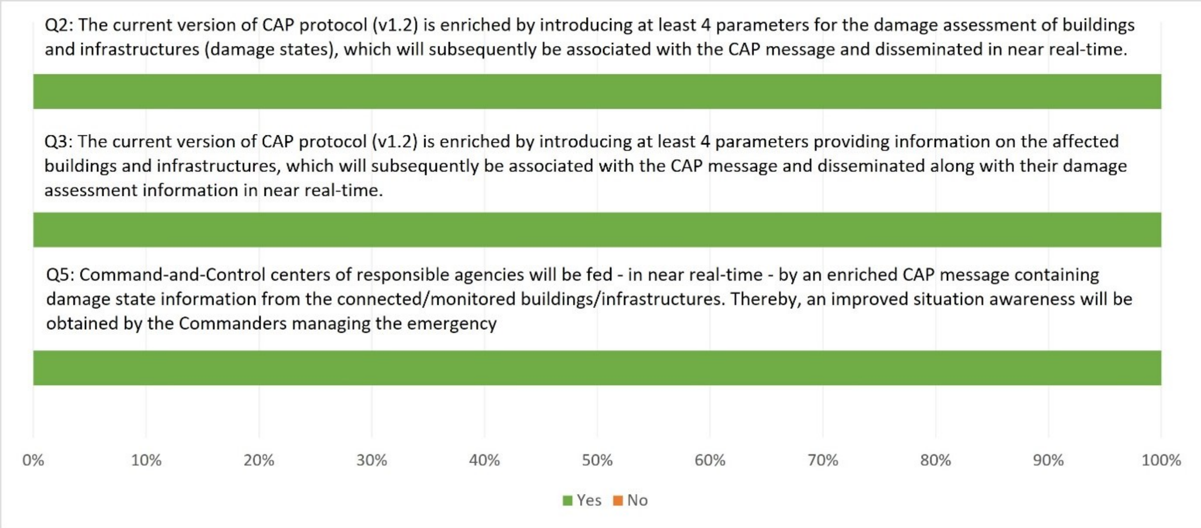


Figure 33: Answers for all the YES/NO statements in Stream 5, CWA 2 questionnaires (2nd part)

More specifically, respondents provided their answers to the dedicated questionnaires which are linked to the criteria and KPIs as follows:

Criterion 1: Decision-making support

The proposed communication protocol aims to provide decision-making support by enhancing the availability and dissemination of crucial information related to structural and operational/maintenance data of buildings and infrastructure assets and damage alerts after a hazard event. The KPI for this criterion evaluates the added value that the communication protocol brings to the decision-making process.

Based on the results of Q2, where 100% of the respondents agreed, it is evident that reliable and high-quality data about the condition of buildings and infrastructure assets and their health status following a disaster is crucial for effective response activities. This indicates the importance of having access to accurate information for making informed decisions. Furthermore, according to the responses to Q1, with an average score of 4.7 out of 5, the dissemination of damage alerts related to various types of hazards is highly valued. This underscores the significance of timely and accurate information about the extent of damage caused by different events.

Criterion 2: Completeness

The second criterion emphasizes the significance of conveying complete information regarding damages. This entails sharing various data points such as building and/or infrastructure details (location, type, floors, capacity, vulnerability, etc.), hazard characteristics (intensity, size, etc.), and damage assessment (level of damage, loss of integrity, etc.). Stakeholders have acknowledged the value of enhancing the CAP protocol with additional parameters pertaining to building or infrastructure detailed characteristics (Q3 with unanimous agreement of 100%) and damage stage (Q2 100% agree).

Criterion 3: Situational awareness

Based on the strong performance of Q5 (100% agree), stakeholders have acknowledged that implementing the proposed communication protocol, which includes comprehensive damage state information as outlined in the CWA 2, will significantly enhance the situational awareness of emergency commanders. This improved situational awareness brings numerous benefits to the management of emergencies, enabling the commanders to have a more holistic understanding of the current state of the emergency, including the extent and nature of the damages incurred. By having access to detailed information about the damage state, commanders can make more informed decisions regarding resource allocation, prioritization of response efforts, and coordination of personnel on the ground.

Criterion 4: Interoperability

Given the diversity of command-and-control (C2) centres and incident management systems (IMS), achieving appropriate and comprehensive alerting is challenging, the CWA also examines the interoperability in alerting. It is crucial for authorities to obtain timely, accurate, and informed information following a disaster, enabling them to warn individuals (who are in danger) through various communication channels. Both Questions 6 (related to the improvement of alerting in situations of potential damage) and 8 (regarding the information sharing among multiple first responders), with average scores of 4.0 and 4.7 (out of 5) respectively, indicate the enhancement of interoperability using the proposed CAP profile.

Criterion 5: Exchange of operational information

In summary, the proposed communication protocol demonstrates its ability to support the exchange of operational information by facilitating the availability and dissemination of reliable data on building and infrastructure assets, as well as providing timely damage alerts. The positive responses from the survey participants (Q7 average score 3.7 out of 5) further validate the importance and value of this protocol in supporting the exchange of operational information in the aftermath of disasters.

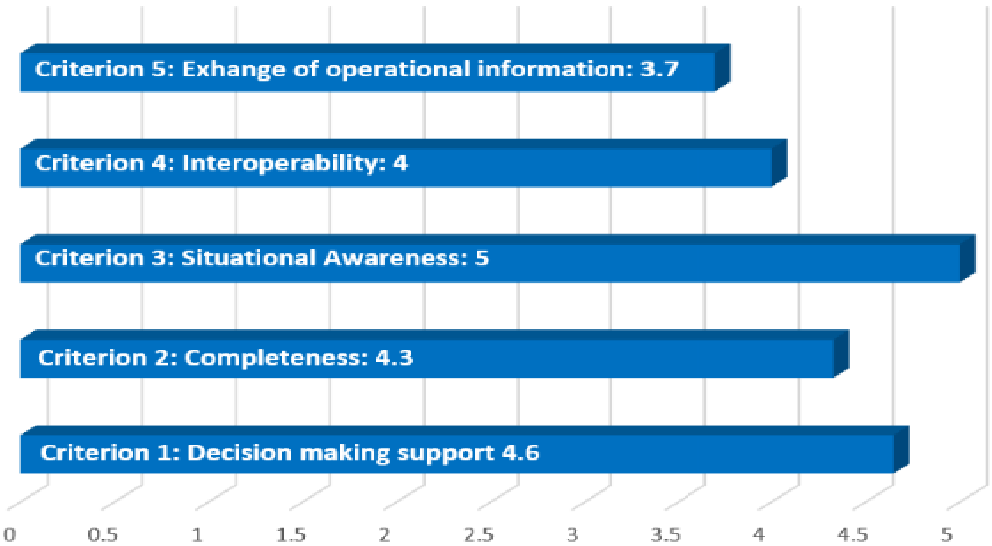


Figure 34: KPI values averages per criterion.

3.5.2.4 Results from Formal Aspects Evaluation

A formal aspects evaluation was conducted by Cristina Popa (ASRO). Positive results were received, and the CWA 2 was evaluated as well structured with clear scope and high quality of clauses and subclauses. Some comments related to the formatting of the Table of Contents and Annexed received and updated to the document following the evaluator's recommendations.

3.5.2.5 Results from RAF Tool Analysis

During the STRATEGY project the stream 5 CWA 2 standardization activities were assessed also with the help of the ResiStand Assessment Framework (RAF), see D2.5 [8]. For the final evaluation the RAF supported analysis was reconsidered and updated. The figures below illustrate the output from the Excel tabs labeled "INTAKE" (Figure 35) and "ASSESSMENT" (Figure 36). The "Intake" sheet serves as the initial data input, providing an understanding of the necessity and potential impact of the CWA 2 on different stakeholder groups. This data forms the basis for further analysis. The scope of the CWA 2, along with an illustrative example of its practical use and its target groups, are described and serve as the groundwork for the subsequent analysis. Furthermore, detailed information on the expected benefits for end-users, industry, and researchers is provided to emphasize the impact of the proposed CWA 2. To conduct a comprehensive analysis of the CWA 2's impact, specific questions related to potential benefits for end-users/practitioners and economic benefits/technological advancements for industry and/or research organizations are addressed on dedicated tabs within the Excel workbook. These questions help to map out the potential benefits and assess the impact in-depth.

Based on the assessment, the CWA 2 on "Exchanging building and infrastructure damage information with Common Alerting Protocol" is anticipated to have a significant positive impact on disaster resilience and crisis management. It is expected to enhance communication efficiency by enabling the timely transmission of damage-related information following a disaster. This will improve the situational awareness of the emergency management agency responsible. Additionally, the CWA 2 is expected to provide valuable support for search and rescue operations by allowing first responders to prioritize their activities based on early knowledge of the damage status. Consequently, it is believed that the implementation of this CWA will lead to a substantial reduction in the loss of life and injuries caused by disasters.

Intake

ResiStand

Title of the proposed standard: CWA-RDA

Identification number:

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
Satways	Industry
	-
	-
	-
	-

Type of standard: Workshop Agreement

Scope of the standard: The proposed CWA specifies a protocol that aims to cover the dissemination of the status of buildings, after the occurrence of a natural hazard or a man-made incident, in a form of an alert. The related results/alerts are communicated to the command-and-control centres of the agencies that manage the emergency.

Example or illustration 1: A critical infrastructure facility (i.e. a hospital) had deployed state-of-the-art sensors to estimate the damage induced by the seismic event and assess the extent of potential structural damage.The deployed sensors

Example or illustration 2:

Compliance with legislation

	Compliant (Y/N)?	Explanation
with European legislation:	Yes	
with national and regional legislation:	Unknown	

Target groups for applying the standard

	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Firefighters, Police, Rescue teams,
Governmental organisations:	Yes	Civil protection agencies, ministries, municipalities
NGOs:	Unknown	
Industry/SMEs:	Yes	Companies building sensors and software for C2
Consultancy organisations:	Unknown	
Research institutes:	Yes	Research centers could use data for research, and/or provide data for improving sensors
Standardisation bodies:	Yes	OASIS Committee
Others:	Unknown	

Potential impact and urgency

Benefits for stakeholder categories

	Description of who will benefit and in what way
End-users:	Establishing faster response system to prioritize intervention to buildings after a disaster, by having a first automated rapid damage assessment estimation shortly after the communication
Industry:	Companies could be able to build sensors and systems interoperable and that could easily integrated
Research:	Researchers could benefit from the data and use the standard to collect data
Policy makers:	
Citizens:	

Urgency of the standard: (when is it needed for implementation?) Moderate (< 2 yrs)

Development

Required types of stakeholders

	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	As main user, they would provide requirements and needs: firefighters, civil rprotection
Governmental organisations:	Yes	Also, as possible users and relevant actors, they proivide requirements
NGOs:	Unknown	
Industry/SMEs:	Yes	As producers of sensors and software developers responsible also for C2 systems
Consultancy organisations:	Unknown	
Research institutes:	Yes	Research centers focusing on studying (Rapid) Damage Assessment, remote sensing,
Others:	Yes	OASIS Committee

Preferred leading type of stakeholder: Industry/SME — As a more technology driver stantard, it could help in having a leading from industry

Expected barriers and constraints:

Other information

Additional remarks:

ResiStand Assessment Framework 2.0 TNO

Acknowledgements | INTAKE | IMPACT END-USERS | IMPACT INDUSTRY&RESEARCH | ETHICAL-SOCIETAL-LEGAL | FEASIBILITY

Figure 35: RAF 2.0 – Intake tab filled for the CWA 2 on "Exchanging of building and infrastructure damage information with Common Alerting Protocol".

Assessment

ResiStand

Title of the proposed standard: CWA-RDA
Identification number:

Proposed standardisation activity and its perspectives

Proposing organisations or projects:	Organisation or project consortium	Stakeholder category
	Satways	Industry
		-
		-
		-
		-

Scope of the standard: The proposed CWA specifies a protocol that aims to cover the dissemination of the status of buildings, after the occurrence of a natural hazard or a man-made incident, in a form of an alert. The related results/alerts are communicated to the command-and-control centres of the agencies that manage the emergency.

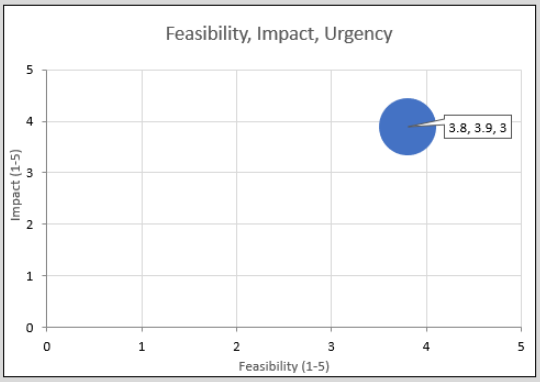
Type of standard: Workshop Agreement

Urgency (when needed?): Moderate (< 2 yrs)

Overall impact: Considerable

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Considerable
Improvement of responder safety:	Moderate
Cost savings for end-user organisations:	Moderate

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Considerable
Innovation progress:	Great
Improvement of business functions:	Moderate

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Insufficient
Anticipated drawbacks and constraints:	Amply sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	Earthquake Flash flood -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Changes in crises, disasters and their impact due to e.g. increasing number of natural disasters due to climate change (forest fires, extreme rainfall, etc.), increasing number of physical attacks, increasing number of cyber incidents/attacks, increase of
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	- Sensing technology - Command, Control and Communication technology - Protection of the public in general

ResiStand Assessment Framework 2.0 TNO

Figure 36: RAF 2.0 – Assessment tab filled for the CWA 2 on “Exchanging of building and infrastructure damage information with Common Alerting Protocol”.

3.5.2.6 Conclusion

Based on the feedback obtained through questionnaires and discussions, it is evident that the participants of the FSX have provided positive evaluations of the KPIs. The evaluation, along with the comments and remarks received during the exercise, served as valuable input for enhancing the content of the CWA 2 under discussion. The working group approved the proposed new parameters, recognizing their significance in improving the communication protocol developed within the CWA 2.

As a result, the document has been enriched and updated accordingly. In general, the proposed structure of the CWA 2 received positive feedback for its effectiveness in facilitating coordination and collaboration among relevant authorities during emergencies. It was also commended for enhancing the decision-making process and enabling a more efficient and timely response following a disaster.

3.6 Final Evaluation of Stream 6 CBRNE

3.6.1 Basis of the Final Evaluation

The Subject of the final evaluation in Stream 6 – CBRN-E were the concepts presented in the two parts of the

> Technical Specifications (TS) "Societal and citizen security — Digital Chain of Custody for CBRN-E Evidence":
>
> Part 1: Overview and Concepts
>
> Part 2: Data Management and Audit

Many suggestions made previously during the TTX in Lisbon and during other discussions with experts were taken into account, such as DCM (digital custody metadata), which is one of the main outcomes. The following points from the TTX in Lisbon (see STRATEGY deliverable D4.3 [35], Chapter 4.6.1.2) were not be implemented due to different reasons:

Suggestion: use existing standards, namely for interoperability, symbols used (e.g., symbols in figures), and data/time definition (e.g., date format according to ISO 8601).

Authors comment: Not applicable: ISO 8601 – this standard is covering communication of date and time-related data. It is out of the scope of the CBRNE dCoC.

Suggestion: Compare NATO forms (handbook for sampling and identification for Chemical, Biological or Radiologic agents AEP-66) for CoC with TS2, page 11, draft of GUI

Authors comment: Not applicable or partially covered: The dCoC is about the chain of custodianship not about how to collect or package CBRNE samples

Suggestion: Add samples into the figure at page 19

Authors comment: Not applicable: The TS are under the CEN/TC Editorial team and no changes to the TS content are accepted/possible, aside from this a TS should avoid examples - when they are provided they should be included into an Annex

Suggestion: The pictures and names of the owner and receiver (i.e., the intervenients in the CTP procedures) should be removed and replaced by a Personnel ID number for data protection purposes.

Authors comment: This makes no sense and interferes with the requests by key stakeholders.

Suggestion: CTP (Custody Transfer Point) might be replaced by CT (Custody Transfer), since "point" indicates a location, where the transfer takes place, and "transfer" is the correct wording for an action as a part of a process.

Authors comment: The suggested renaming was not adapted because CTP refers to a point in the dendrogram and therefore the word "point" is in order.

The final evaluation concerning stream 6 – CBRNE was performed mainly on the base of events during the FSX. Details about this can be found in STRATEGY deliverable D4.1 [34] "Full scale exercise demonstration report" and especially for the activities regarding stream 6 in D3.7 [31] "Standards selection, implementation, testing and interoperability assessment for CBRN-E". Therefore, here just a short summary on the stream 6 activities:

On the 3rd day of the FSX week an expert session (15 attendants) was dedicated to the stream 6 standardisation activities. The scope of the two TS was presented to ensure a shared understanding, and to pave the way for expert discussions. Furthermore, a special technical solution (demonstrator) was presented, where a transfer of a CBRNE evidence item was simulated with the help of a GUI. After some general discussions a role play exercise was introduced where different experts were assigned to special stakeholders in the chain of custody, so that there were several teams:

1. Mission commander team
2. Reconnaissance team
3. Sampling team
4. Carrier team
5. Laboratory team/Storage team

The stakeholder teams were asked to discuss internally their tasks, needs, and the relation of these to the content of the TSs. Then they should introduce their thoughts, doubts, ideas etc. to all participants and the whole process was simulated in a kind of role play and lively discussed.

During the field exercise in the next day a live demonstration was performed to test and validate Stream 6's Technical Specifications. Thus, a specific incident was created, which was conceptualized as a sub-scenario of Scenario A – "Earthquake causes the partial collapse of a dike element".

The CBRNE Incident storyline was as follows: The collapse of the dike generated a flood, where some industrial drums were transferred to the area of Valfabbrica stadium. The drums were labelled with chemical danger pictograms, its content being unknow. Following the earthquake, a team of CBRNE experts from the Portuguese Army (represented by UMLDBQ), which was in Umbria to attend a training event, expressed their availability to support any operations that might be necessary. After the required authorizations, the Portuguese team was mandated to support local authorities (represented by CNNVF) in the collection of samples in the Valfabbrica stadium hazmat incident. In this context the concept of dCoC as described in the TS (part 1 and part 2) was tested in comparison with the hitherto procedure for sample collecting and transport.

During the FSX, participants had access to the digital version (i.e. PDF file) of the FSX playbook. The digital versions (i.e. PDF files) of the TS were shared with STRATEGY partners in the project repository. The affiliations of the participants and their role in the FSX event can be seen in the table below.

Table 23: List of organisations with attendants in the FSX related to Stream 6

Organisation Name	Type/Description	Role in the FSX
ATOS (Spain)	Industry/Technology Provider	Observer
Hellenic Police (Greece)	HEADQUARTER OF THE HELLENIC POLICE LEA Law enforcement and Civil Protection	Observers
HMOD- Military/Hellenic Ministry of Defence (Greece)	Military CBRNe Military company	Observers

Organisation Name	Type/Description	Role in the FSX
-Trilateral Researsh (United Kingdom)	Academia & Research	Observers
SINTEF-SINTEF AS (Norway) Academia & Research	Academia & Research	Observer
CINAMIL Centro de Investigação, Desenvolvimento e Inovação da Academia Militar (Portuguese)	Military Portuguese Acedemia & Researchj	Observer/Players
PdS- Polizia di stato (Italy)	LEA	Observers
FhG-INT Fraunhofer INT (Germany)	RTO	Observers
CNVVF Corpo Nazionale Dei Vigili del Fuoco (Italy)	FR	Observers
INOV-INOV Inesc Inovação (Portuguese)	RTO	Observers
LEA - Law Enforcement Agency; NSB - National Standardization Body; RTO - Research and Technology Organisation; SME - Small and medium-sized enterprise; FR - First Responders		

Several data were collected previous to and during the FSX as a base for the final evaluation:

- Input from the live discussion between the participants of the exercise
- Small group discussions (expert session and role play)
- Notes collected by evaluators
- Registration forms with information on the respondents
- Video and audio recordings of the exercises
- Questionnaires, filled in by the participants remotely after the event.

To support the data collection and evaluate the CBRNE FSX results, a specific evaluation team was set up, see Table 24.

Table 24: Stream 6 – CBRNE - FSX Evaluation team

Full Name	Affiliation	Role / Required expertise
Tiago Rocha da Silva	INOV	Lead Evaluator
Sebastian Chmel	FhG-INT	Evaluator
Shenae Lee	SINTEF	Evaluator

Full Name	Affiliation	Role / Required expertise
Luis Carvalho	CINAMIL	Evaluator

3.6.2 Results from Experts Discussions and Testing during the FSX

During the expert session described above a vivid discussion showed a high interest of the participants in the problems addressed by the TSs. Especially the role play allowed a deeper understanding and investigation of the procedures and the needs to be considered in a dCoC. Special situations like a broken seal or a splitting of a sample by a laboratory decision and therefore a splitting of paths in the CTP dendrogram were re-enacted and scrutinized. The discussion focused on the advantages of implementing the digital CBRNE and the role and competencies of the Mission Commander. The information security aspects of the digital CBRNE tools were also discussed. In comparison with the results of the discussions at the TTX in Lisbon now the experts didn't have many new improvement suggestions. It seems that the TSs were consolidated well enough so far.

The main suggestions to improve the TS were:

1. Suggestion: Replace the "Accept"-button with a button that states clearly, what is accepted (namely custodianship, not just validation of the data or reading)

Authors comment: This suggestion refers to the demonstrator, the specific technical solution, which was shown and used at the FSX, and could be easily implemented, but it is not subject of the TS.

2. Suggestion: At CTP, personal data (i.e. photo and name) must not presented.

Authors comment: This is not applicable in dCoC CBRNE, in emergency context the GPDR is not applicable (since they are involved as workers) and only authorized persons have access to information.

3. Suggestion: Allow access to the dCoC application without requiring special equipment

Authors comment: This is ensured via the possibility to use a web-page tool, as done for the demonstrator. Again, it should be noted that the TSs do not indicate which tools should be used, this is the responsibility of the implementer.

Some more remarks were given in the free text part of the Questionnaires, which were filled in after the FSX remotely (see next subchapter).

The CBRNE incident in the framework of the FSX revealed also positive insights: To address the demonstration requirements, UMLDBQ and INOV decided that it was important to perform the chain of custody (CoC) procedures in duplicate, in order to compare the actual procedures, i.e., manual version with paper-based forms, with the solution proposed in the TS, i.e., digitalized version of the CoC using a tablet. Thus, during the demonstration, UMLDBQ performed with two compilers (and at same time). The compiler is the team member responsible to collect and manage the information regarding the samples, see Figure 37.

Figure 37: Sampling team operating with two compilers – manual procedures (paper-based forms) on the left side and digitalised procedures (using a tablet) on the right side.

After the demonstration, the main outcomes identified by UMLDBQ's team was:

- dCoC improves the information management, as it simplifies the process - there is no need to manually copy the documents, and eliminates the errors that might be originated due to copying;
- with a dCoC communication is faster and easier, as the information on the custody receiver identification was directly available through the software, whereas in the current procedures such information is required by the compiler to its team commander through VHF radio; and
- dCoC reduces the time needed to complete the mission, as the 'dCoC compiler' delivered the samples to CNVVF mobile laboratory team faster than the 'manual compiler'.

3.6.3 Questionnaire Results

After the TTX in Lisbon the Technical Specifications were developed further and, therefore, two additional KPIs, KPI 9 and KPI 10, were identified by the Stream 6 KPI Group. They refer to the alert concept, which was developed after the TTX in Lisbon. KPI 9 and 10 and the related questions in the questionnaire were also described in STRATEGY deliverable D4.3 [35], Chapter 3.6, since they were already developed at the time when D4.3 [35]was written. For more details on the criteria, KPIs and Questionnaires please see D4.3 [35], Chapter 3.6.

Figure 38 shows the scores' distribution of the questions included in the questionnaire. . An Average Score per Question was obtained using Equation (1) in D4.3 [35], Chapter 2.2.3.

It should be noted that the respondents have by and large positively rated the TS; all questions received a score of > 3.75 on the Likert scale. The highest scores (≥ 4.25) were received for questions Q1, Q2, Q4, Q5, Q7, Q8 and Q10, the lowest scores (< 4.00) were received for questions Q6 and median scores (between highest and lowest score) were received for Q3 and Q9. The answers "strongly agree" plus "agree" represent 100% on Q2, Q3, Q4, Q5, Q8 and Q10 more than 87.5% in all of the questions, including in the only question that has received one "Disagree" classification.

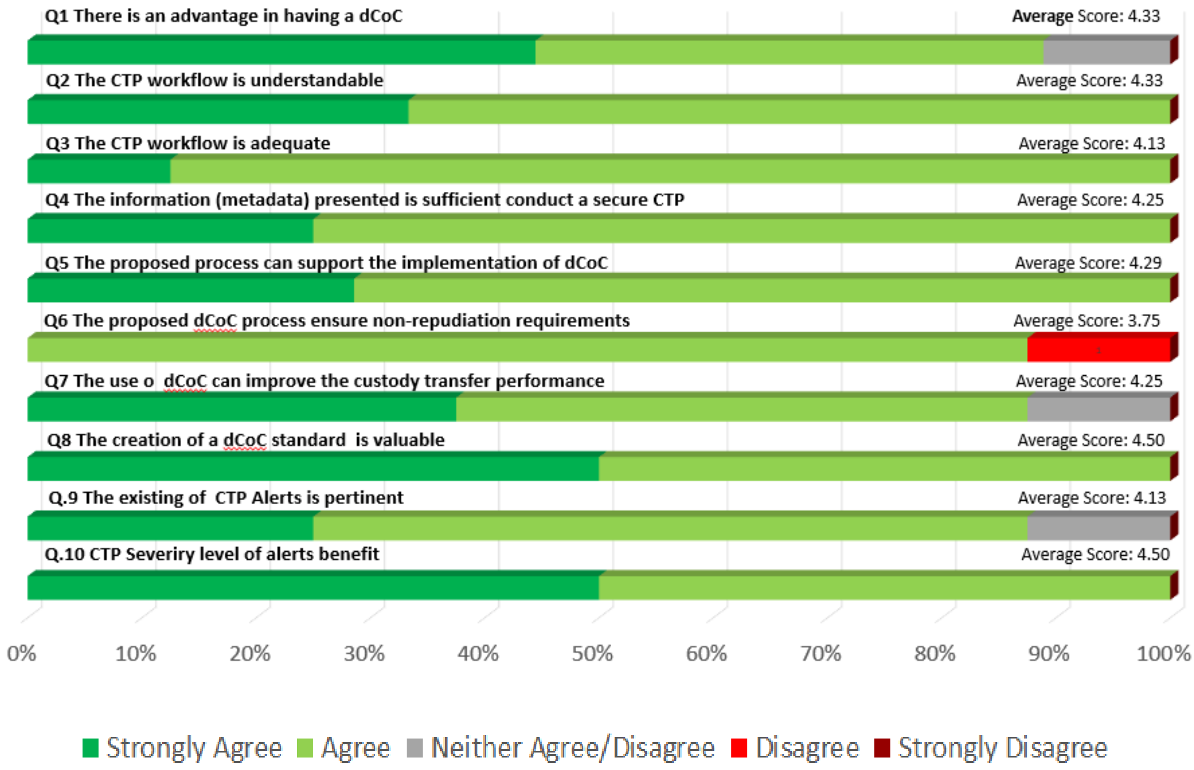


Figure 38: Scores' distribution obtained for all the statements in Stream 6 – CBRNE questionnaires.

Figure 39 provides the KPI values, calculated as described in D4.3 [35], chapter 2. Note that "No answer"- responses have not been considered. If respondents answered the questions as "neither agree nor disagree" and add the comment "not qualified", the answers were reclassified as "unanswered" for calculation of question score and KPI values. The values presented in this table take into consideration the reclassification of these answers. The KPI values were obtained using Equation (2) as described in deliverable D4.3 [35], Chapter 2.2.3.

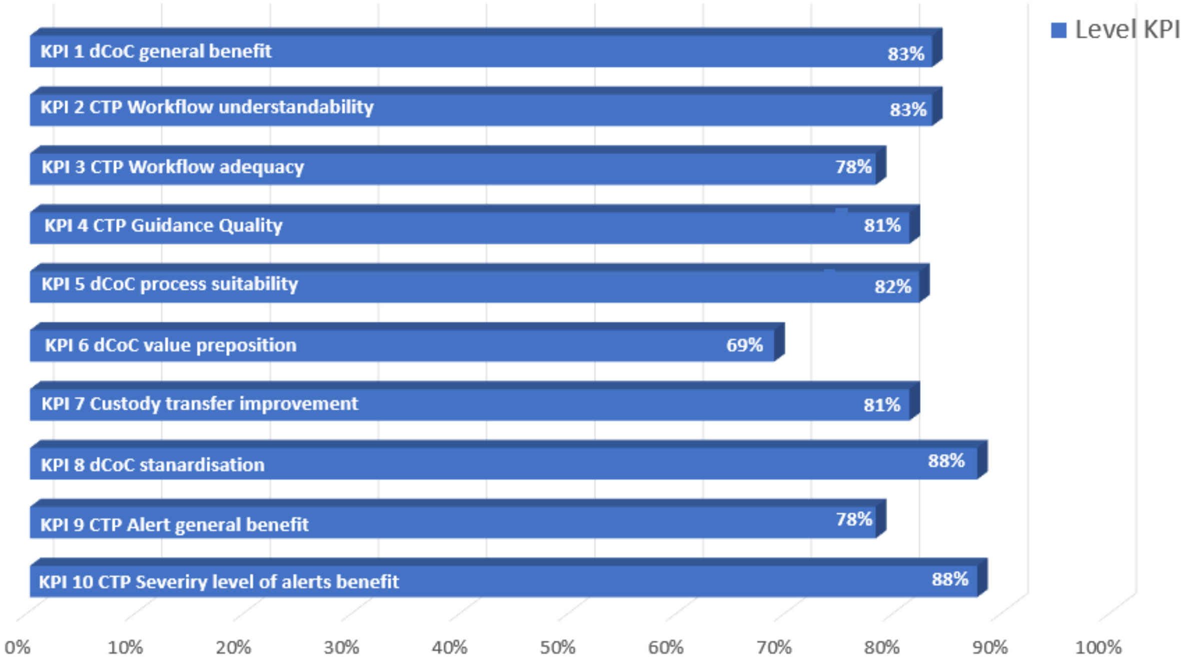


Figure 39: KPI values, i.e. summary of the assessment obtained for each KPI.

In particular, it should be noted all statements got a very positive assessment, in almost all cases better than in the intermediate evaluation. But one has to take into consideration that a lotless respondents were available during the FSX. More details to the different questions, especially also the comments by the respondents (C1, C2 etc.) are given below.

Q.1 There is an advantage in having a dCoC

A high average score was reached (4.33). Comments provided in the free text field were:

C1. The key should not be generated only by the crisis Commander

C2. It is very useful, when it is "digitilized"

C3. You have the ability not only to check the inconsistency of the package but also to locate the evidence at any time which is the solution for the major problem, when there several departments that have the custody of several multiple evidencies;

C4. Having a digital CoC seems like a great way to have.

Q.2 The proposed CTP workflow is understandable

A high average score was reached (4.33). Comments provided in the free text field were:

C1: It is very easy, even for novice

C2. CTP workflow was explained clearly

Q.3 The proposed CTP workflow is adequate

A high average score was reached (4.13). Comments provided in the free text field were:

C1: Mentioned issue on keys;

C2: It is as adequate, as it needs

Q.4 The information (metadata), i.e. DCM, presented is sufficient to conduct secure CTP operations

A high average score was reached (4.25). Comments provided in the free text field were:0

C1. According to those with the experience in the group, it is sufficient;

C2, Details about metadata information model were not discussed on the table. For this reason, it is difficult to say the sufficiency related to this particular subject matter.

Q.5 The proposed process can support the implementation of dCoC

A high average score was reached (4.29). Comments provided in the free text field were:

C1. It can cover all processes;

C2. We have not discussed in details about the implementation

Q.6 The proposed dCoC process ensure non-repudiation requirements

Here only an average score of 3.75 was achieved.

One single respondent contributed a "Disagree" and commented "Issue on keys". A further clarification revealed that the remark is related to the technical implementation of the dCoc, not the concept itself. It was discussed how an encryption support process could be implemented in a concrete technical realization. It makes sense to analyze this from the point of view of the technology that could be used for the implementation of the TS, but the TS is not referring to a special technology, it is just supposed to provide the general framework.

Q.7 The use of a dCoC can improve the custody transfer performance

A high average score was reached (4.25). Comments provided in the free text field were:

C1. I belive that it can help the custody to be safe

Q.8 The creation of a dCoC standard is valuable

A high average score was reached (4.50). Comments provided in the free text field were:

C1: "Don't have an opinion on the subject" (this person ticked "Neither agree-nor disagree", answer reclassified to "No Answer")

C2: In general terms, the dCoC is great;

C3: It is strongly valuable, because a creation is "wrong", it drives;

C4: Definitely but not necessarily in the context of a CBRN case;

C5: I think that dCoC is important for enhancing interoperability in CoC processes.

Q.9 The existing of CTP Alerts is pertinent

A high average score was reached (4.13). Comments were not given.

<u>Q.10 The creation of a specific alert when the sample's package is compromised is valuable</u>

A high average score was reached (4.50). Comments provided in the free text field were:

C.1 Not discussed; (the person ticked "no answer", it should be noted that several use cases were discussed, one of which was the violation of a seal on a sample package).

C2. CBRN alerts itself

Some other comments, which were given in free text field not related to a specific question, were:

C1. I was thinking that the whole process could be exploited by the DVI teams. It can be introduced and used in case of sampling during a major incident with multiple unidentified victims. Can the content of the proposed CWA be expanded to collecting evidence generally?

C2. CTP data governance process is so abstract that it can be applied to any type of evidence handled by the Police and can solve chronic problems. A major problem is the countless evidences that pile up on services due to the time-consuming processes until they end up at endpoints. By using the CTP, the search time to find where the evidences exist is reduced

3.6.4 Results from Formal Aspects Evaluation

The evaluation regarding formal aspects was conducted by Diana Iorga (ASRO) with the help of the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3 [35], chapter 2.1.9) and gave the following results:

In both parts of the TS a high Likert scale score was reached almost throughout: As an average 4.9 for part 1 and 4.8 for part 2. In detail there was 40 times (39 times for part 2) "5" (strongly agree), in one case (Q.7 in both parts) "4" ("Agree") and in one case (Q.25 in both parts) "3" ("Neither agree or disagree"). The latter was related to the question Q.25: If used, are "may" and "can" used correctly so that they do not communicate obligation but alternativeness? In this case it was not possible for the reviewer to assess this aspect, because as a formal aspects reviewer she was not familiar in detail with the subject of the TS and it was not obvious, where alternativness was intended. The authors of the TS noted the issue, checked it carefully and assured that they have used the wording correctly.

The only "disagree" referred to the subtitle "Part 2: Data management and audit". It was stated that it should indicate more clearly whether the document provides guidelines or requirements. Since the title was suggested by end-users without the phrase "Guidelines for..." and since submitted to TC391/WG2 (CBRNE) and the title was approved, a change to the subtitle is not possible anymore at that stage of the process.

The formal Quality of Foreword was not yet checked, because the evaluation was based on the available version of the TS, which did not yet contain the necessary information.

Therefore, the TS passed the formal aspect evaluation threshold preliminarily.

3.6.5 Results from RAF Tool Analysis

During the STRATEGY project the stream 6 standardisation activities were assessed also with the help of the ResiStand Assessment Framework (RAF), see D2.4 [7] Annex 4 "Completed RAF intake sheets for dCOC". For the final evaluation the RAF supported analysis was reconsidered and updated. More details about the RAF in general can be found in chapter 2.4 of the present deliverable or in STRATEGY deliverable D1.2 [2] "Project Handbook".

The RAF Tool Intake sheets for dCoC presented in this section, are an update of the RAF Tool Intake sheet presented in D2.4 [7] Annex 4 "Completed RAF intake sheets for dCOC". During the preparation of the TS the document was split and TC391/WG2 (CBRNE) approved the development of two TS a CEN Technical Specifications "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" composed by Part 1: "Overview and Concepts" and Part 2: "Data Management and Audit" an overview of the completed RAF intake sheets for both TSs are presented respectively in:

1. Figure 40, CEN Technical Specification 1 "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" composed by Part 1: "Overview and Concepts": Intake Sheet.
2. Figure 41, CEN Technical Specification 1 "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" composed by Part 1: "Overview and Concepts": Assessment Sheet.
3. Figure 42, CEN Technical Specification 2 "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 2: "Data Management and Audit": Intake Sheet
4. Figure 43, CEN TS 2 "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 2: "Data Management and Audit": Assessment Sheet.

It can be seen that the final assessment shows very good results in terms of feasibility (4), impact (4,75) and especially urgency (5) for both parts of the TS. It should be noted that the assessment of the feasibility has been raised in comparison with the former RAF tool analysis. The reason is that "Implementation and follow up perspective" is not any more to rate as "insufficient", but as "sufficient" due to promising late developments: The work on improving and revising the TS has been finalised under CEN/TC391/WG2 (CBRNE) and both TSs are in the process of being voted on by CEN, pending the outcome.

Intake — ResiStand

Title of the proposed standard:	Societal and citizen security — dCoC for CBRNE Evidences Part1 Overview
Identification number:	t.b.d.

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
INOV	Research
CINAMIL/UMLDBQ (End-User & Research)	Hybrid
Fraunhofer INT	Research
SFS (Standardisation Body)	-
ASI (Standardisation Body)	-

Type of standard:	Standard
Scope of the standard:	Digital Chain of Custody (dCOC) for Evidence Part 1: Overview and Concepts, provides guidance for technical and non-technical personnel of senior management within the organisation, including those responsible for compliance with regulations and regulatory requirements and industry standards. It describes how such personnel can identify and audit the custody ownership at each transfer point in the dCoC; set policies and
Example or illustration 1:	
Example or illustration 2:	

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Unknown	
with national and regional legislation:	Unknown	

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries (& Military)
NGOs:	Yes	WHO, OIE, FAO
Industry/SMEs:	Yes	Technology companies developing efficient tools, incident management and decision
Consultancy organisations:	Yes	CBRN specialists and decision-makers
Research institutes:	Yes	Research Centers when developing new solutions/applications related to CBRN dCoC
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protecion Machanism, Military

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Improve the Chain of Custody process; speed up turnaround times; Prevent common user-errors associated with paper forms; Less paper work; Improve multi-agency interoperability in first response to a CBRN incident
Industry:	Global market.
Research:	Laboratorial results speed up; improving laboratory management and response.
Policy makers:	Strengthening civil protection capacities in the partner countries; Strengthening cooperation between international CBRN CoC stakeholders
Citizens:	Better public service.

Urgency of the standard: (when is it needed for implementation?)	Very high (ASAP)	dCoC for CBRNE is a tool which will improve the response in CBRNE scenarios, especially in those, where criminal actions are suspected to triggered the event. The dCoC will speed up the CBRNE forensics and tactical investigations, reducing the

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	Criminal police and other law enforcement authorities
NGOs:	Yes	Interpol; UNSGM; OPCW; OIE; FAO; IAEA
Industry/SMEs:	Yes	IT companies
Consultancy organisations:	Unknown	
Research institutes:	Yes	Security, defence and law research centers
Others:	Unknown	

Preferred leading type of stakeholder:	First Responders	
Expected barriers and constraints:	Legal constraints specific from each country.	

Other information

Additional remarks:	The chain of custody is established whenever the intervenient actors (e.g., collection team, escort team, laboratory) take custody of evidence

ResiStand Assessment Framework 2.0 — TNO

Figure 40: CEN TS 1 "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 1: "Overview and Concepts": RAF Tool Intake sheet

Assessment — ResiStand

Title of the proposed standard: Societal and citizen security — dCoC for CBRNE Evidences Part2 Data Mng
Identification number: t.b.d.

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
INOV	Research
CINAMIL/UMLDBQ (End-User & Research)	Hybrid
Fraunhofer INT	Research
SFS (Standardisation Body)	-
ASI (Standardisation Body)	-

Scope of the standard: Digital Chain of Custody (dCOC) for Evidence Part 1: Overview and Concepts, provides guidance for technical and non-technical personnel of senior management within the organisation, including those responsible for compliance with regulations and regulatory requirements and industry standards. It describes how such personnel can identify and audit the custody ownership at each transfer point in the dCoC; set policies and follow good practices for data governance workflow automation, and raise awareness at each

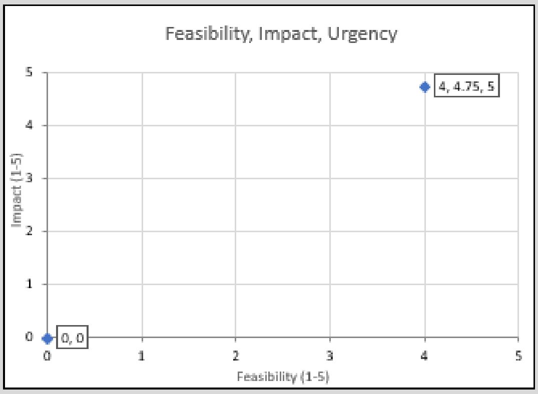
Type of standard: Standard

Urgency (when needed?): Very high (ASAP)

Overall impact: Great

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Great
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Great
Innovation progress:	Great
Improvement of business functions:	Considerable

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard: Considerably

Benefit of the standard to types of incident: Incidents in general
Specifically for the following incidents: CBRN attack; Nuclear accident; Epidemics/Pandemics

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard: The relevance of Non-repudiation – ensure that the sender of data is provided with proof of delivery and the recipient is provided with proof of the sender's identity; neither (the sender nor the receiver) can deny having processed the data.

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:
- Command, Control and Communication technology;
- Technological progress in CoC migration to dCoC.

ResiStand Assessment Framework 2.0 — TNO

Figure 41: CEN TS "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 1: "Overview and Concepts": RAF Tool Assessment sheet

Intake

ResiStand

Title of the proposed standard: Societal and citizen security — dCoC for CBRNE Evidences Part2 Data Mng
Identification number: t.b.d.

Proposed standardisation activity

posing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
INOV	Research
CINAMIL/UMLDBQ (End-User & Research)	Hybrid
Fraunhofer INT	Research
SFS (Standardisation Body)	-
ASI (Standardisation Body)	-

Type of standard: Standard

Scope of the standard: Digital Chain of Custody (dCOC) Part 2: Data Management and Audit, contributeto replace the manual Chain of Custofy (i.e., paper-based forms completed by hand), electronic of information workflow, focus on describing describes how such personnel can identify and audit the custody ownership at each transfer point of the dCoC for CBRNE evidences, set data management policies, comply with good practices for

Example or illustration 1:

Example or illustration 2:

Compliance with legislation

	Compliant	Explanation
with European legislation:	Unknown	
with national and regional legislation:	Unknown	

groups for applying the standard

	Target group	Free space for additional comments
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries (& Military)
NGOs:	Yes	WHO, OIE, FAO
Industry/SMEs:	Yes	Technology companies developing efficient tools, incident management and
Consultancy organisations:	Yes	CBRN specialists and decision-makers
Research institutes:	Yes	Research Centers when developing new solutions/applications related to CBRN
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protecion Machanism, Military

Potential impact and urgency

nefits for stakeholder categories

	Description of who will benefit and in what way
End-users:	Improve the Chain of Custody process; speed up turnaround times; Prevent common user-errors associated with paper forms; Less paper work; Improve multi-agency interoperability in first response to a
Industry:	Global market.
Research:	Laboratorial results speed up; improving laboratory management and response.
Policy makers:	Strengthening civil protection capacities in the partner countries; Strengthening cooperation between international CBRN CoC stakeholders
Citizens:	Better public service.

Urgency of the standard: (when is it needed for implementation?) Very high (ASAP)

dCoC for CBRNE is a tool which will improve the response in CBRNE scenarios, especially in those, where criminal actions are suspected to triggered the event. The dCoC will speed up the CBRNE forensics and tactical investigations,

Development

Required types of stakeholders

	Required	Free space to explain why their participation in the development is
First Responders:	Yes	To test the proposed standard by providing feedback and implementing table top
Governmental organisations:	Yes	Criminal police and other law enforcement authorities
NGOs:	Yes	Interpol; UNSGM; OPCW; OIE; FAO; IAEA
Industry/SMEs:	Yes	IT companies
Consultancy organisations:	Unknown	
Research institutes:	Yes	Security, defence and law research centers
Others:	Unknown	

rred leading type of stakeholder: First Responders

xpected barriers and constraints: Legal constraints specific from each country.

Other information

Additional remarks: The chain of custody is established whenever the intervenient actors (e.g., collection team, escort team, laboratory) take custody of evidence

ResiStand Assessment Framework 2.0 — TNO

Figure 42: CEN TS "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 2: "Data Management and Audit" RAF Tool Intake sheet.

Assessment

ResiStand

Title of the proposed standard: Societal and citizen security — dCoC for CBRNE Evidences Part2 Data Mng

Identification number: t.b.d.

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
INOV	Research
CINAMIL/UMLDBQ (End-User & Research)	Hybrid
Fraunhofer INT	Research
SFS (Standardisation Body)	-
ASI (Standardisation Body)	-

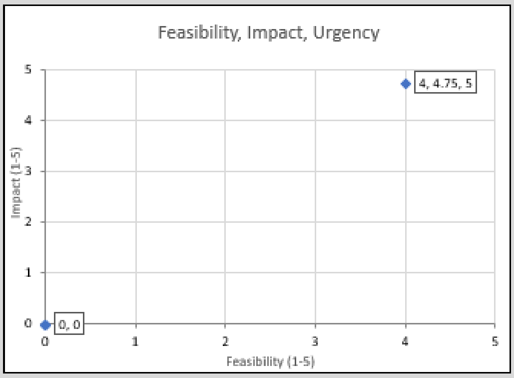
Scope of the standard: Digital Chain of Custody (dCOC) Part 2: Data Management and Audit, contributeto replace the manual Chain of Custofy (i.e., paper-based forms completed by hand), electronic of information workflow, focus on describing describes how such personnel can identify and audit the custody ownership at each transfer point of the dCoC for CBRNE evidences, set data management policies, comply with good practices for data governance workflow automation and raise awareness at each custody transfer point. It also suggests the

Type of standard: Standard

Urgency (when needed?): Very high (ASAP)

Overall impact: Great

Feasibility: High



Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Great
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Considerable
Improvement of business quality management:	Great
Innovation progress:	Great
Improvement of business functions:	Considerable

Feasibility

Foundation:	Sufficient
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard: Considerably

Benefit of the standard to types of incident: Incidents in general

Specifically for the following incidents: CBRN attack; Nuclear accident; Epidemics/Pandemics

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard: The relevance of Non-repudiation – ensure that the sender of data is provided with proof of delivery and the recipient is provided with proof of the sender's identity; neither (the sender nor the receiver) can deny having processed the data.

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:
- Command, Control and Communication technology;
- Technological progress in CoC migration to dCoC.

ResiStand Assessment Framework 2.0 — TNO

Figure 43: CEN TS "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" Part 2: "Data Management and Audit" RAF Tool Assessment sheet (overview).

3.6.6 Conclusions

By and large, the results show that the TS “Societal and citizen security — Digital Chain of Custody for CBRNE Evidence” is on the right track and that the improvements introduced after the mid-term evaluation and the discussion within CEN/TC 391/WG2 - Societal and citizen security - CBRNE , have borne fruit. For the usage of a digital chain of custody, data security standards need to be respected which is feasible with the approach of the CWA. An important step on the way of the TSs to become standards was that stream 6 was invited to the plenary session of CEN/TC 391 and there was a slot where the TSs could be presented. The very positive appreciation of the dCoC demonstration in FSX is also a good indicator and was an incentive for both TSs to be submitted to CEN/TC for voting (no changes are foreseen or allowed at this stage). Given this development there is a high probability that the TSs indeed become standard soon.

3.7 Final Evaluation of Stream 7 Training

3.7.1 Final Evaluation of Stream 7, CWA 1

3.7.1.1 Basis of the Final Evaluation

The evaluation of the standardization activities conducted within Stream 7 were mainly based on three pillars: the TTX evaluation, the FSX evaluation and comments received through the public. The results from the TTX have already been reported in D4.3 "Intermediate Evaluation and Validation of Crisis Management solutions". As such, this section will focus on the final evaluation of Stream 7,

CWA "Specifications for Digital Scenarios for Crisis Management Exercises''

which was carried out during the FSX week in March 2023 (27-31/3/2023) in Gualdo Tadino, Italy. The aim of the FSX was to showcase all the integrated systems and methodologies of the different pre-standardization items, evaluate the concepts of the CWAs/TSs in realistic operational conditions. Stakeholder from FRs, Emergency Management Centres, Crisis Management Response teams across different threat scenarios participated to the event and gave welcome feedback on the CWA. Details about this event can be found in D4.1 "Full scale exercise demonstration report".

Stream 7 participated in the FSX with the two standardisation items entitled "Specifications for Digital Scenarios for Crisis Management Exercises'' and ''Evaluation of exercises – Implementation Guidelines'', hereafter refereed as CWA 1 and CWA 2 respectively. The target group of invitees to the FSX included STRATEGY partners and external experts from the following relevant organisations:

- Crisis and disaster management Practitioners
- First Responders
- Standardization Experts
- Civil protection agencies' representatives
- Technical experts
- Industry representatives

The Stream 7 involvement in the FSX week took place in 2 phases. Firstly during 29/3/2023 with a 2 hours slot dedicated to present the current status of each CWA documents, a demonstration of a potential solution adapted to the propositions of the CWA and an open discussion based on the end-user questions. In the end of the discussions a questionnaire shared among the participants to provide input. Secondly during 30/3/2023 the CWA 1 of Stream 7 played a main role in the exercise by putting into practice the incorporation of both FSX scenarios into the proposed data model through the application of TMT software tool.

3.7.1.2 Results from Expert Discussions and Testing during the FSX

During the FSX week, a scheduled 2 hour discussion with experts in the domain of planning and execution of exercises in crisis management took place. The discussion included 5 relevant authorities representing the prefecture of Perugia, the civil protection and first responders. Everyone seemed intrigued by the idea of a digital scenario specifications proposed by the CWA 1, which aims to specify a digital process for the planning of exercises and provide recommendations on the use of data models for exchanging scenario characteristics. This will enable a common digital process used by relevant authorities to plan a crisis management exercise. Several questions and issues were brought up during the session and main points are summarized below:

1. Scenario aspects which required more effort and work: It was explained that the main problem was that a lot of tools for running the exercises were available, but what was missing was a unique data model structure.
2. Main gap addressed by the proposed standardisation activity: It was explained that the main gap addressed was the lack of a unique data model which can be used and transferred to different solutions and tools. The proposed CWA aims to provide the interoperability specifications that digital scenarios need to have in order to be able to be implemented by different scenario building tools.
3. Simplifying procedures: it was underlined that the proposed pre-standardisation activity aims to provide a way to distinguish the different elements of an exercise and a common way to incorporate scenario characteristics in to the proposed data model that would simplify the process of planning of crisis management exercises.

All the participants were interested and engaged in knowing more about the standardisation document presented. At the end of the round table discussion, participants were asked to fill in the evaluation questionnaire reporting their feedback.

3.7.1.3 Questionnaire Results

Based on the criteria and KPIs are listed in D4.3 at section 3.7. the original questionnaire that was created for the TTX was updated with minor modifications based on the experience gained at the TTX and the progress of the CWA 1. In the figure below are available the feedback collected after the presentation and the open discussion conducted in a dedicate slot for CWA 1 of Stream 7. In the round table joined representative’s consisted of prefecture of Perugia, the civil protection and Italian firefighters.

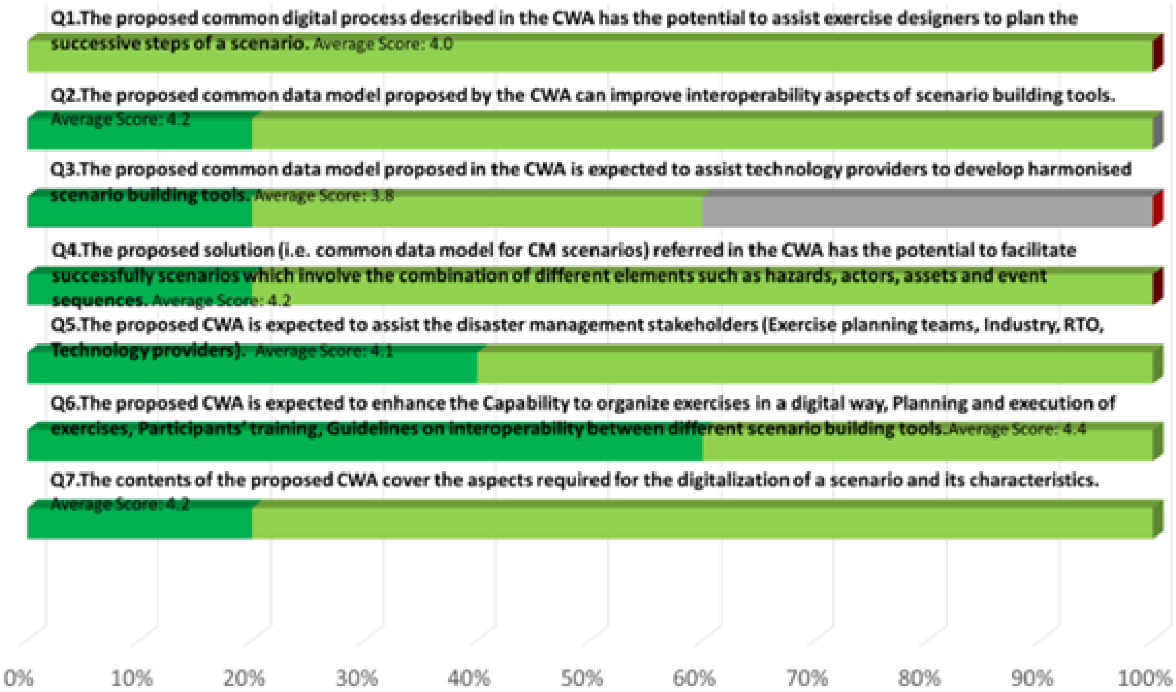


Figure 44: Scores' distribution obtained for all the statements of CWA 1: Specifications for Digital Scenarios for Crisis Management Exercises'' of Stream 7: Training questionnaire.

The figure above summarizes the results of the evaluation received from the participants of the round table. The overall score ranging from 3.8 to 4.4 (out of 5) and the average score is 4.1 which indicates that the participants find the CWA 1 activities promising and well-structured.

3.7.1.4 Results from Formal Aspects Evaluation

Based on the D4.3, section which is entitled "criteria and KPIs related to formal aspects" provide on more evaluation point to be assessed in the final version of the CWA. From Stream 7 and CWA 1 a member that was not involved in the development of the CWA assessed to provide feedback on the quality of the document. The CWA 1 of Stream 7 received average score more than 4 in all the points proving the overall good quality of the document which has now been submitted for publication at CEN/CENELEC.

3.7.1.5 Results from RAF tool Analysis

RAF tool was used in the beginning of the project for the selected standardisation items identified in Stream 7. In this section are listed two out of seven tabs that are available in the excel file for CWA 1. The file updated at the completion of the standardization activities of this Stream based on our experience of the TTX, FSX and the actual document developed and is published at CEN/CENELEC. The tabs that are available below are the intake and the assessment. In the assessment tab the scores obtained are:

- Feasibility: 3.8 out of 5
- Impact: 4.5 out of 5

- Urgency: 3 out of 5

Intake

ResiStand

Title of the proposed standard: Digitising specifications for exercise scenarios for SaR activities

Identification number:

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
STRATEGY	Hybrid
ICCS	Research
CNVVF	End-users
SAMU	End-users
ASI	Policy makers

Type of standard: Workshop Agreement

Scope of the standard: Main scope is the development of an interoperable and harmonised process that aims to digitise specifications of exercise scenarios and assist exercise designers in SaR training related activities. This will assist exercise designers to precisely plan all the successive steps of a preparedness scenario and will enable the effective preparation of large-scale exercises involving multiple agencies, cross-border

Example or illustration 1:

Example or illustration 2:

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Unknown	Existing standards (ISO/IEC 19510:2013) provide general information and are not focused on scenario buidling activites in SaR training
with national and regional legislation:	Unknown	

Target groups for applying the standard	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Firefighters, Police, Crisis Management Operation Centers etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	Unknown	
Industry/SMEs:	Yes	Technology companies developing scenario building tools
Consultancy organisations:	Unknown	
Research institutes:	Yes	Research institutes developing new solutions for search and rescue operations
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protection Mechanism

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Governmental organisations (e.g. CPA) and FRs will be benefited by putting into practice an easy to use digital process to enhance the planning and execution of SaR training related activities
Industry:	Technology providers of scenario building solutions will be benefited by the development of a digital process to assist with incorporation of input data that can be consumed by their tools.
Research:	Technology and research providers will develop a customised solution to the requirements of end-users building innovative pathways for other SaR related activities.
Policy makers:	Improved planning and execution of exercises will lead to potential new practices that will be used to update the current policy in SaR related activities.
Citizens:	Enhanced training in SaR related activities will develop improved preparedness to efficiently manage disaster risk and avoid preventable deaths of citizens.

Urgency of the standard: (when is it needed for implementation?) Moderate (< 2 yrs) — The proposed standard is expected to play a significant role in scenario simulation area by providing an interoperable and harmonised process that aims to digitise specifications of exercise scenarios and assist exercise designers in

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	Digitalisation of scenario building will be customised to enhance their training
Governmental organisations:	Yes	Main stakeholders organising and planning exercises
NGOs:	Unknown	
Industry/SMEs:	Unknown	Potential tech providers
Consultancy organisations:	Yes	Potential consultancy services to stakeholders
Research institutes:	Yes	Technology and research providers enhancing the process of training through
Others:	Unknown	

Preferred leading type of stakeholder: First Responders — Firefighters, Police, Crisis Management Operation Centers etc.

Expected barriers and constraints:

Other information

Additional remarks:

ResiStand Assessment Framework 2.0 TNO

Figure 45. RAF final assessment for CWA 1 - Intake sheet.

Assessment

ResiStand

Title of the proposed standard: Digital Scenario Specifications for Crisis Management Exercises

Identification number: TR-7

Proposed standardisation activity and its perspectives

Proposing organisations or projects:	Organisation or project consortium	Stakeholder category
	STRATEGY	Hybrid
	ICCS	Research
	CNVVF	End-users
	SAMU	End-users
	ASI	Policy makers

Scope of the standard: Main scope is the development of an interoperable and harmonised process that aims to digitise specifications of exercise scenarios and assist exercise designers in SaR training related activities. This will assist exercise designers to precisely plan all the successive steps of a preparedness scenario and will enable the effective preparation of large-scale exercises involving multiple agencies, cross-border locations and data sources. Digitising specifications of an exercise scenario will involve the combination

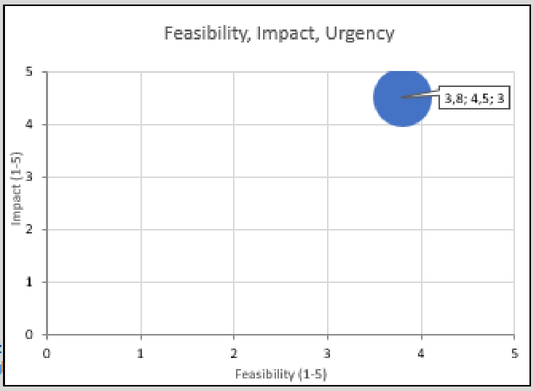
Type of standard: Workshop Agreement

Urgency (when needed?): Moderate (< 2 yrs)

Overall impact: Great

Feasibility: High

Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very hig



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Great
Improvement of responder safety:	Great
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Moderate
Improvement of business quality management:	Moderate
Innovation progress:	Great

Feasibility

Foundation:	Amply sufficient
Development perspectives:	Amply sufficient
Implementation and follow-up perspectives:	Sufficient
Anticipated drawbacks and constraints:	Not at all

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Completely
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	-

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard:	Developments in disaster resilience and crisis management due to increasing involvement of society (building on societal potential is required because the size of public services is decreasing), towards Network-Enabled Capabilities of
Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:	Technical trend related with the standardisation of a digitised process to assist SaR training related activities - Non-technical trend related with improvement of SaR related training enhancing preparedness of first responders and end users

ResiStand Assessment Framework 2.0 TNO

Figure 46 RAF final assessment for CWA 1 - Assessment sheet.

3.7.1.6 Conclusion

The results show that the CWA 1 “Specifications for Digital Scenarios for Crisis Management Exercises” which was assessed is the final draft version is on the right track. Operational stakeholders participated in the round table presentation, demonstration and open discussion identify the standardization activity useful and promising to incorporate proposed data model to modernize the planning process crisis management exercises. The CWA, which has now been submitted for publication at CEN/CENELEC, was discussed from different angles such as the perspective of civil protection authorities but also of first responder teams as potential users but also the perspective of SMEs/RTOs as developers of similar solutions.

The high average score of all questions indicates that the CWA objectives have been met while the feedback from the key stakeholders assisted towards the next steps of the pre-standardization activities. During the FSX, the CWA used an existing solution (i.e. TMT) which was developed in the context of Driver+ project to demonstrate the incorporation of scenario inject using the data model specification proposed by the standardization activity.

The Technical Specifications under Stream 7 CWA 1 namely "Specifications for Digital Scenarios for Crisis Management Exercises" supports a shared strategy aimed at enhancing the effectiveness of exercises. While this could necessitate practitioners to modify their existing protocols, it's crucial, much like in Stream 4, to establish uniform methodologies to achieve common objectives.

3.7.2 Final Evaluation of Stream 7, CWA 2

3.7.2.1 Basis of the Final Evaluation

Considering the progress of the CEN Workshop on

CWA “Evaluation of exercises – Implementation guidelines”

that has taken place following the intermediate evaluation of the document during the TTXs (please refer to D4.3 for further details) and additional exchanges among the workshop participants, the updated concepts /outcomes of the said pre-standardization activity were again assessed as part of the Full Scale Exercise (FSX) that was held in Gualdo Tadino in the province of Perugia in northeastern Umbria in Italy during March 27th – 31st 2023. This event was, hosted by the Italian National Fire and Rescue Service and entailed the operational evaluation of all the pre-standardisation activities of STRATEGY project on the basis of actual operational conditions (see D4.1). Specifically with respect to the thematical stream 7 of STRATEGY on Training and in particular the CWA “Evaluation of exercises – Implementation guidelines”, the final evaluation was realised in two sequential steps. These were:

a) Evaluation by participants at the technical session of the FSX, held on the 29th of March 2023:

During this session, the scope of the CWA , the gaps it addresses in existing standardization of crisis management, its main concepts and contents were presented by the CWA leader. To the session, a number of stakeholders from different organizations (see Table 23) in charge of the planning of exercises and the development of evaluation methodologies have participated. A thorough discussion has taken place on issues highlighted by the CWA leader and other raised by the participants. The scope of the current session was to obtain feedback on the structure, content, terminology, implementation possibilities, etc of the current CWA by exerts with operational and scientific background.

Table 25 List of organisations with attendants in the dedicated workshop for CMEx

Organisation Name	Type/Description
KEMEA (Greece)	RTO - Research and Technology Organisation;
CNVVF (Italy)	End-user (Firefighter)
POLIZIA (Italy)	End-user (Law Enforcement)
Hellenic Police (Greece)	End-user (Law Enforcement)
AIT (Austria)	RTO - Research and Technology Organisation;
ARPA UMBRIA (Italy)	End-user (Regional Agency for the protection of the Environment)
Prefettura – Ufficio Territoriale del Governo di Perugia (Italy)	End-user (Local Government Agency Prefecture)
STWS (Greece)	SME - Small and medium-sized enterprise

During the final stage of the technical session, in consideration of any outcomes and conclusions that resulted and in order to allow space and time for a more detailed evaluation, participants were engaged in providing feedback though a structured questionnaire that was prepared specifically for the occasion. The questionnaire was distributed in hard copies to help the participants keeping notes throughout the session and guide them through the discussions and was also available online (https://forms.office.com/e/qtbNtKpsgt). This was prepared based on the guidelines for the creation of the “Participant Feedback Form” as presented in the current CWA, and more specifically focusing on solution’s evaluation topics. For facilitating the feedback collection process, the session was further supported though team of specifically appointed evaluators that focused on collecting additional data / feedback as part of a parallel “note keeping” process. This team appears in Table 24. Results are presented during the subsequent sections. This evaluation/ note-taking team is provided in the following table.

Table 26 FSX Evaluation team for CMEx

Full Name	Affiliation	Role / Required expertise
Alessia Golfetti	Deep Blue	Lead Evaluator – note-taker
Katerina Valouma	STWS	Evaluator

b) Evaluation by CWA leaders:

The current CWA due to its nature couldn’t be evaluated by the exercise participants, as also explained in D4.3, D3.8 and 5.7. It was implemented horizontally by all CWA leaders who aimed to test and evaluate their solutions during the FSX (and previously the various TTXs). The CWA leaders were informed about the content of the current CWA providing guidelines on the creation of Participant Feedback Forms and Evaluators’ Forms. As far as the former is concerned, the CWA leaders and the FSX manager were guided for the preparation of the evaluation questionnaires, focusing on the evaluation of their solution tested and for the evaluation of the exercise itself. It should be noted that per the current CWA, “solution” is any “technological tool, doctrine, standard process, concept, methodology or any combination thereof” and thus the testing of the CWAs within an operational environment are considered exercises with scope to evaluate solutions, thus “trials”.

Subsequently the CWA leaders were asked to provide their feedback, on the content and their satisfaction on the usability of the proposed CWA, via the same aforementioned online questionnaire.

3.7.2.2 Results from Expert Discussions and Testing during the FSX

As discussed above, on the second day of the FSX period (29th of March) a special session dedicated to the presentation and discussion of the CWA content took place, with the participation of a number of experts of operational and scientific background.

Following a presentation of the overall structure of the draft, some topics, that the CWA leader wished to have clarifications, were raised and some other comments emerged from the discussions. At the end of the sessions the following significant elements were commented, which were subsequently discussed during the last CWA meeting so as to reach final conclusions of common approval among the workshop participants.

- According to ISO 22398 and other literature on exercise planning (e.g. USDHS, 2020), evaluation composes the last phase of the exercise cycle. In the current CWA, the evaluation is introduced as a process that is ongoing throughout the entire exercise cycle, it is prepared during the planning phase, data to be sued for evaluation is collected during the conduct phase, and it is finalised during the review and improvement last phase. It has been discussed that this “innovation” in the positioning of the evaluation in the exercises cycle needs to be communicated clearly in the CWA to avoid any potential definition conflicts and contradictions.

- According to the current CWA, a trial (i.e. events aiming to the validations of solutions by practitioners in the context of realistic situations and operational environment) is considered as a subtype of exercise which scope is the evaluation of solutions rather than the player’s performance. It has been discussed that it needs to be clear that there is no conflict with the CWA 17514 “Systematic assessment of innovative solutions for crisis management – Trial Guidance Methodology” which described the whole methodology for the testing and evaluation of solutions (trials). The current CWA is complementary to CWA 17514, focusing on the evaluation process, especially for the case that an operational exercise includes also the validation of solutions, but not only. It is then clarified that CWA 17514 is the most comprehensive and complete as far as the development of a “trial” is concerned.
- It has been discussed that clear distinction of “evaluation” and “validation” is necessary and the participants have agreed on the definitions proposed.
- Discussing on the content of the Post-exercise debrief and whether more recommendations should be added, it has been commented that the performance of a Strength-Weaknesses-Opportunities-Threats (SWOT) analysis, even orally, would be beneficial.
- The structure and content of the Evaluator’s form raised some discussions and an interesting operational experience was shared and evaluated. It would be useful to recommend the inclusion of a numerical scale per capability, with a pre-defined rating scale, which will also lead to an overall score. Each capability may have a different weight factor according to the exercise’s scope and specific objectives. The overall score may be also reflected and reported in the After-Action report and proposed summary by the current CWA (After-Action Summary).
- It has been commented that the ethical, environmental and safety considerations mentioned in the CWA needs to be clear that they refer explicitly to the evaluation process of the exercise.

3.7.2.3 Questionnaire Results

Following the technical session, the participants were asked to respond to the online questionnaire. Moreover, the CWA leaders that read and used the CWA for the preparation of the evaluation of their own CWAs for the FSX, were also asked to provide their feedback. The questionnaire is developed in the principles of the “Participant Feedback Form”, presented by the current CWA. This is composed by questions aiming to the validation of the current CWA as a proposed methodological solution and has a set of questions focusing on the usability of the draft, the effectiveness and efficiency of the its content and the satisfaction of the practitioners that make us of it. In addition to this evaluation, questions more directly content-related, questioning e.g. the validity of a statement or the incorporation of some elements in the draft, have been also included. The latter are not considered directly related to the evaluation of the CWA but they are of outmost importance for the development of the CWA and the collection of direct input from experts. The questionnaire consists of closed-ended questions, with yes/no answers, rating scale (1 to 5) and Likert scale questions, while open-ended questions are also included only to allow for explanations and/or commenting should the responded need it.

a) Questions without evaluation score needed for feedback on CWA content

Q1. Are all different data collection methods well explained?

All respondents agreed that the Evaluators' form and the Participant Feedback Form are well explained. Some missing answers appear only at other means of data collection, including interviews and the post-exercise debrief.

Q2. Is there any method missing that can provide valuable input? Please explain

The vast majority of the respondents cannot think of any other method. A comment about the incorporation of a qualitative and a quantitative scale that can lead to more objective evaluation, appear. The numerical scale, with qualitative or qualitative explanation, is also added in the Evaluator's form, while a recommendation for its consideration already exists in the guidelines for the Participant Feedback Form.

Q3. Would you recommend providing more guidance on the interviews and the debriefing session?

The answers to this question were divided. More guidance is recommended for the inclusion of a numerical scale also for these sessions, what will also allow the creation of statistics.

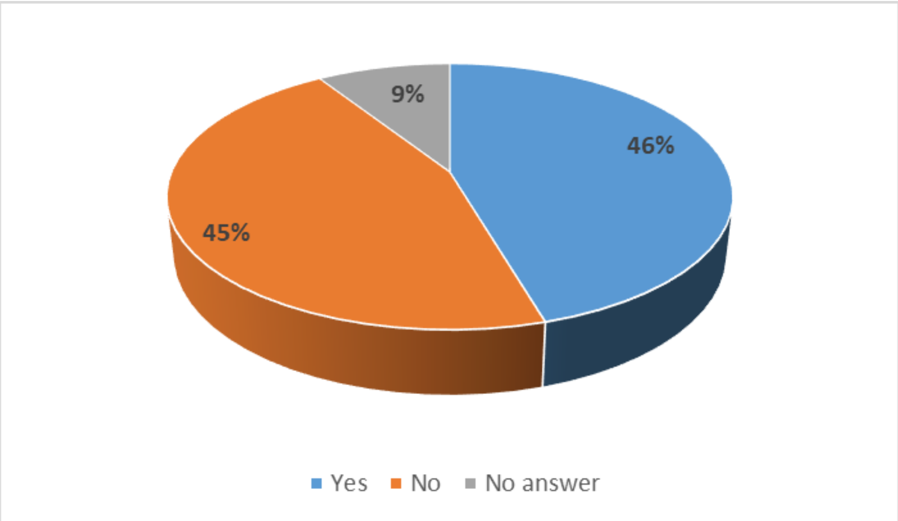


Figure 47 Answers to Q3 to CMEx evaluation questionnaire

Q4. Do you agree that the evaluator's form (not being in the form of a traditional questionnaire) better allows the expression of the evaluators' view in a guided yet more flexible manner, directly revealing the strengths and areas for improvement of the exercise?

The answers to this question vary between "agree" and "strongly agree". The evaluator's form remains as proposed.

Q5. How do you rate the content and the completeness of the Guidelines for the evaluation of the exercise conduct within the Participant Feedback Form? (For FSX organizers please explain how did the Guidelines help you for the evaluation of the FSX PFF)

This was a rating scale question varying from 1-poor to 5-excellent. The vast majority of the answers rate as “very well” to “excellent” the completeness of the content of this CWA.

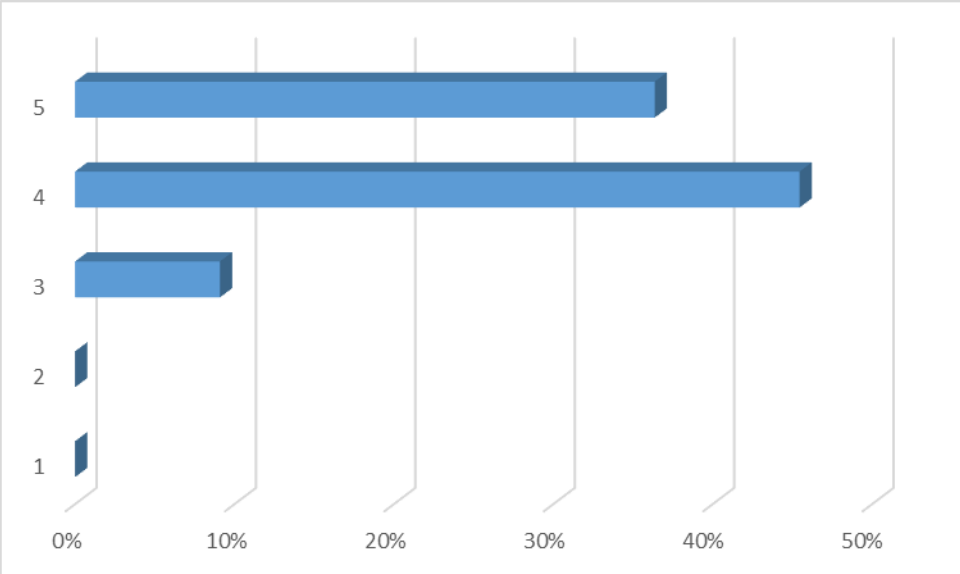


Figure 48 Answers to Q5 to CMEx evaluation questionnaire

<u>Q6. Based on your operational experience, how do you rate the topics proposed for the evaluation of the players performance?</u>

The topics proposed in the current CWA have obtained a very high score (1-poor, 5-excellent) (Figure 4).

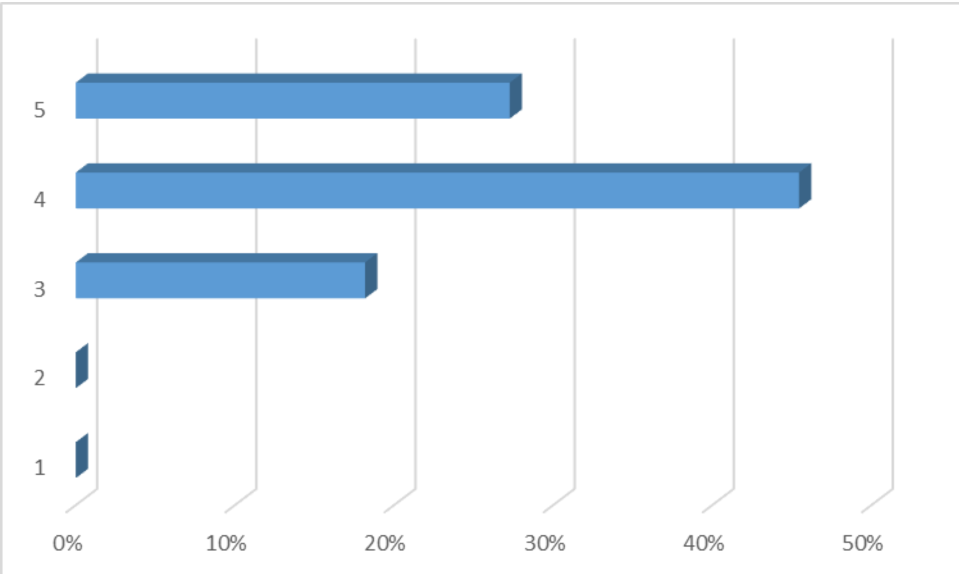


Figure 49 Answers to Q6 to CMEx evaluation questionnaire

<u>Q7. Do you agree about the integration of the chapter "Evaluation and validation of solution" in the current CWA aiming to the evaluation of exercises? (The evaluation of solution is addressed for trials, i.e. exercises that test technological and methodological solutions)</u>

The vast majority of the respondents agree or strongly agree about including the evaluation of solutions as special subchapter of the current CWA for which specific recommendations for the evaluation of the usability of the solutions exists. The evaluation and validation of solutions is considered as objective of an exercise developed and conducted for that scope.

Q8. In the chapter "Evaluation and validation of solution" most focus is given to the end-user perspective (usability and performance). Do you think other dimensions should be equally evaluated? (Note that some extended reference is made to the overall impact of the solution in its operational environment, e.g. organizational, political factors, etc)

The majority of the answers are negative, considering that the focus of the guidelines should be given to the suability of the solution and the practitioners' perspective. An interesting comment observes the fact that should the solution tested be a technical standard, then the validation is not about the usability of the solution and it is rather the contribution of the standard (here "solution") to the technical interoperability. As a matter of fact, the recommendations included in the validation of solution highlight that the validation topics and criteria strongly depend on the solution itself, its functionalities and the purpose it serves. The successful integration of technical solutions in the operative environment is recommended to be an additional element to be questioned in the validation, what is a partially already included.

Additionally, another participant recommended to follow closer the Trial Guidance Methodology of CWA 17514 and question the crisis management dimension and the solution dimensions separately. The former includes:

— the influence the crisis management (SOPs, roles, responsibilities, etc.) has on the trial;

— the impact that a solution has on the performance of the CM organization.

However, this is considered to be partially included in the evaluation of exercise conduct section, as in the current CWA, the trial is considered to be a type of exercise with given focus on the solutions tested.

Q9. Do you agree with the differentiation of the terms "validation" and "evaluation"?

All respondents agree or strongly agree with this differentiation and the definitions provided. Some comments are included:

- The use of an additional term for validation, more directly related to technical issues (e.g. verification) is suggested, however it was considered that this is out of the scope of the current CWA
- The evaluation and validation especially for the case of technical solutions are not necessarily related to usability
- To justify the use of alternative definitions when not adopted the ones from international documents (e.g. ISO 22300). The definition of "evaluation" is adopted from ISO 22398 with an additional exercise-related comment and the "validation" definition is created for this document based on the experience of the workshop participants and input from international literature.

Q10. Do you find useful the After-Action Resume (renamed to After-Action Summary)?

Everyone but for one participant found useful the After-Action Summary, for which the structured template is proposed.

Q11. How do you see the intended use of the After-Action Resume (renamed to After-Action Summary)?

The following comments are communicated:

- It is very helpful and it should be used generally.
- The resume provides helpful summarized strategic opinions for the future!!
- A good and standard report of exercise expectations
- It is a useful document to discuss and reflect about the exercise objectives, results and main feedback received
- Makes a lot of sense, it could make sense to ask for more detailed information here and there, but this is for sure a very good starting point

Q12. How much do you agree with the content of the After-Action Resume (renamed to After-Action Summary)?

As depicted in Figure 5 the majority of the respondents agree with the content of the After-Action Summary, while the comment about including a qualitative result, also discussed during the technical session, has been adopted.

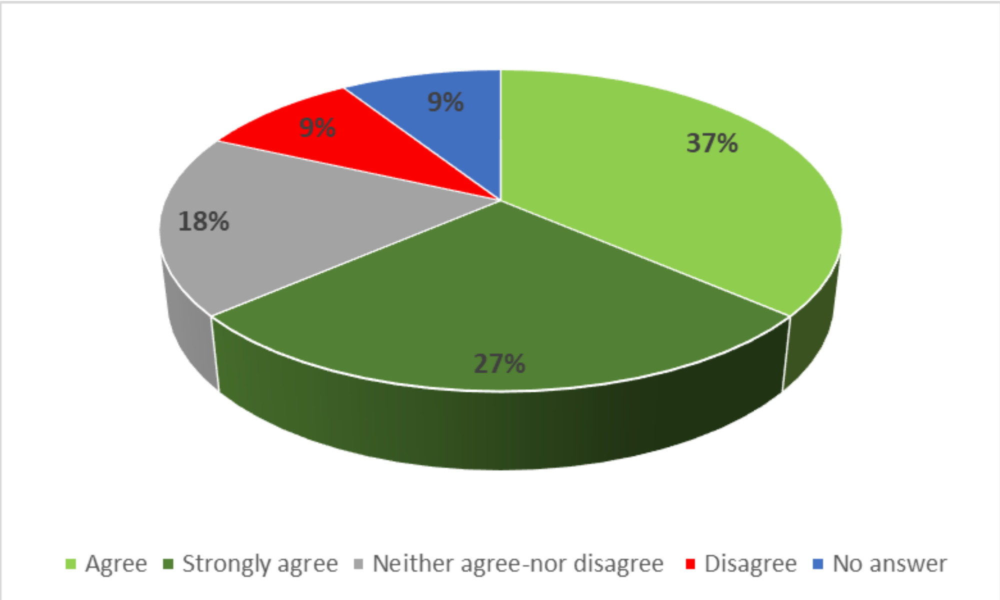


Figure 50 Answers to Q12 to CMEx evaluation questionnaire

Q13. In overall, in the CWA on Evaluation of exercises, would you expect and recommend to have additional aspects (e.g. training of an evaluator), rather than concentrating mainly on implementation guidelines for the data collection and analysis?

The answers to this question are split in two, as some participants would rather see some additional elements in the CWA. Some participants would like to see additional elements of the evaluation process into a standardization document of evaluation of exercises. A participant would prefer to differentiate the evaluation of trials and keep in the same document only evaluation of performance of players and solutions takes place simultaneously. This comment after discussions has been rejected as the trial has been recognized as a type of exercise.

Generic comments

Considering the existence of the CWA 17514 which focuses on the conduct and subsequently evaluation of trials, as well as the existence of standards on technical evaluation of systems, its has been commented that the current CWA focuses more on operational type of exercises. It should be noted that this is a valid comment and the trials have been included into the draft as a type of exercise with the participation of operational stakeholders aiming to validate the solutions tested only from their perspective and practical experience. There has been no deep dive on the particularities of the planning and conduct of a trial, nor on the technical verification of systems; both are out of the scope of the current CWA.

b) Questions with score evaluation score

The following questions are considered to provide evaluation of the CWA as solution tested, focusing on the satisfaction of the user and the applicability of the CWA content. Most of the questions directly respond to KPIs initially set. A numerical score is also computed according to the formula horizontally implemented for all CWA evaluations aiming to the quantification of the evaluations for comparative and demonstration purposes. All positive and respectively negative types of answer are grouped together e.g. Great/strongly agree depending on the form of the question.

The full questions, their short version and corresponding average score are enlisted in Table 25. In Figure 46 the distribution of the answers per question are illustrated. An overall satisfaction and positive response to all questions can be observed.

Table 27 Evaluation questions and average score for CMEx

Question	Short question	Average score
Q1. The Guidelines are addressed to all types of stakeholders of which mandate is the organization of and participation to exercises for operational training.	Q1. Guidelines addressed to all stakeholders	4.09
Q2. The guidelines are useful and applicable to technology providers that wish to test their solutions in an operational environment, allowing for the targeted development and eventual implementation of innovative solutions	Q2.Useful and applicable to technology providers	4
Q3. To what extent do you think that the current Guidelines cover satisfactorily a wide spectrum of the evaluation process (preparation, conduct and analysis), taking into account the available information in ISO 22398:2017?	Q3.A wide spectrum of evaluation process is covered satisfactorily	4.5

Question	Short question	Average score
Q4. The Guidelines are easily read and directly applicable, focusing on their implementation primarily by the evaluation leading team	Q4. Easily read and directly applicable	3.8
Q5. The Guidelines successfully consider a wide range of aspects to be evaluated during operational exercises and trials, spanning from players performance and organizations capabilities to technology performance and methodological procedures	Q5. A wide range of aspects is successfully considered	4
Q6. The Guidelines equally address the evaluation needs of all types of exercises (incl. trials) and of different scopes	Q6. The evaluation needs of all types of exercises are equally addressed	4.1
Q7. How likely would you recommend for implementation the current Guidelines in your everyday operational life?	Q7. Recommendation of Guidelines for operational everyday life	3.5
Q8. How likely would you recommend the current CWA to become a standard, taking into consideration the completeness of its content and the existing standardization gaps it addresses?	Q8. Recommendation for the CWA to become a standard	4.2

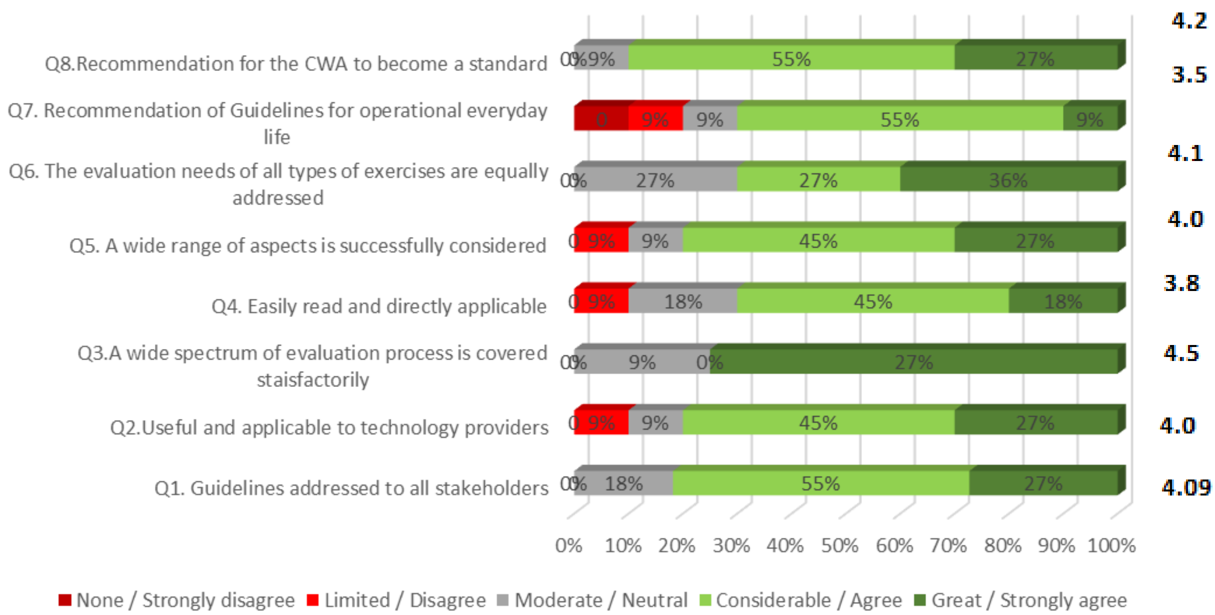


Figure 51 Distribution of responses at the evaluation questions of the questionnaire for CMEx

In Table 26 the KPIs are enlisted together with the matching evaluation questions of Table 25 that are considered to respond as well as possible to the satisfaction of the KPIs. The average score is the average of all contributing questions and the level of KPI satisfaction, also illustrated in percentage in Figure 47 demonstrates in average the level the KPIs have been met. In overall there is a very good satisfaction of the KPIs, over 80%.

Table 28 Matching of KPIs with evaluation questions and relevant score for CMEx

Key Performance Indicators (KPIs)	Question responding	Average score	Level of KPI satisfaction
Enhancement of FRs awareness for the need of improvement of their operational capabilities	Q7	3.5	70%
Relevance to all types of relevant stakeholders	Q1, Q6	4.095	82%
Evaluation of operational exercises	Q3, Q6	4.3	86%
Evaluation of tools/systems	Q2, Q3, Q6	4.2	84%
Usability	Q4, Q6	3.95	79%

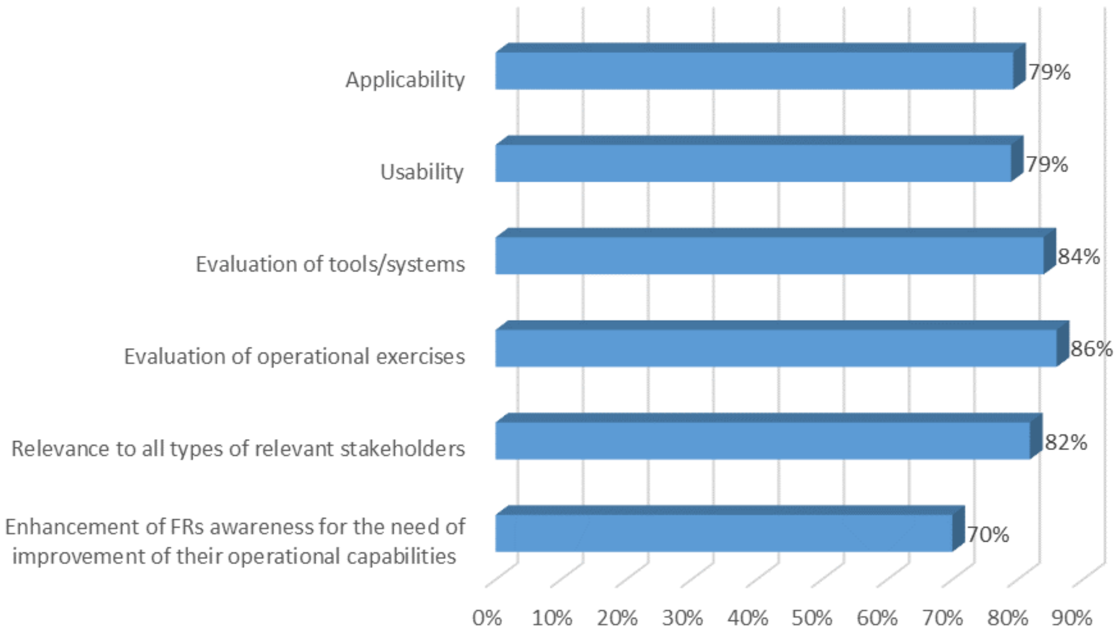


Figure 52 Level of satisfaction of each KPI for CMEx

3.7.2.4 Results from Formal Aspects Evaluation

The evaluation regarding formal aspects was conducted by Diana Iorga (ASRO) with the help of the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3, chapter 2.1.9). The majority of the types of questions were Likert scale (0 – most negative to 5- most positive answer), while there were also some questions with Yes/No answers which were also treated with a numerical scale: Yes-5 / No-0. In Table 27 the main criteria considered for the formal aspects evaluation have been enlisted together with the average score of answers. The average is computed from the quantified evaluation of the KPIs per criterion. Where the KPIs is scored with less than 5 leading to a reduction of the final score to the given criterion, the KPI has been marked in the comments column with the appointed score.

Table 29 Results of formal aspects evaluation for CMEx

Formal Criteria	Answers	Comments
C1. Using plain language	5	
C2. Quality of title	5	
C3. Quality of Table of contents	5	
C4. Quality of Foreword	5	The text is in Introduction as decided / Not all the participants of the KOM are yet listed
C5. Quality of Introduction	5	
C6. Quality of scope	5	2: Does the normative reference clause include only references cited in the text in such a way that some or all their content constitutes a requirement?
C7. Quality of Normative references	4	
C8. Quality of Terms and definitions	5	
C9. Quality of clauses and subclauses	4.4	4: The clauses are brief enough and include information related to one major idea
C10. Quality of Notes and examples	5	
C11. Quality of Tables and Figures	5	Where applicable
C12. Quality of Annexes	5	
C13. Quality of Bibliography	3.5	2: There is no confusion between bibliography and normative references
C14. Quality of Graphical symbols	not applicable	

In addition, two comments were included. The first refers to the difference of Normative references and Bibliography and provides specific recommendations about moving elements from Normative references to the Bibliography as they are not indispensable for the application of the document. Documents from which examples are derived, recommended options or are quoted as sources of definitions or recommendations, should be included in Bibliography.

The low scoring of the two criteria of Table 27 refer also to the small confusion between normative references and bibliography.

Following the comments above and direct contact with Diana Iorga (ASRO), the situation has been clarified and the relevant documents have been properly placed in Normative references, when indispensable, and to the Bibliography.

An additional comment refers to the title of Clause 4 of the CWA Symbols and Abbreviated term. It is recommended to be renamed to Abbreviated terms or Abbreviations as no symbols are contained. The comment has been adopted.

Moreover, there has been effort in making briefer and clearer some long clauses, per the comment of the Criterion 9.

Although the initial scoring of the formal evaluation was very high, it can be considered that after adopting all the comments, significant improvements were conducted to the document, at least as far as formal aspects of a standard are concerned.

3.7.2.5 Results from RAF tool Analysis

During the STRATEGY project the feasibility, impact and urgency of the standardisation activities of Stream were assessed also with the help of the ResiStand Assessment Framework (RAF), see D2.4. This has been populated by the CWA leader together with the end-0users of STRATEGY project that participate into the CWA development. For the final evaluation of the CWA “Evaluation of exercises – Implementation guidelines, its impact and importance for crisis management, the RAF supported analysis was reconsidered and updated.

In Figure 48 and Figure 49 the main information on the scope of the standard, the stakeholders it is addressed to, the impact to them and the importance of their participation in the development of the CWA are included in the “Intake” of the tool. In Figure 50, the overall assessment of the standardization activity scores very high in all its three dimensions: 4.3 in feasibility (high), 4.4 in impact (considerable) and 4 in urgency (high)

The current CWA has been supported by a number of participants, with operational, scientific and technical background, who contributed actively to the drafting of the CWA, recognizing the importance and necessity of having a standardization document clarifying and supporting the evaluation phase of the exercises, including trials for validating solutions. The wish for proceeding to the proposal of a New Working Item for standardization on the topic of evaluation of exercises, complementing ISO 22398, has been strongly expressed by workshop participants and other external experts.

Intake

ResiStand

Title of the proposed standard:	Evaluation and assessment reporting of exercises
Identification number:	

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):	Organisation or project consortium	Stakeholder category
	STRATEGY	Hybrid
	ICCS	Research
	KEMEA	Research
	CNVVF	End-users
	HP	End-users

Type of standard:	Workshop Agreement
Scope of the standard:	The main scope is the standardisation of procedures related with the evaluation and concise assessment reporting of exercises. This will enable the identification of the most important conclusions of exercises and will ensure that beneficial information is factored for continuous improvement and enhanced preparedness practices of incidents in future exercises, as well as operational capability building.
Example or illustration 1:	
Example or illustration 2:	

Compliance with legislation	Compliant (Y/N)?	Explanation
with European legislation:	Unknown	Compliance with ISO 22398:2013 and US Homeland Security Exercise and Evaluation Program (2020). Compliance with CWA 17514:2020 Trial Guidance Methodology
with national and regional legislation:	Yes	Compliance at least with the guidelines of the Greek Civil Protection Authorities.

arget groups for applying the standard	Target group	Free space for additional comments
First Responders:	Yes	Firefighters, Police, Crisis Management Operation Centers etc.
Governmental organisations:	Yes	Civil Protection/ Public Safety Agencies/Ministries
NGOs:	Yes	e.g. Red Cross and others participating in and organizing exercises
Industry/SMEs:	Yes	For the case of exercises developed as trials for the testing and validation of technological tools
Consultancy organisations:	Unknown	Possibly, for organizations organizing exercises
Research institutes:	Yes	Research institutes developing new solutions for exercise evaluation and developing technological and other solutions they wish to validate (trials)
Standardisation bodies:	Yes	National SBs, CEN, International Standardisation bodies
Others:	Yes	European Commission, EU Civil Protection Mechanism

Figure 53 RAF tool Intake sheet (1/2) for CMEx

Potential impact and urgency

Benefits for stakeholder categories	Description of who will benefit and in what way
End-users:	Governmental organisations (e.g. CPA) and FRs will be benefited by putting into practice standardised evaluation and structured reporting format of the outcomes of exercises.
Industry:	The outcomes of the proposed standardisation item will lead to a better understanding of the pros and cons of exercises and potentially provide the opportunity to industry partners to develop customised solutions for crisis management exercises based on the evaluation outcomes. Moreover, the industry may develop technological solutions to assist evaluation on digital format and calulculation of KPIs. Finally, should exercises (trials) are developed for the testing and validation of tools towards their operationalization, the CWA support targeted evaluation by end-users, allowing for their improvement.
Research:	The outcomes of the proposed standardisation item will support and guide research organization on the development of evaluation methodologies and in collaboration with FRs to quantify thresholds on required KPIs. Additionally, will allow the preparation of the evaluation process of a trial, developed for the testing and validation of research solutions.
Policy makers:	Improved understanding of planning and execution of exercises, through the outcomes obtained by the evaluation and the concise reporting, will lead to potential new practices that will be used to improve the future crisis management exercises and most importantly to work towards enhancement of operational preparedness and response and the main objectives of the exercises.
Citizens:	Voice will be given to citizens, through the participation to the exercises and their evaluation, to express their risk perception and needs on targeted actions towards their risk awareness and preparedness, in collaboration with FRs

Urgency of the standard: (when is it needed for implementation?)	High (within 1 year)	The proposed standard is expected to play a significant role in the evaluation of exercises and its outcomes potentially can enhance the planning and purpose of future exercises,aiming to the enhanced preparedness of all actors in crisis/disaster

Development

Required types of stakeholders	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	Main end-users that have experience in the implementation and evaluation of
Governmental organisations:	Yes	Main stakeholders organising, planning and taking part in the evaluation of exercises
NGOs:	Yes	NGOs that act as volunteer first responders having the experience of FRs
Industry/SMEs:	Yes	Being interested and having the experience of development of trials for the evaluation
Consultancy organisations:	Unknown	
Research institutes:	Yes	For the compilation of evaluation methodologies currently in use and improvement
Others:	Unknown	
Preferred leading type of stakeholder:	First Responders	To take advantage of their hands on experience and identified needs, also based on

Expected barriers and constraints:

Other information

Additional remarks:

ResiStand Assessment Framework 2.0

TNO

Figure 54 RAF tool Intake sheet (2/2) for CMEx

Assessment

ResiStand

Title of the proposed standard: Evaluation and assessment reporting of exercises
Identification number:

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
STRATEGY	Hybrid
ICCS	Research
KEMEA	Research
CNVVF	End-users
HP	End-users

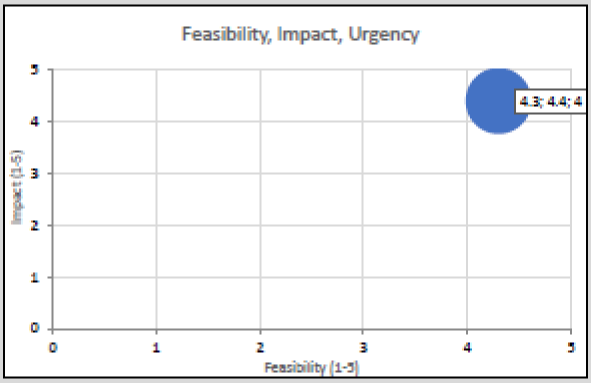
Scope of the standard: The main scope is the standardisation of procedures related with the evaluation and concise assessment reporting of exercises. This will enable the identification of the most important conclusions of exercises and will ensure that beneficial information is factored for continuous improvement and enhanced preparedness practices of incidents in future exercises, as well as operational capability building.

Type of standard: Workshop Agreement

Urgency (when needed?): High (within 1 year)

Overall Impact: Considerable

Feasibility: High



Legend (scores presented in the rectangle)
1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high
2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great
3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high

Impact

End-users

Improvement of DR and CM capabilities (functions/tasks): Great
Improvement of the safety of society: Great
Improvement of responder safety: Considerable
Cost savings for end-user organisations: Considerable

Industry & Research

Increase of business opportunities: Moderate
Improvement of business quality management: Considerable
Innovation progress: Great
Improvement of business functions: Great

Feasibility

Foundation: Sufficient
Development perspectives: Amply sufficient
Implementation and follow-up perspectives: Sufficient
Anticipated drawbacks and constraints: Sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard: Completely

Benefit of the standard to types of incident: Incidents in general
Specifically for the following incidents: -
-
-

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard: Change in crises, disasters and their impact (increasing number of natural disasters due to climatechange, increasing number of physical attacks, increasing number of cyber incidents/attacks, increasing number of cascading effects due to interdependencies).

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard: Regarding the technical trends, a significant percentage of exercises focuses on testing tools, systems and software. The assessment of the of the tools themselves and of the exercise players' performance could potentially pinpoint strengths and weaknesses both

ResiStand Assessment Framework 2.0

TNO

Figure 55 RAF tool assessment sheet for CMEx

3.7.2.6 Conclusion

The evaluation of the CWA "Evaluation of exercises – Implementation guidelines" went through different paths and processes, allowing for a number of, internal and external to the consortium, experts to provide their input and feedback on different elements of the draft, in addition to evaluating the usability of the CWA as a solution. The scope of the evaluation through the TTXs and the FSX was to understand the applicability of the CWA to multiple actors of crisis management, the ease of use and inclusivity of the document, the willingness of the end-users to professionally adopt it, etc. Meanwhile, the technical session of the FSX and the interaction with the professionals during the TTXs and the FSX, allowed for thorough discussions on several elements of the CWA draft, taking advantage of the operational experience of the participants. Necessary adaptations, alterations and upgrades have taken place following each phase of evaluation and experience exchange. The evaluation of the formal aspects allowed the upgrade of the draft according to the formal requirements of the standardization documents, while the update of the RAF tool was necessary in order to confirm the feasibility, impact and urgency for issuing CMEx as pre-standardization item. Following to that, proceeding to a New Working Item proposal for standardization has been strongly expressed, as it also revealed from the results the standardization assessment per RAF tool, and this is currently under consideration.

The Technical Specifications under Stream 7, namely "Specifications for Digital Scenarios for Crisis Management Exercises" and "Implementation Guidelines for Exercise Evaluation," embody a shared strategy aimed at enhancing the effectiveness of exercises. While this could necessitate practitioners to modify their existing protocols, it's crucial, much like in Stream 4, to establish uniform methodologies to achieve common objectives.

3.8 Final Evaluation of Stream 8 Terminology

3.8.1 Basis of the Final Evaluation

Subject of the final evaluation in Stream 8 – Terminology is the approach to map and match symbols presented in the

> CWA "International and interinstitutional crisis and disaster management - Guideline for the mapping of terminology and icons" [REF1].

This document was developed in a CEN Workshop with partners of the STRATGY project as well as external partners. In this document, adjustments, suggested during the intermediate evaluation based on the TTXs in Athens in June 2022 and in Lisbon in October 2022 were integrated. Further information on the TTXs may be found here [REF2] and [REF3].

The final evaluation was proceeded in the context of the FSX. As the CWA addresses the preparedness phase and not the response phase, the testing was not embedded in the exercise itself but ahead of the exercise. A group of four experts applied the methodology of the CWA on a prepared selection of symbols from ENGAGE system (of Satways) and from the SITAC symbology. Three of the experts were firefighters, one was a practitioner in civil protection.

Several data were collected before and during the FSX as a base of the final evaluation:

- Input from the live discussion between the participants of the exercise
- Notes collected by evaluators
- Registration forms with information on the respondents
- Notes provided by the participants through Mentimeter[1]
- Questionnaires, filled in by the participants at the end of the expert session

Due to the small number of participants and the wide options for the participants to provide feedback (Mentimeter, questionnaires), one evaluator was foreseen. This role was taken over by Trilateral. The session was led by Fraunhofer INT.

Additionally, feedback was constantly collected during the phase, when CWA was developed, in a CEN Workshop with participants which were members of the project consortium as well as externals. During the project, the CWA was also presented in STRATEGY dissemination and communication events, where experts such as practitioners gave their feedback. These events were:

- STRATEGY Interoperability Event, Rome, 15.02.2023-16.02.2023
- 2nd STRATEGY Workshop on Crisis Management and Civil Protection, Berlin, 26.04.2023-27.04.2023

In advance of the finalisation phase of the CWA, which took place in June and July 2023, the CWA was published for a public commenting period, where three comments have been received, which were all integrated to the document.

[1] Mentimeter is a tool for interactive presentations where the audience can participate, e.g. through surveys, free text comments, https://www.mentimeter.com/

During the TTXs, during the commenting period and the events mentioned above, participants discussed the approach of the CWA and suggested improvements. In the commenting period, three comments have been received. The comments were on missing descriptions and a missing figure number and have all been integrated. Constant feedback was that an approach of mainstreaming the symbols instead of developing this CWA for translating the symbols is preferable. The authors of the CWA admitted that this is true as the usage of the same symbols is the best way to avoid misunderstandings. But they pointed out that several attempts to mainstream the symbols exist. However, these attempts can only be realised in long time frames and are therefore not helpful for the current needs of interoperability. Other feedback concerned the comprehensibility of the approach. This feedback was integrated in some cases, where additional explanations were added.

3.8.2 Results from Experts Discussions and Testing during the FSX

During the session, experts discussed the approach while proceeding through the steps. The participants proceeded to the steps individually and submitted their result via Mentimeter. Then, differences in the results and interim results of the individual steps became visible. With that procedure, the common problem was addressed, that the answers of opinion leaders are more considered, while the opinion of rather quiet participants are overseen in group discussions [REF4]. After each round of submitting interim results, the results were discussed. Some differences can be led back to the limited time and therefore limited knowledge of the participants of the methodology described in the CWA and the symbology. Other than that, the discussion showed, that even the same symbology might not necessarily be used exactly in the same way. It might, for example, be the case that a symbology that is designed to communicate with different target groups might be used only to communicate with a small number of these different target groups. That leads to the conclusion, that the application of the methodology needs to be accompanied by experts not only for the symbology but also for the specific organisation. Furthermore, these findings show that the CWA is also useful as a tool to discuss the understanding and usage of symbols.

Participants also gave the feedback, that an example would help to understand the methodology. Therefore, an example was added as an Annex in the CWA.

3.8.3 Questionnaire Results

The small number of participants had the advantage that the discussion was very detailed and helpful findings were made but had the disadvantage that the results of the questionnaires are not sufficient for a quantitative analysis. Therefore, only the cumulative results of both TTXs (Athens in June 2022, 15 participants and Lisbon in October 2022, 21 participants) and the FSX are shown in Figure 56. Overall, the CWA reaches KPIs, which are above 3 out of 5 which means that in average participants would rather agree than disagree that the CWA fulfils the requested goals. The participants shared the understanding, that the CWA in general is needed with an average score of 4.21 out of 5 (Mapping of symbols is needed). Also, participants agree that the methodology is comprehensible (Comprehensibility of the methodology) with an average score of 3.66 out of 5 and that the formula is easy to apply (average score of 3.88 out of 5). With regard to the KPIs Accuracy of mapping, Compatibility, Applicability, Speed of communication it can be stated that in all cases more participants agree or strongly agree than disagree or strongly disagree. In these cases, numerous participants said that they neither agree or disagree. A reason for that might be that not all aspects are relevant for all practitioners, for example the question whether the solution is compatible with the systems in use can only be answered if you have knowledge about the technical architecture of a system.

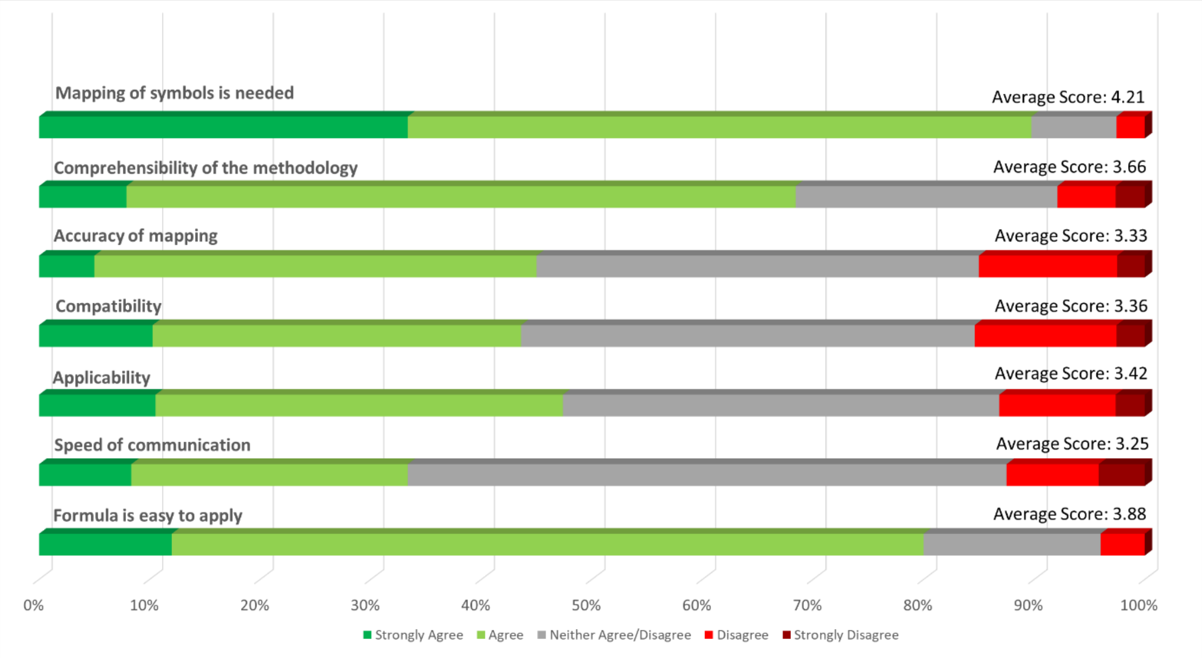


Figure 56: Scores' distribution obtained for all the statements in Stream 8 - Terminology questionnaires.

Additional explanations were given for the KPI Applicability and Formula is easy to apply:

- „You need a mathematical background to understand linear problems and their mathematical solution"
- "A computer can easily compute the solution"

Even though these answers were given for different questions, both concern the formula. These contradicting answers confirm the thesis that the high number of "neither agree/disagree" answers are caused by the different backgrounds of the participants.

3.8.4 Results from Formal Aspects Evaluation

The evaluation regarding formal aspects was conducted by Antti Karppinen (SFS) using the Formal Aspects Questionnaire (see STRATEGY deliverable D4.3 [35], chapter 2.1.9) and yielded the following results:

The CWA was given a very high score (average 4,96/5,00) by the evaluator. The answers (according to the Likert scale) received the value "5" (strongly agree) except Q.13, which received only the value "4" (agree). This means, that the CWA passed the formal Aspect Evaluation with high points.

According to the evaluator (who is a very experienced expert and executive in standardisation), the CWA is "an excellent example of a well-done pre-standardisation deliverable; on the one hand, it shows that the Workshop participants have really been committed to the drafting of the Agreement, and on the second hand, it underlines the impact of STRATEGY's iterative process—including the TTX and FSX exercises—on the quality of the resulting document".

3.8.5 Results from RAF Tool Analysis

During the STRATEGY project the stream 8 standardisation activities were assessed also with the help of the ResiStand Assessment Framework (RAF), see D2.4 [7] Annex 4. For the final evaluation the RAF supported analysis was reconsidered and updated. More details about the RAF in general can be found in chapter 2.4 of the present deliverable or in STRATEGY deliverable D1.2 [2] "Project Handbook".

The RAF Tool Intake sheets for the CWA of Stream 8 presented in this section, are an update of the RAF Tool Intake sheet presented in D2.4 [7] Annex 4. During the preparation of the CWA some adjustments have been made which made the updates necessary. An overview of the completed RAF intake sheet for the CWA is shown in Figure 57.

The results of the RAF tool based on the input is shown in It can be see that the final assessment shows very good results in terms of feasibility (3.8), impact (4) and high urgency (4) for the CWA.

Intake

ResiStand

Title of the proposed standard: Guideline for the mapping of terminology and icons

Identification number: 18023

Proposed standardisation activity

Proposing organisations or projects (at least 1 and up to 5):

Organisation or project consortium	Stakeholder category
Satways	Hybrid
INOV	Research
CINAMIL	End-users
Fraunhofer INT	Research
SFS (Standardisation Body)	-
ASI (Standardisation Body)	-

Type of standard: Workshop Agreement

Scope of the standard: This document provides recommendations for the mapping of different sets of terminology and icons used in international or inter-institutional crisis and disaster management. It provides an ontology for existing terminologies and taxonomies but will not develop a new set of terminologies and icons or provide linguistic translation. This document is applicable to all kind of crisis and all actors of crisis response across Europe that either support or get support by other actors from the same or another state.

Example or illustration 1: The problem occurs e.g. in wildfire hazards requiring the involvement of international resources. Institutions of different nations have to cooperate in fighting forest fires, support each other with resources, but use different terminology and symbols.

Example or illustration 2: In dealing with CBRN-E accidents, different institutions, such as civil and military, must cooperate. Usually, these instituions use different terminology and symbology. The mapping of the different sets of terminology and symbology can ensure that the cooperation is efficient and fast.

Compliance with legislation

	Compliant (Y/N)?	Explanation
with European legislation:	Yes	
with national and regional legislation:	Yes	

Target groups for applying the standard

	Target group (Y/N)?	Free space for additional comments
First Responders:	Yes	Not in first place responsible for applying the CWA, but users of translations of sets of
Governmental organisations:	No	
NGOs:	Yes	Relevant for NGOs that support or are First Responders
Industry/SMEs:	No	
Consultancy organisations:	Yes	To some extent - maybe for the mapping von sets of terminology/symbology
Research institutes:	Yes	Review and mapping of existing Terms and icons; standard development
Standardisation bodies:	Yes	Standard development
Others:	Unknown	

Potential impact and urgency

Benefits for stakeholder categories

	Description of who will benefit and in what way
End-users:	The international and interinstitutional interoperability in crisis and disaster management can be improved. This can simplfy the work of first responders
Industry:	Solutions can be more easy implemented in different countries and institutions
Research:	Deriving and documenting a knowledge base on terms and icons to be used in further interoperablity research
Policy makers:	Recuction of losses through more efficient and safe response operations
Citizens:	Recuction of losses through more efficient and safe response operations

Urgency of the standard: (when is it needed for implementation?) High (within 1 year) — Disasters - especially natural hazards as large wildfire events and heavy rain - are likely to increase with climate change

Development

Required types of stakeholders

	Required (Y/N)?	Free space to explain why their participation in the development is required
First Responders:	Yes	Knowledge base (mapping of existing terms abd icons), needs assessment, evaluation
Governmental organisations:	Yes	Political support for the development of a shared terminology
NGOs:	No	
Industry/SMEs:	No	
Consultancy organisations:	No	
Research institutes:	Yes	Implementing and testing the CWA
Others:	Unknown	

Preferred leading type of stakeholder: First Responders — Primary stakeholder and beneficiary of standard

Expected barriers and constraints: Multitude of terminologies and icons could complicate the identification of a semantic service applicable to more than two countries
Political resistance to adapting national (sovereing) approaches to civil protection

Other information

Additional remarks:

ResiStand Assessment Framework 2.0 TNO

Figure 57: RAF Tool Intake sheet for CWA on "International and interinstitutional crisis and disaster management – Guideline for the mapping of terminology and icons"

Assessment

ResiStand

Title of the proposed standard: Guideline for the mapping of terminology and icons

Identification number: 18023

Proposed standardisation activity and its perspectives

Proposing organisations or projects:

Organisation or project consortium	Stakeholder category
Satways	Hybrid
INOV	Research
CINAMIL	End-users
Fraunhofer INT	Research
SFS (Standardisation Body)	-

Scope of the standard: This document provides recommendations for the mapping of different sets of terminology and icons used in international or inter-institutional crisis and disaster management. It provides an ontology for existing terminologies and taxonomies but will not develop a new set of terminologies and icons or provide linguistic translation. This document is applicable to all kind of crisis and all actors of crisis response across Europe that either support or get support by other actors from the same or another state.

Type of standard: Workshop Agreement

Urgency (when needed?): High (within 1 year)

Overall impact: Considerable

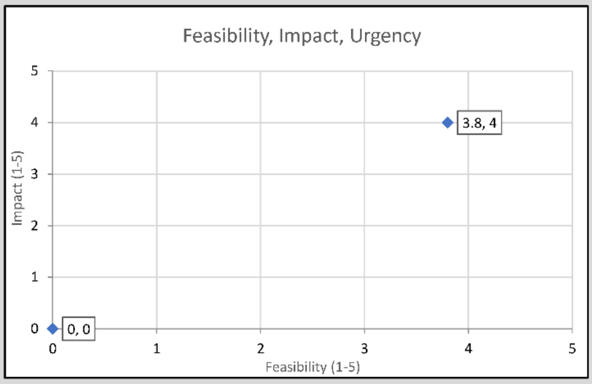
Feasibility: High

Legend (scores presented in the rectangle)

1st Feasibility: 1=Very low; 2=Low; 3=Medium; 4=High; 5=Very high

2nd Impact: 1=None; 2=Limited; 3=Moderate; 4=Considerable; 5=Great

3rd Urgency: 1=Very limited;2=Limited;3=Moderate;4=High;5=Very high



Impact

End-users

Improvement of DR and CM capabilities (functions/tasks):	Great
Improvement of the safety of society:	Considerable
Improvement of responder safety:	Considerable
Cost savings for end-user organisations:	Great

Industry & Research

Increase of business opportunities:	Great
Improvement of business quality management:	Moderate
Innovation progress:	Considerable
Improvement of business functions:	Limited

Feasibility

Foundation:	Moderate
Development perspectives:	Sufficient
Implementation and follow-up perspectives:	Moderate
Anticipated drawbacks and constraints:	Amply sufficient

Other issues

Potential ethical, social and/or legal effects of the proposed standard:	Considerably
Benefit of the standard to types of incident:	Incidents in general
Specifically for the following incidents:	CBRN attack Wildfire -

Relevant trends

Trends in society, in incidents, and/or in disaster resilience and crisis management that are typically anticipated by the standard: The relevance of interoperability in crisis and desaster management is increasing due to two general trends. The globalisation leads to the challenge that crisss and desasters do not only have impacts on a regional or national level. E.g. the spread of the corona

Technical and non-technical trends of interest for industry and research that potentially are addressed by the standard:

ResiStand Assessment Framework 2.0 TNO

Figure 58: Assessment of the RAF tool for the CWA "International and interinstitutional crisis and disaster management – Guideline for the mapping of terminology and icons"

3.8.6 Conclusions

The CWA "International and interinstitutional crisis and disaster management – Guideline for the mapping of terminology and icons" provides a sufficient solution for an urgent challenge. As a further development of an existing standardisation item (CWA 17335), it is suitable for the existing standardisation landscape. As the FSX session had only four participants, it was not possible to measure the improvements quantitively but feedback was taken seriously and implemented whenever meaningful. Overall, the high RAF tool results and the predominantly positive answers in the questionnaire confirm that. With the submission to CENCELEC and the publication of the CWA the CEN workshop has been successfully completed. In order to transform the CWA from a pre-standard to a standard, the dissemination of the standard, which was already done during the project and during the development of the CWA, needs to be continued. Practitioners, as a first step STRATEGY partners, need to start implementing the CWA into their work. This needs to be accompanied by all CEN workshop participants.

4 General Conclusions and Outlook

In the STRATEGY project, the following aims were achieved: Crisis Management related gaps were identified **Error! Reference source not found.**, solutions were proposed, and a considerable number o f pre-standardisation documents, related to these solutions, were developed, for at least one of each of the eight streams of the project (see STRATEGY deliverables D2.1 [4] to D2.8 [11], D3.2 [26] to D3.9 [33] and D5.1 [36] to D5.8 [43]).

It turned out that for all streams the intermediate evaluation (see D4.3 [35]) was an important stopover, which provided valuable feedback and suggestions for improvements. For all pre-standardisation documents developed in the framework of the STRATEGY project the related KPI group leaders could state that the documents were on the right track and now it can be stated that all CWAs/TSs still were substantially enhanced after the related TTXs, and that the results of the intermediate evaluation have bene taken into account.

The final evaluation was mainly based on the activities held during the Full-scale exercise, but also on interviews, expert discussions, results from interoperability workshops, a second ResiStand Framework assessment and a formal aspects evaluation. By and large a positive evaluation of the CWAs/TSs could be attested. Positive assessments were achieved for almost all different KPIs measured by questionnaires in the context of the FSX and the formal aspect evaluation. The intensity of the expert discussions showed that the standardisation activities arouse great interest and acknowledgment. In one or the other aspect, suggestions for improvement were received, which could at least partly be included in the very last versions of the pre-standardisation and standardisation documents.

The following table provides a summarizing overview of main results which enables the reader to compare the status of the various CWAs/TSs.

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
1 Search & Rescue	CWA Requirements for acquiring digital information from victims during Search and Rescue operations.	**SA 1**	The findings demonstrate that the CWA "Requirements for acquiring digital information from victims during Search and Rescue operations" is on the right track. Operational stakeholders view the CWA as both valuable and promising for integrating advanced technology into search and rescue efforts following mass casualty incidents. The CWA underwent thorough examination and testing by First Responders in their user capacity, as well as from the viewpoint of SMEs/RTOs, who develop similar solutions. The notable high average rating across all questions attests to the successful achievement of the CWA's defined objectives. Feedback from key stakeholders has been instrumental in shaping the subsequent steps for pre-standardization activities. The CWA document presents crucial guidelines for the functionality of digital victim tracking solutions, without imposing limitations on potential technologies or approaches for their implementation. Moreover, the CWA seamlessly aligns with various aspects of standardization, such as data traffic standards, data sharing, and triage rules. These topics were extensively deliberated in expert discussions and thoughtfully integrated by the CWA developers.
2 Critical Infrastructures	CWA Emergency management – Incident situational reporting for Critical Infrastructures	**SA 2.1**	During the evaluation, some attendees suggested adding terms for the number of available hospital beds in the semantic layer, although this is already addressed by the existing EDXL-HAVE standard. Valuable feedback led to the identification of 14 missing or needing-amendment concepts in the semantic layer, with only 12 concepts truly absent. Notably, the inability to express building collapse status and the absence of values for CBRNE event types in the semantic layer prompted significant attention, resulting in revising these concepts. Additionally, the need for clarity regarding the crisis management phases covered by EDXL-CAP and EDXL-SitRep was addressed, with attendees expressing agreement on the evaluation's quality and outcomes in CWA 1.
	CWA Semantic layer definition and suitability of EDXL-CAP+EDXL-SitRep	**SA 2.2**	The Field Simulation Exercise (FSX) has demonstrated that the CWA titled "Emergency management – Incident situational reporting for Critical Infrastructures" is on the right trajectory. The document's structure and content effectively serve the purposes of incident notification, reporting, and improving situational awareness. Notably, this progress is evidenced by the incorporation of additional fields, clearer content descriptions, and the potential to drive technological advancements. The successful implementation of an online form and the effective demonstration of an incident notification and reporting command-and-control system during the FSX highlight this potential. The participants'

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
			understanding of the CWA's compatibility with other standards improved, albeit this understanding varies based on the respondent's awareness of existing technical standards—a subjective aspect. As indicated by the CWA's title, the relevance of preexisting standards like EDXL-CAP and EDXL-SitRep is duly acknowledged and addressed within the context of the CWA.
3 Response Planning	CWA Structuring an emergency response plan for crisis management stakeholders	**SA 3**	The validation steps following the tabletop exercises highlighted the need for standardizing the response planning approach for plan document structure. Feedback collected through surveys and discussions indicated favorable assessments of Key Performance Indicators (KPIs) by participants and project event attendees. These evaluations, along with exercise-related comments, guided improvements in the Content Work Area (CWA). The CEN workshop participants found these contributions valuable in enhancing the CWA's utility for organizing plans, resulting in enriching and updating the document accordingly. The proposed structure of the CWA received positive feedback for effectively promoting coordination, collaboration, and decision-making among authorities during emergencies, facilitating efficient and timely disaster response. Overall, the CWA offers a common approach that aids practitioners in harmonizing planning processes, enhancing coordination among crisis management stakeholders, and optimizing response operations' efficiency.
4 Command & Control	CWA Collaborative emergency response – Common addressing format and emergency identification protocol	**SA 4.1**	The proposed common identification protocol and naming service have the potential to enhance information sharing among relevant agencies during situations. This approach addresses existing gaps that hinder efficient information exchange among these agencies' command and control (C2) systems during emergencies. Consequently, it is recommended that the proposed CWA (Common Working Approach) be finalized and formally submitted. This requires refining the protocol's structure, hosted information, and supported operations. It's important to incorporate insights from collaborative practices utilized by other organizations like INSARAG.
	CWA Management of forest fire incidents – SITAC-based symbology	**SA 4.2**	The feedback collected through both distributed questionnaires among FSX participants and open discussions consistently conveyed the perception that the proposed symbol set for wildfire management is highly valuable and holds significant importance for stakeholders engaged in such

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
			incidents. Participants emphasized that the primary objective and priority lie in employing this symbol set on paper maps during field operations, allowing for swift visualization of the ongoing situation and effective communication among stakeholders. The preference for paper maps stems from the practical challenges faced by operational personnel, rendering digital maps and associated tools less practical. Consequently, the key takeaway is that the CWA should be finalized and submitted, while incorporating final adjustments and refining its categories and symbols in specific instances. These adjustments should be made while considering the alignment with existing national-level symbols, standard operating procedures, and best practices, which can be contributed and deliberated upon by members of the CWA.
5 Early Warning	CWA Requirements and recommendations for social media early warning messages in crisis and disaster management	**SA 5.1**	The evaluation process, including both the intermediate evaluation and subsequent assessments, highlights the effectiveness of the efforts put into CWA 1 titled "Requirements and recommendations for social media early warning messages in crisis and disaster management." The initial need identified by Stream 5 has been successfully addressed, as evidenced by positive feedback and valuable input received during testing, including the Tabletop Exercise (TTX) and Field Simulation Exercise (FSX). The CWA 1's value in standardizing social media early warning messages is underscored by its alignment with existing warning standards. The evaluation process, conducted in two stages with different perspectives, allowed for comprehensive integration of viewpoints. The TTX involved practitioners and emergency management experts, while the FSX engaged communication experts and operational actors, affirming the CWA 1's applicability and effectiveness. This collective effort, coupled with extensive prior work, led to a well-received CWA 1 validated by diverse stakeholders. Moreover, the RAF assessment reveals potential business opportunities for market players to support first responders and local authorities. The publication of CWA 1 in July 2023 creates further potential for interested parties to develop it into a full-fledged standard.

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
	CWA Exchanging of building and infrastructure damage information with Common Alerting Protocol	**SA 5.2**	The feedback collected from questionnaires and discussions underscores the positive evaluations provided by FSX participants regarding the Key Performance Indicators (KPIs). This evaluation, coupled with comments and insights gathered during the exercise, has been instrumental in refining the content of the ongoing discussion around CWA 2. The working group's approval of the proposed new parameters acknowledges their importance in enhancing the communication protocol established within CWA 2. Consequently, the document has been updated and enriched accordingly. Overall, the suggested structure of CWA 2 received favorable feedback for its efficacy in promoting coordination and collaboration among pertinent authorities during emergencies. Additionally, it received praise for its contribution to improving decision-making processes and enabling a more efficient and timely disaster response.
6 CBRN-E	TS Societal and citizen security — Digital Chain of Custody for CBRN-E events — Part 1: Overview and concepts	**SA 6.1**	The results indicate that the Technical Specification (TS) titled "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence" is making substantial progress, with evident improvements made following the mid-term evaluation and discussions within CEN/TC 391/WG2 - Societal and citizen security - CBRNE. The adoption of a digital chain of custody necessitates adherence to data security standards, a requirement well addressed by the approach outlined in the Conformity Assessment Work Area (CWA). A significant milestone was the inclusion of Stream 6 in the plenary session of CEN/TC 391, providing an opportunity to present the TSs. The highly positive reception of the digital Chain of Custody (dCoC) demonstration during the Field Simulation Exercise (FSX) serves as an encouraging indicator and motivator for submitting both TSs to CEN/TC for voting, with no further changes permitted at this stage. Given these advancements, the likelihood of the TSs transitioning into standards in the near future is notably high.
	TS Societal and citizen security — Digital Chain	**SA 6.2**	Overall, the findings point to substantial advancements in the Technical Specification (TS) titled "Societal and citizen security — Digital Chain of Custody for CBRNE Evidence," notably benefiting from

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
	of Custody for CBRN-E events — Part 2: Data management and audit		enhancements made after the mid-term evaluation and discussions within CEN/TC 391/WG2 - Societal and citizen security - CBRNE. The integration of a digital chain of custody necessitates adherence to data security standards, a requirement effectively addressed by the approach outlined in the Conformity Assessment Work Area (CWA). A pivotal step was Stream 6's inclusion in the plenary session of CEN/TC 391, providing a platform to showcase the TSs. The enthusiastic response to the digital Chain of Custody (dCoC) demonstration during the Field Simulation Exercise (FSX) serves as a promising sign and motivation for submitting both TSs to CEN/TC for voting, with no further modifications permitted at this stage. Given these significant strides, the prospects of the TSs evolving into standards in the near future are considerably promising.
7 Training	CWA Specifications for Digital Scenarios for Crisis Management Exercises	**SA 7.1**	The assessed final draft of CWA 1 titled "Specifications for Digital Scenarios for Crisis Management Exercises" is progressing positively, with operational stakeholders recognizing its value in modernizing crisis management exercise planning. The standardization effort received favorable feedback during round table presentations, demonstrations, and discussions, reflecting its promising potential for integration. Submitted for publication at CEN/CENELEC, the CWA was evaluated from various perspectives, including civil protection authorities, potential user first responder teams, and developer SMEs/RTOs. The high average score obtained from assessments indicates successful achievement of objectives, with stakeholder feedback guiding pre-standardization efforts. In the Field Simulation Exercise (FSX), the CWA effectively incorporated an existing solution (TMT) to showcase scenario inject integration using the proposed data model specification. This contribution in Stream 7 supports a collective strategy for enhancing exercise effectiveness, akin to the importance of uniform methodologies in Stream 4, even if adjustments to existing protocols are required for practitioners.
	CWA Evaluation of Exercises - Implementation Guidelines	**SA 7.2**	The evaluation aimed to gauge the CWA's applicability to different crisis management actors, its inclusivity, user willingness for adoption, and more. Technical sessions during the FSX and interactions with experts throughout the TTXs and FSX facilitated in-depth discussions on multiple aspects of the draft, leveraging participants' operational experience. Iterative adaptations, alterations, and enhancements were made based on each evaluation phase and experience exchange. The formal evaluation informed draft upgrades in accordance with standardization document requirements, and RAF tool updates confirmed the feasibility, impact, and urgency for pre-standardization issuance of

Stream	Pre-Standardisation Document	Standardisation Activity ID	Main Result
			"CMEx" (Crisis Management Exercises). Expressions for proposing a New Working Item for standardization emerged strongly from the results and the RAF tool assessment, currently being deliberated. The Technical Specifications within Stream 7, particularly "Specifications for Digital Scenarios for Crisis Management Exercises" and "Implementation Guidelines for Exercise Evaluation," embody a collective strategy to elevate exercise effectiveness, potentially involving modifications to existing protocols for practitioners. Like in Stream 4, establishing uniform methodologies remains vital for achieving shared objectives.
8 Terminology	CWA Guideline for the mapping of terminology and icons	**S A8**	The CWA titled "International and interinstitutional crisis and disaster management – Guideline for the mapping of terminology and icons" presents an effective response to an urgent challenge. Building upon the foundation of an existing standardization item (CWA 17335), it harmonizes well with the current standardization landscape. Although the FSX session had limited participation with only four attendees, improvements were made based on their meaningful feedback, contributing to overall enhancement. The notably high RAF tool results and the predominantly positive questionnaire responses validate this. The successful completion of the CEN workshop has been achieved through the submission to CENCELEC and the subsequent publication of the CWA. To transition the CWA from a pre-standard to a standard, sustained dissemination efforts are essential, building on the dissemination conducted during project development. Practical implementation of the CWA into work processes, beginning with STRATEGY partners as the initial step, should be supported by all participants from the CEN workshop.

In addition to the positive assessment of the pre-standardisation and standardisation items in the course of the exercises, the expert sessions and other events, the publication of the all of the CWAs and TSs by CEN-CENELEC can be seen as validation of the high quality of all CWAs and TSs, which were developed in the STRATEGY project:

Table 30: CWA Publications

Reference	Title	Status	Responsible Body	Last Milestone Name	Dav
CWA 18013:2023	Collaborative emergency response - Common addressing format and emergency identification protocol	Published	CEN/WS CER	DAV/Definitive text available	05/07/2023
CWA 18009:2023	Evaluation of exercises - Implementation Guidelines	Published	CEN/WS CMEx	DAV/Definitive text available	19/07/2023
CWA 18019:2023	Specifications for Digital Scenarios for Crisis Management Exercises	Published	CEN/WS DigScen	DAV/Definitive text available	19/07/2023
CWA 18004:2023	Requirements for acquiring digital information from victims during Search and Rescue operations	Published	CEN/WS DIV	DAV/Definitive text available	05/07/2023
CWA 18018:2023	Structuring an emergency response plan for crisis management stakeholders	Published	CEN/WS ERP	DAV/Definitive text available	26/07/2023
CWA 18023:2023	International and interinstitutional crisis and disaster management - Guideline for the mapping of terminology and icons	Published	CEN/WS IICDM	DAV/Definitive text available	16/08/2023
CWA 18028:2023	Semantic layer definition and suitability of OASIS EDXL-CAP and OASIS EDXL-SitRep	Published	CEN/WS IPCI	DAV/Definitive text available	23/08/2023

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Reference	Title	Status	Responsible Body	Last Milestone Name	Dav
	standards for crisis management in critical infrastructures				
CWA 18024:2023	Emergency management - Incident situational reporting for critical infrastructures	Published	CEN/WS IPCI	DAV/Definitive text available	23/08/2023
CWA 18022:2023	Exchanging of building and infrastructure damage information with Common Alerting Protocol	Published	CEN/WS RDA	DAV/Definitive text available	23/08/2023
CWA 18017:2023	Management of forest fire incidents - SITAC-based symbology	Published	CEN/WS SITAC	DAV/Definitive text available	26/07/2023
CWA 18005:2023	Requirements and recommendations for social media early warning messages in crisis and disaster management	Published	CEN/WS SMCD	DAV/Definitive text available	05/07/2023

Over and above 2 CWAs were finally selected to be promoted as New Work Item Proposals (NWIP): Stream 3 CWA "Structuring an emergency response plan for crisis management stakeholders" and Stream 4 CWA 2 "CWA Management of forest fire incidents – SITAC-based symbology". Last but not least the two TSs (Stream 6, TS "Societal and citizen security — Digital Chain of Custody for CBRN-E events", part 1 and 2) are already submitted as NWIPs to CEN/TC 391 for voting. This demonstrates that the STRATEGY project will have visible impact beyond the project lifetime.

Relating to the future of the standardisation activity on the project, it is worth drawing attention to a relevant conclusion from D1.7 [3] "Ethical, Societal, and Privacy Framework": There is an important distinction between (a) ethical and social impacts that fall within the scope of a standard, and (b) those that are relevant to the standard's implementation but fall outside of its scope. Several of such issues within the scope of the standards are touched upon implicitly here and are reported on systematically in D1.7[3] . Furthermore, as a number of the standards focus on facilitating digital interoperability, there are recurrent themes in privacy and especially data security that fall outside of the scope of these standards but nevertheless should be considered closely by those implementing them.

References

[1] STRATEGY deliverable D1.1 “Standardization landscape: Gaps and Opportunities”

[2] STRATEGY deliverable D1.2 “Project handbook: The project handbook describes the structures and practices of the standardization framework and defines the project’s approach, semantics and processes”

[3] STRATEGY deliverable D1.7 “Ethical, Societal, and Privacy Framework

[4] [STRATEGY deliverable D2.1 “Use Cases and TTXs for Search and Rescue interoperability standards”

[5] STRATEGY deliverable D2.2 “Use Cases and TTXs for Critical Infrastructure protection interoperability standards”

[6] STRATEGY deliverable D2.3 “Use Cases and TTXs for Response Planning interoperability standards”

[7] STRATEGY deliverable D2.4 “Use Cases and TTXs for Command and Control interoperability standards”

[8] STRATEGY deliverable D2.5 “Use Cases and TTXs for Early warning and RDA interoperability standards”

[9] TRATEGY deliverable D2.6 “Use cases and TTXs for CBRN-E related interoperability standards”

[10] STRATEGY deliverable D2.7 “Use Cases and TTXs for Training interoperability standards (crosscutting topic)”

[11] STRATEGY deliverable D2.8 “Use cases and TTXs Terminology interoperability standards”

[12] STRATEGY deliverable D2.10 “TTXs for Search and Rescue interoperability standards”

[13] STRATEGY deliverable D2.11 “TTXs for Critical Infrastructure protection interoperability standards”

[14] STRATEGY deliverable D2.12 “TTXs for Response Planning interoperability standards”

[15] STRATEGY deliverable D2.13 “TTXs for Command and Control interoperability standards”

[16] STRATEGY deliverable D2.14 “TTXs for Early warning and RDA interoperability standards”

[17] STRATEGY deliverable D2.15 “TTXs for CBRNE related interoperability standards”

[18] STRATEGY deliverable D2.16 “TTXs for Training interoperability standards”

[19] STRATEGY deliverable D2.17 “TTXs Terminology interoperability standards”

[20] STRATEGY deliverable D2.18 “FSX scenarios planning report”

[21] STRATEGY deliverable D3.1 “Initial version of Common Information and Integration Environment

[26] STRATEGY deliverable D3.2 “Standards selection, implementation, testing and interoperability assessment for Search and Rescue”

[27] STRATEGY deliverable D3.3 “Standards selection, implementation, testing and interoperability assessment for CIP M32”

[28] STRATEGY deliverable D3.4 “Standards selection, implementation, testing and interoperability assessment for Response Planning”

[29] STRATEGY deliverable D3.5 “Standards selection, implementation, testing and interoperability assessment for Command and Control”

[30] STRATEGY deliverable D3.6 “Standards selection, implementation, testing and interoperability assessment for Early Warning and RDA”

[31] STRATEGY deliverable D3.7 “Standards selection, implementation, testing and interoperability assessment for CBRN-E”

[32] STRATEGY deliverable D3.8 “Standards selection, implementation, testing and interoperability assessment for Training”

[33] STRATEGY deliverable D3.9 “Standards selection, implementation, testing and interoperability assessment for Terminology”

[34] STRATEGY deliverable D4.1 “Full scale exercise demonstration report”

[35] STRATEGY deliverable D4.3 “Intermediate evaluation and validation of crisis management solutions and pre-standardization items (from all user groups)”

[36] STRATEGY deliverable D5.1 “Pre-standardisation activities for Stream 1: Search and Rescue”

[37] STRATEGY deliverable D5.2 “Pre-standardisation activities for Stream 2: CIP”

[38] STRATEGY deliverable D5.3 “Pre-standardisation activities for Stream 3: Response Planning”

[39] STRATEGY deliverable D5.4 “Pre-standardisation activities for Stream 4: Command and Control”

[40] STRATEGY deliverable D5.5 “Pre-standardisation activities for Stream 5: Early warning and RDA”

[41] STRATEGY deliverable D5.6 “Pre-standardisation activities for Stream 6: CBRN-E”

[42] STRATEGY deliverable D5.7 “Pre-standardisation activities for Stream 7: Training”

[43] STRATEGY deliverable D5.8 “Pre-standardisation activities for Stream 8: Terminology”

[44] The ResiStand project, funded by EU, Frame Program “Horizon 2020”

[45] CEN/TS N582 Societal and citizen security — Digital Chain of Custody for CBRNE evidence — Part 2: Data Management and Audit, Version 1.2 (CEN/TC 391 meeting jan.2023)

[46] CEN, «Semantic layer definition and suitability of EDXL-CAP+EDXL-SitRep standards for crisis management in Critical Infrastructures,» 2023. [En línea]. Available: https://www.cencenelec.eu/media/CEN-CENELEC/News/Workshops/2023/2023-05-02/cen-wsipci_draft-cwa-1-for-public-commenting-v6-2.pdf.

Annex 1

Guidance for Evaluation Teams

appointed for the STRATEGY FSX

in Gualdo Tadino 27.-30.3.2023

V03

This guidance should assist the different Evaluation Teams (as announced by each Stream-KPI-Group in STRATEGY Task 4.3) with their work.

For general guidance see "Guidance for the Stream KPI Groups".

It is still a draft. Suggestions are welcome!

Working Steps:

1. **Check of available tools and background**

The Lead Evaluator should check, if the relevant tools, which you need for evaluation, are available:

- Participant Feedback Form (PFF) online version (provided by the KPI group leader or the T4.3 Leader)
- Participant Feedback Form (PFF) as a word document (provided by the KPI group leader)
- Participant Feedback Form (PFF) printed out at least 3 times (depending on the number of people in the discussion session)
- Evaluators Forms (as far as provided by the KPI group leader), see description in Stream 7 CWA 2 "Evaluation of exercises – Implementation guidelines", 1 printout for each evaluator
- Evaluation Plan. If there is no specific evaluation plan for the CWA, the present guidance may be considered as "Evaluation Plan"
- paper/tablet for own notes
- plane paper for participants (so that they can write their comments here)
- check with the CWA-session leader, if breakout sessions with small groups are planned. Then it might make sense, to ask for more evaluators, at best 1 for each group, but this will only be feasible, if there are just a few groups.

All evaluators in the Evaluation Team should be familiar to some extent with the CWA, which is subject of the CWA session. In case of need they should ask for an extract or a draft before, to read it.

2. **During the CWA-sessions on Tuesday and Wednesday:**

- all evaluators should take notes during these discussion-based exercises
- all evaluators should fill in their Evaluators Forms (as far as available)

- in case of small group discussions evaluators should join one group or observe the groups in turns
- the Lead Evaluator should take care that a reasonable time slot is available at the end to fill in the PFF

3. **At the end of the CWA-session:**

- The Lead Evaluator should explain the procedure with the PFFs.

Especially:

(1) participants feedback is highly appreciated

(2) participants should connect via their mobile phone or laptop or … (QR Code or Link)

(3) At the beginning there are questions about the participants. No one is forced to give his name or other personal details, but it would be helpful. E.g. one could group the answers according to professional background.

(4) instead of a name a code name may be given

(5) All feedback will be treated confidentially. In the report no names will be quoted

(6) participants should answer the online questions just by ticking. If they want to write comments and they find it cumbersome to do so on their mobile phone, they are invited to use the paper sheets (which are distributed). In this case everyone should write her/his name (or code name) on the paper. Also the number of the question, to which she/he wants to give a comment, has to be given for each comment. The name/codename is necessary to match the online PFF with the paper.

(7) The Lead Evaluator will collect the papers at the end of the CWA-session. All participants who will attend the FSX on Thursday are invited to get their notes on Thursday again (by the Lead Evaluator) and add something or change something after they have observed the trial. Also, they can fill in then a new online PFF – but they must write here the same name or code name as in the CWA session. Else they would be count wrongly as a new participant in the evaluation analysis.

(8) If participants have the feeling (A) they cannot decide what to tick, because of lack of experience or lack of knowledge about the CWA, they should tick “no answer”, NOT “neutral” or something like this.

If participants have the feeling (B): I see reasons for it and against it, in an equal way, then they should tick “neutral”.

(9) In case there is no “no answer” box: Participants should simply NOT tick anything in case (A), and may provide a comment “no answer” for clarification.

- if necessary the Lead Evaluator should show the PFF on a screen and go through the questions one by one (e.g. in case people have problems or if translation is necessary)
- the evaluators should give background information to the questions (which is e.g. available at the Evaluators Form), if there are problems
- The participants should have enough time to fill in the PFF.

- After the sessions all evaluators and CWA leaders should check, if they want to have the one or other question included in a general PFF, which could be presented on Friday. This may depend also on the situation on Friday, i.e. if and what kind of feedback may be expected by the participants on Friday (a related list of participants will be available then, and it should be checked, how many of them were present during the other days).
- Provide the selected questions or some new created questions to the T4.3 Task Leader. But keep in mind that these questions should be limited, so that we don't have 1000 questions in the PFF at the end (having in mind that there are 8 streams!).
- if some preliminary analysis of the online PFFs shows some results, you (and the CWA leader) want to share, please prepare some slide to do so during the plenary discussion on Friday – or tell the T4.3 Task Leader, if you want him to present it.

4. **During the FSX on Thursday**

- all evaluators should take notes, mainly related to "their" CWA. They should pay attention to strengths and weaknesses; references to plans, procedures should be listed; actions and consequences should be described and discussed with respect to the CWA. Also recommendations for improvements are welcome, of course.
- the evaluators should consider the FSX as trial concerning the CWA content. It is not task of the CWA Evaluation Team to evaluate the FSX as itself, they should just help to evaluate and validate the solutions, the content of the CWAs.
- all evaluators should fill in their Evaluators Forms (as far as available)
- all evaluators, at least the Lead Evaluator should be open to answer questions by the participants

5. **During the debriefing at the end of each scenario**

- all evaluators should take notes
- all evaluators should fill in their Evaluators Forms (as far as available)

6. **After the debriefing at the end of each scenario**

- evaluators should exchange their impressions, possibly share notes with the CWA leader/KPI leader.
- if some preliminary analysis of the online PFFs shows some results, you (and the CWA leader) want to share, please prepare some slide to do so during the plenary discussion on Friday – or tell the T4.3 Task Leader, if you want him to present it.

7. **During the final discussion/post exercise debriefing on Friday**

- all evaluators should take notes
- all evaluators should fill in their Evaluators Forms (as far as available)
- all evaluators should submit their Evaluator's Form and/or their notes to the KPI group leader or CWA leader.

Guidance for the Formal-Aspects-KPI-Group

V03

This guidance should assist the Formal-Aspects-KPI-Group in STRATEGY Task 4.3 with her work. It is still a draft. Suggestions are welcome!

The envisaged output and the document notations might be considered as mandatory. A compilation of input and output is given at the end of this guidance.

Working Steps:

1. **Get the Standardisation Documents, which are subject of the project STRATEGY from bitrix (https://satways.bitrix24.eu/~0mlNP or https://satways.bitrix24.eu/~zjmHL) or ask the stream leaders to provide it. They may serve as inspiration for the formal aspects KPIs.**

2. **Get the completed RAF tools for the standardization documents from bitrix (https://satways.bitrix24.eu/~0mlNP or https://satways.bitrix24.eu/~zjmHL) or ask the stream leaders to provide it. The completion of the RAF tool was/is to be done in the framework of WP2. The RAF tool is quite essential and it can provide quite a few ideas for KPIs.**

3. **Ensure that competencies related to the standardisation documents are in your group.**

4. **Identify criteria to evaluate the standardisation document(s) with respect to its structure and language and formal design like included figures etc.. Take into consideration the scope, objectives and characteristics of the standardisation document(s). Note that there is a difference between criteria, PIs and KPIs.**

Output: List of criteria (named: formal-aspects-evaluation-criteria-v01 and similar, always with version number, please)

What is the difference between formal and general criteria?

Formal criteria refer to the form of the standardisation document(s), i.e. to the structure, the clearness, the inclusion of supportive graphs etc. General criteria refer to the content of the standardisation document(s), but are not stream-specific.

Examples:

Derived from the ISO document “Guidance for writing standards taking into account…” possible criteria related to the **form** could be “Conciseness” or “Understandability”, and KPIs therefore could be:

- Is the language understandable by the expected users?

  This is something different from the content. The content might be great, but if the language is too complicated, e.g. with many technical terms, which might be not known by the users of the document – then formally the standardisation document is not as good as with respect to the content.

- Is the number of referenced standards minimized?

  The proposed procedures in the standardisation document might be very good, just the number of referenced standards is abundant and makes it too complicate to understand.

- Are supportive graphs included?

  The described procedure might help to overcome interoperability problems, but without graphs it is not well understandable.

- Is the standardisation document as short as possible?

  Also this is independent of the quality of the content.

All criteria related to the form, to the structure of the standardization document could be transferred into KPIs and they refer just to the document, to the text itself, not to the content.

A **general** criterium on the other hand could be “user friendliness” with a possible KPI:

- Is the proposed solution in the standardisation document user friendly?

  In this case the solution might be well described (form is good). But in practice it is not user friendly, because, e.g. firemen cannot handle a mobile phone in the described manner with protective cloves… user friendliness is important for all streams in general (maybe the restrictions like “protective cloves” are different).

  So this KPI-Question should be answered by end users (maybe in a questionnaire, where the end users may give a score from 1 to 10 or something similar)

A **special** criterium could be the “acceptance of suggested symbols”. It is special, because not in all standardization documents are symbols suggested. The KPI might be: Percentage of the end users from X different organisations and Y countries, which agree with the symbols - and this KPI should be 100% fulfilled (else symbols should be changed until all accept them). This could be checked with a survey, where the symbols are listed and the users are requested to tick YES or NO or MAYBE or whatever (not necessarily restricted to persons present at the TTX). How the KPIs are evaluated after they have been defined is subject of the data collection plan, but it might be helpful to write it to each KPI directly in the list of KPIs.

5. **Identify SMART KPIs to evaluate the standardisation documents in formal respect (see e.g. also ISO Guidance for writing standards)**

   - Output: List of SMART KPIs (named formal-aspects-evaluation-KPIs-v01, always with version number, please)

   - And be sure, that the SMART criteria are fulfilled. Be aware: Criteria are not KPIs and PIs are not KPIs. KPIs must be “Key”.

6. **Identify how the data should be collected, which allow to determine, if and how well the KPIs are fulfilled: Questionnaires, Feed-Back session, observation (time-measurements etc.)…**

Set up a data collection plan together with the supervising group and possibly other stream-KPIs-Groups. It is not necessary to do the formal evaluation during a TTX. It might be good to plan „Small scale validation events" (see proposal, online possible): 2 end user/1 tech partner for each . Cooperation with T4.2 and T4.5. A repeated evaluation may be considered. The "grow" could be documented that way. E.g. in the first evaluation the standardisation documents just gets 50% in KPI X. But in the second evaluation it gets 90%. This could be documented in the D4.3 then with coloured bars in a graphic figure.

Output: Data collection plan.

7. **Discuss criteria/KPIs/data collection within the streams for feedback. Share it with partners, which you expect to make valuable contributions.**

8. **Coordinate data collection with supervising group doing:**

Organize events for validation (online possible):

2 end user/1 tech partner for each std. item. Cooperation with T4.2 and T4.5.

9. **Analyse the collected data and provide a contribution to deliverable D4.3 to the supervising group. The deliverable D4.3 has to be finished after the FSX in 2023.**

INPUT:

- standardisation documents
- RAF tools completed
- Possibly list of KPIs from research leader
- data collection plan draft?

OUPUT:

1. List of criteria (named formal-aspects-evaluation-criteria-v01 and similar, always with version number, please)
2. List of SMART KPIs (named formal-aspects-evaluation-KPIs-v01 and similar, always with version number, please)
3. Data collection plan
4. Contribution to deliverable D4.3

Guidance for Stream KPI-Groups:

TTX/FSX result analysis

V08

This guidance should assist the different Stream-KPI-Groups in STRATEGY Task 4.3 with their work, especially after the related TTXs/FSX are completed. For general guidance see "Guidance for the Stream KPI Groups". It is still a draft. Suggestions are welcome!

The envisaged output of a Stream-KPI-Group and the document notations might be considered as mandatory. A compilation of input and output is given at the end of this guidance.

Definitions:

The following terms and definitions refer to the use of the related words in the project STRATEGY. They are no general definitions.

Standardisation Document: CWA or TS or existing standard

Working Steps:

1. **Check of available data**

   - Check, if you have the data, which you need for analysis.

Available Data (principally):

(1) Filled-in Questionnaires

(2) Notes by the evaluation team about discussions/hot wash during the TTX/FSX

(3) Notes by the evaluation team about their own observations and impressions

(4) (Tele-)Interviews of participants

(5) Other

To 1):

They should be collected by the Lead Evaluator. Check it with your Lead Evaluator. Check, if they are complete – it the Lead Evaluator has not yet done this: Tick it off at a participants list. Maybe there are still some filled-in Questionnaires at the host.

Some participants may be more involved in the project implementation – since they are strongly involved in writing the standardisation document or they are a member of the related KPI Group or of the KPI Supervising Group. Their contribution could be treated in consideration with their engagement in the standardisation document process separately.

If the number of Questionnaires is to small one should consider writing emails to ask for filling in.

To 2) and 3):

These information could be gathered at a tele-conference with KPI Group and evaluation team. Or you get written notes by the evaluation team.

To 4):

It might be decided during the evaluation process that single or group interviews would be good. They could be aligned to the Questionnaire, but could also include further questions. They should be documented by the interviewer in a written way.

To 5):

Whatever you find valuable in addition to 1) – 4) to be analysed, might be collected.

2. **Check the KPI Thresholds**

Assign a threshold to each KPI, which indicates, what level must be reached by the KPI (100% or 90% or 80%...). Include persons which are not so much involved in the development of the standardisation document (e.g. First Responders) to avoid to be based.

3. **Quantitative Analysis of the Questionnaires**

Analyse the answers of the **section in the Questionnaires, which refers to the evaluation of the standardization document** (in almost all questionnaires section 1) in a quantitative way and document it graphically (bar diagram and possibly additionally in another way, if some interesting could be illuminated in that way). The other sections of the Questionnaire referring to the performance of the TTX/FSX itself will be analysed separately (WP2).

Discuss the distribution of work with the stream leader and the person responsible for the standardisation documents (CWAs or TCs).

Distribution of results to deliverables:

- The results of standardisation-document-section of the Questionnaire will be presented in D4.3
- the other results (of the TTX/FSX-sections - organizational aspects - in the Questionnaire) in WP2 or T4.1 (also the After-Action-Report).
- The Questionnaires itself will be part of WP2 deliverables. That means, it is not necessary to repeat them in D4.3, we could just refer to WP2.

4. **Qualitative Analysis of the Questionnaires**

Collect the free text answers of the **section in the Questionnaires, which refers to the evaluation of the standardization document**. Discuss the distribution of work with the stream leader and the person responsible for the standardisation documents (CWAs or TCs). The results of standardisation-document-section will be presented **in D4.3/D4.4**, the **other results of the TTX/FSX-section (organizational aspects) in WP2 (also the Questionnaire in the After-Action-Report).**

5. **Circulation of Questionnaire Results**

Circulate the quantitative and qualitative analysis of the Questionnaires among your evaluation team and your KPI Group. And also among the **supervising group**. Deadline: 6 weeks after TTX/FSX. If this seems not to be possible, please contact the T4.3 leader.

6. **Fix content of immediate evaluation text**

Collect suggestions by the KPI Group members and the evaluation teams about the content of the paragraph which should be part of D4.3. This might be done via email, but probably at least at the end of this process a tele-meeting will be helpful. This is especially the case, if the notes by the evaluators during the TTX/FSX are not given to the KPI Group in written form. During the tele-meeting the presence of the task leader of T4.3 or at least someone from the supervising group is mandatory along with the stream leader and the person responsible for the standardisation documents (CWAs or TCs).

7. **Further Data collection**

In a tele-meeting (mentioned above in 6.) it should be decided, if more data would be desirable at the present stage. Keep in mind that formal aspects will be evaluated later by the Formal Aspects KPI Group. This is not your task.

If necessary, plan interviews or other events to get more data/more feedback. Discuss this with other KPI group leaders.

The data collection may also be postponed to a later phase of the project. Keep in mind that data will be collected also during the FSX, but it might be good to get some information earlier. Be aware, that the FSX might be not focused to your standardisation document. You may plan to test specifically the KPIs which were still under your threshold in this first round.

8. **Draft of intermediate evaluation text**

Write a draft of the paragraph in D4.3, which refers to the standardisation document and the evaluation based on the related TTX/FSX and possibly some further data collection (7.). For this purpose you will get a form. They main author should be **NOT** the stream leader or the person responsible for the standardisation documents (CWAs or TSs) to demonstrate that the evaluation is to some extent really independent. The main author should be on the other hand, involved and competent enough to write the paragraph. This might be the Lead Evaluator as appointed for the TTXs/FSX. Anyway, the author should integrate the content collected in step 6. Have in mind, that there might be more possibilities to test and evaluate the standardisation document in the framework of STRATEGY. At the present stage suggestions for improvements are certainly welcome and should be included into the paragraph in D4.3.

If then the standardisation document has been improved and is tested again (during the FSX e.g.), a further paragraph in D4.4 will be needed to describe the progress and evaluate the standardisation document finally.

Circulate the draft among your Evaluation Team and your KPI Group and also among the Supervising Group.

After confirmation by these groups the draft will be sent to the stream leader. Also he/she has the possibility to comment on the draft.

9. **Planning further data collection**

Discuss with the stream leader how the standardisation document could be improved and how the improvements could be evaluated (tele-meetings. FSX etc.). Plan data collection in that sense.

Overview of Input and Output:

INPUT:

(1) Filled-in Questionnaires

(2) Notes by the evaluation team about discussions/hot wash during the TTX/FSX

(3) Notes by the evaluation team about their own observations and impressions

(4) Possibly (Tele-)interviews of participants

(5) Other

OUTPUT:

(1) Analysis of Questionnaire Results (see step 3. and 4.)

(2) Intermediate evaluation text

Guidance for the Stream-KPI-Groups

V13

This guidance should assist the different Stream-KPI-Groups in STRATEGY Task 4.3 with their work. It is still a draft. Suggestions are welcome!

The envisaged output of a Stream-KPI-Group and the document notations might be considered as mandatory. A compilation of input and output is given at the end of this guidance.

Definitions:

The following terms and definitions refer to the use of the related words in the project STRATEGY. They are no general definitions.

Standardisation Document: CWA or TS or existing standard

Working Steps:

1. **Preparation of the Stream-KPI-Group:**

Identify TTXs in STRATEGY (and FSX), which are related to standardisation documents of your stream (see definition above).

2. **Contact the "Problem Owner" (see deliverable D3.1) of the related TTX:**

2.1 In XGM (see deliverable D3.1) these documents are mentioned in phase 1:

- "description of scope and context" (not the scope of the standardization document)
- "exercise objectives"

Get these documents by the "problem owner".

2.2 Ask for other documents from phase 1, that could be useful to define first criteria, and then (!) KPIs according to the SMART method.

2.3 Ask for criteria suggestions and KPI suggestions.

2.4 Also ask in general, what kind of improvement he/she wants to see, how the standardisation should improve the situation.

3. **Contact the "research leader" (see deliverable D3.1) of the related TTX:**

3.1 In XGM (see deliverable D3.1) these documents are mentioned in phase 1:

- "problem analysis"
- "definition of solutions"

Get these documents by the "research leader".

3.2 Ask for other documents from phase 1, that could be useful to define criteria, and then (!) KPIs according to the SMART method.

3.3 In phase 2 there should be the following documents:

- "mission criteria" and
- "solution criteria" and a
- list of related SMART KPIs

provided by the "research leader". Ask him for this.

There seems to be overlap with Task 4.3. You should take into account what was done before. You can use the KPIs. Focus in Task 4.3 is not the exercise itself, just the standardisation document.

3.4 In phase 2 there should be created a

- data collection plan by the "research leader".

There seems to be overlap with Task 4.3. You should take into account what was done before.

4. **Get the completed RAF tools for the respective standardization document from bitrix (https://satways.bitrix24.eu/~0mlNP or https://satways.bitrix24.eu/~zjmHL) or ask the stream leader to provide it. The completion of the RAF tool was/is to be done in the framework of WP2. The RAF tool is quite essential and it can provide quite a few ideas for KPIs.**

5. **Ensure that competencies related to the standardisation document are in your group.**

6. **Ensure that at least one practitioner/end user (first responders or other) is a member of your Stream-KPI-Group. In addition you should have close contact to other practitioners to show them the criteria and KPIs for reviewing. In case you can't win a practitioner over to be member you should have at least some contacts for the reviewing. Practitioners play a significant role in our project, because they are the target group, they are the one, whose work should be enhanced by the standardisation document and they can evaluate the solution.**

7. **Identify criteria to evaluate the standardisation document(s). Take into consideration the scope, objectives and characteristics of the standardisation document(s). Note that there is a difference between criteria, PIs and KPIs.**

Output: List of criteria (named: stream-1-evaluation-criteria-v01 and similar, always with version number, please)

Criteria which refer to the content (the solution) of the standardisation document(s), but are more general and could be applied also (or similar) to other standardisation document(s) should be listed in the same list, but separated from the other criteria. The supervising group will collect these criteria and produce a general criteria document.

Criteria which refer to the form/structure of the standardisation document(s) are welcome, too. They should be listed in the same list, but separated from the other criteria. The formal aspect group will collect these criteria and produce a formal criteria document.

What is the difference between formal and general criteria?

Formal criteria refer to the form of the standardisation document(s), i.e. to the structure, the clearness, the inclusion of supportive graphs etc. General criteria refer to the content of the standardisation document(s), but are not stream-specific.

Examples:

Derived from the ISO document "Guidance for writing standards taking into account…" possible criteria related to the **form** could be "Conciseness" or "Understandability", and KPIs therefore could be:

- Is the language understandable by the expected users?

  This is something different from the content. The content might be great, but if the language is too complicated, e.g. with many technical terms, which might be not known by the users of the document – then formally the standardisation document is not as good as with respect to the content.

- Is the number of referenced standards minimized?

  The proposed procedures in the standardisation document might be very good, just the number of referenced standards is abundant and makes it too complicate to understand.

- Are supportive graphs included?

  The described procedure might help to overcome interoperability problems, but without graphs it is not well understandable.

- Is the standardisation document as short as possible?

  Also this is independent of the quality of the content.

All criteria related to the form, to the structure of the standardization document could be transferred into KPIs and they refer just to the document, to the text itself, not to the content.

A **general** criterium on the other hand could be “user friendliness” with a possible KPI:

- Is the proposed solution in the standardisation document user friendly?

  In this case the solution might be well described (form is good). But in practice it is not user friendly, because, e.g. firemen cannot handle a mobile phone in the described manner with protective cloves… user friendliness is important for all streams in general (maybe the restrictions like “protective cloves” are different).
  So this KPI-Question should be answered by end users (maybe in a questionnaire, where the end users may give a score from 1 to 10 or something similar)

A **special** criterium could be the “acceptance of suggested symbols”. It is special, because not in all standardization documents are symbols suggested. The KPI might be: Percentage of the end users from X different organisations and Y countries, which agree with the symbols - and this KPI should be 100% fulfilled (else symbols should be changed until all accept them). This could be checked with a survey, where the symbols are listed and the users are requested to tick YES or NO or MAYBE or whatever (not necessarily restricted to persons present at the TTX). How the KPIs are evaluated after they have been defined is subject of the data collection plan, but it might be helpful to write it to each KPI directly in the list of KPIs.

8. **Identify SMART KPIs to evaluate the standardisation document(s).**

Output: List of SMART KPIs (named stream-1-evaluation-KPIs-v01 and similar, always with version number, please)

For each KPI an evaluation method should be identified. For example: Question to first responders in a questionnaire with quantitative analysis, question to the provider . You may group the KPIs, so that you don’t have to write always the same text.

And be sure, that the SMART criteria are fulfilled. Be aware: Criteria are not KPIs and PIs are not KPIs. KPIs must be “Key”.

KPIs which refer to the content (the solution) of the standardisation document(s), but are more general and could be applied also (or similar) to other standardisation document(s) should be listed in the same list, but separated from the other KPIs. The supervising group will collect these KPIs and produce a formal criteria document.

KPIs which refer to the form of the standardisation document(s) are welcome, too. They should be listed in the same list, but separated from the other criteria. The formal aspect group will collect these KPIs and produce a formal KPIs document.

9. **Identify how the data to determine, if and how well the KPIs are fulfilled, should be collected. Questionnaires, Feed-Back session, observation (time-measurements etc.)…**

Set up a data collection plan together with the research leader (see above).

Discuss, how the comparison baseline – innovation-line is to be implemented into the TTX or FSX concept. Can a certain part of the TTX/FSX be done without the solution proposed by the standardisation document and afterwards again with the new solution?

During each TTX and FSX there should be a **TTX/FSX data collection team**: at least 3 persons who are not involved in actions, who are just observing and taking notes. This might be the Stream-KPI-Group. More observers are invited, but their contribution might be limited.

Output: **Data collection plan**
and announcement of **TTX/FSX data collection team**.

Examples for Questions to end users:

1. Name some features of the presented solution, which address the gap xy (from gap list) and which improve the situation in comparison to the former state!
2. Are these features reflected properly in the requirements written in the standardization item?

KPI could be: at least one feature, properly reflected.

Alternatively features could be named in the questionnaire and just the quality of improvement should be assessed.

Or something like:

To which extend enables the proposed CWA for training the identification of capability needs e.g., operational capacity upgrade, new equipment, upgrade of communications etc.:

None – Limited - Moderate – Considerable - Great

(Thank you, John, for this example!)

10. Discuss criteria/KPIs/data collection within the stream for feedback. Share it with partners, which you expect to make valuable contributions.

11. Coordinate data collection with the TTX/FSX data collection team and the supervising group doing:

    - Organize events for validation (online possible):
      2 end user/1 tech partner for each std. item. Cooperation with T4.2 and T4.5.
    - Coordinate questionnaires, observation with TTX and FSX, T4.1 - T4.5

12. Analyse the collected data and provide a contribution to deliverable D4.3 to the supervising group. The deliverable D4.3 has to be finished after the FSX in 2023.

13. Personal data (e.g. regarding the questionnaires respondents) have to be deleted after the project ends and auditing has been finalised. A maximum period of 5 years after the end of the project is foreseen as a retention period due to potential auditing from the EC until then.

INPUT:

- “problem analysis”
- “definition of solutions”
- “mission criteria”
- “solution criteria”
- List of KPIs from research leader
- data collection plan draft?

OUPUT:

1. List of criteria (named stream-1-evaluation-criteria-v01 and similar, always with version number, please)
2. List of SMART KPIs (named stream-1-evaluation-KPIs-v01 and similar, always with version number, please)
3. Data collection plan
4. contribution to deliverable D4.3